

Beneficiary Yearbook

APP DEVELOPMENT TRACK

A record of the individuals who trained, built, and interned across the Cohorts of 2024 and 2025. Projects that culminated in a working, deployed application.

96

BENEFICIARIES

2

COHORTS

At a Glance

96

Total beneficiaries tracked

2

Program cohorts (2024–2025)

35

Home counties represented

89

Internship placements completed

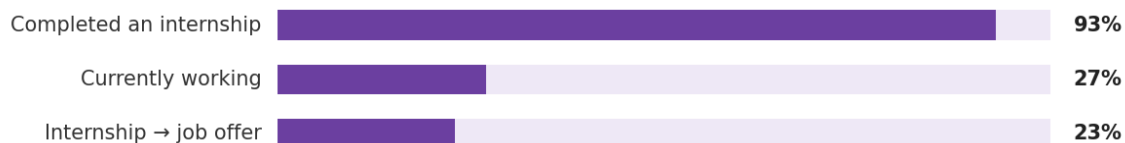
9

Internships that led to a job offer

15

Currently working / employed

Program reach in three numbers



Internship figure is out of all 96 beneficiaries. "Currently working" is out of the 56 who reported a status, and "Internship → job offer" is out of the 39 who answered the follow-up form.

Who Trained Where

Beneficiaries by cohort year



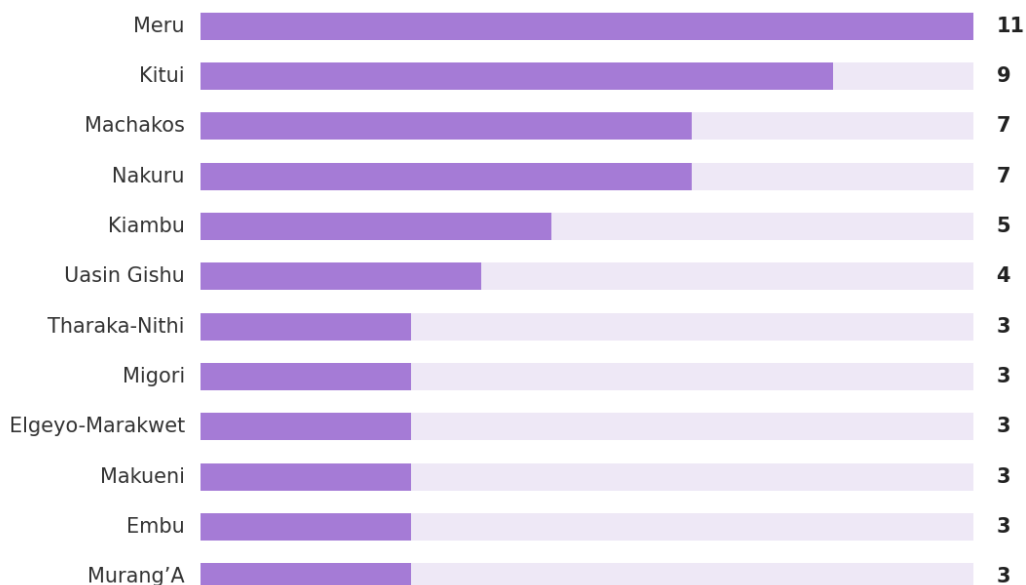
Regional training centre



Program tier



Home counties represented (top 12 of 35)



Where Skills Are Landing

15

Currently working or employed

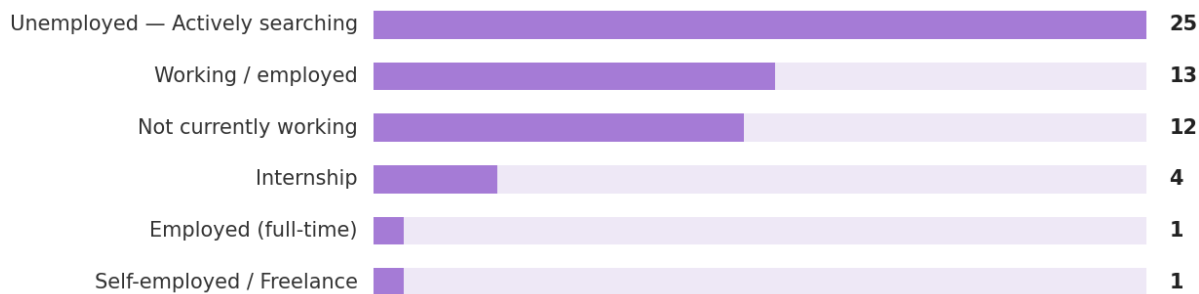
26

In roles related to the ENGAGE sector

9

Internships converted to a job offer

Self-reported current employment status



Role relevance to the ENGAGE sector



Status is reported for 56 of 96 beneficiaries; role relevance and job-offer answers come from the 39 follow-up form responses received so far.

App Development

Projects that culminated in a working, deployed application.

24

beneficiaries

DeKUT (6) · JOOUST (6) · Kabarak (9) · Pwani (3)



CAPSTONE Capstone Project

Predicting Individual Health Insurance Costs to Inform Risk-Adjusted Coverage

BACKGROUND & METHODOLOGY

This project develops and compares machine-learning models to predict individual medical insurance costs using demographic and modifiable risk factors (age, BMI, smoking, region, etc.). By improving performance from linear regression to Random Forest ($R \approx 0.86$), it enables fairer pricing, targeted risk counseling, and scenario testing in population health financing, with a simple Streamlit tool for frontline use.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The capstone project successfully developed and evaluated predictive models for insurance expense estimation. The initial multilinear regression model achieved an R-squared value of 0.7833, explaining 78.33% of the variance in insurance costs, with a Mean Absolute Error of 4,186.94. Model performance was significantly improved by implementing a Random Forest algorithm, which increased R^2 to 0.8629, reduced MSE to 21,274,322.98, and lowered MAE to 2,534.53, achieving an overall accuracy of approximately 86.29%. The trained model was serialized using joblib and successfully deployed as an interactive Streamlit web application, enabling users to input demographic and health characteristics to receive real-time insurance expense predictions. Key findings identified smoking status as the strongest predictor of insurance costs, followed by age and BMI, providing actionable insights for both risk assessment and personalized premium calculation in healthcare financing.

INTERNSHIP Internship Experience

Karatina Sub-county Hospital — HIV patient management, treatment adherence, and health risk stratification

Dataset Used: Comprehensive Care Centre (CCC), Karatina — patient records including treatment regimens, CD4 counts, BMI, blood pressure, age, and sex. The datasets captured detailed clinical and demographic information, enabling analysis of treatment outcomes, immune status, and health risks among HIV patients.

METHODOLOGY

Data cleaning and preprocessing were carried out to address missing values and standardize formats. Descriptive statistics and correlation analyses were applied to identify trends in patient health metrics. Unsupervised hierarchical clustering segmented patients into groups based on age, BMI, CD4 counts, blood pressure, and sex. Visualizations and detailed reports were generated to communicate findings to clinicians and inform intervention strategies.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Four distinct patient clusters were identified: young patients with stable health, middle-aged patients with low CD4 counts, elderly patients with high BMI, and mixed-risk groups. The findings informed targeted interventions such as personalized ART regimens and focused monitoring for high-risk patients. The analysis demonstrated the impact of data-driven approaches in improving treatment adherence, patient care, and allocation of healthcare resources.

CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

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FINDINGS, CONCLUSIONS & RECOMMENDATIONS

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INTERNSHIP Internship Experience

Nyeri County Referral Hospital — Mortality and morbidity trends for patient demographics in a public health setting.

Dataset Used: Nyeri County Referral Hospital — Health Records. Patient information on age, sex, disease type (chronic and infectious), treatment outcomes, and discharge status, covering emergency, pediatric, and chronic disease management units.

METHODOLOGY

Cleaning and preprocessing patient data, handling missing values, encoding categorical features, and performing exploratory data analysis (EDA). Clustering techniques, particularly K-Means, were applied to segment patients based on risk factors, health outcomes, and age groups. Statistical analyses—including correlation, hypothesis testing, and descriptive statistics—were used to identify trends in mortality and morbidity. Visualizations were created using Python libraries (pandas, matplotlib, seaborn) to communicate patterns and findings effectively, supporting targeted public health interventions.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The analysis identified specific high-risk patient groups, enabling targeted public health interventions. Pediatric patients were found to be most affected by respiratory illnesses, while elderly patients exhibited multiple chronic conditions with higher mortality rates. These findings provided actionable insights for resource allocation, prioritization of care, and designing tailored strategies to reduce disease burden and improve health outcomes. Additionally, the work demonstrated the critical role of data-driven approaches in guiding hospital decision-making and optimizing healthcare delivery for diverse patient demographics.

CURRENT STATUS

Property Manager · Working / employed



CAPSTONE Capstone Project

Predicting Thyroid Cancer Recurrence Using Machine Learning for Clinical Decision Support

BACKGROUND & METHODOLOGY

This project developed a logistic-regression model to predict thyroid cancer recurrence using a clinical dataset of 383 patients with 17 features spanning demographics, habits, thyroid function, pathology, staging (T, N, M, Stage), and outcomes. Data exploration and preprocessing included confirming no missing values, encoding categorical variables (one-hot for binary fields; label encoding for ordinal fields), and visualizing distributions and correlations. Chi-square tests indicated significant associations between recurrence and several factors, including gender, smoking, adenopathy, focality, risk level, T, M, and stage. The data were split 70/30 into training and test sets (random_state=42). The trained logistic-regression model achieved ~93.9% test accuracy and an AUC of 0.975, with performance summarized via confusion matrix and classification report. The final model was saved to a .joblib file and deployed in a Streamlit application to enable interactive risk prediction from user-entered clinical inputs. The study recommended further validation, expanded data (e.g., treatment outcomes), and clinician collaboration prior to clinical integration.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The logistic regression model successfully predicted thyroid cancer recurrence with an accuracy of approximately 93.9% and an AUC score of 0.975, demonstrating excellent discriminatory ability between recurrence and non-recurrence cases. Chi-square statistical testing confirmed significant associations between recurrence and key clinical variables including gender, smoking status, adenopathy, focality, risk level, tumor size (T), metastasis (M), and disease stage. The trained model was serialized using joblib and deployed as an interactive Streamlit web application, enabling healthcare providers to input patient clinical features and demographics to receive real-time recurrence risk predictions. The project demonstrated the feasibility of applying machine learning to support clinical decision-making in oncology care, providing a data-driven tool to identify high-risk patients for enhanced monitoring and personalized treatment planning.

INTERNSHIP Internship Experience

Nyeri County Referral Hospital — Cancer patient data analysis and clustering for public health interventions.

Dataset Used: Nyeri County Referral Hospital — Health Records, Cancer Registry. Included patient features such as age, sex, primary site, histology type, cancer stage, and patient status, serving as the foundation for risk assessment and cluster analysis.

METHODOLOGY

Cleaning and preprocessing cancer data, including handling missing values and encoding categorical features. Multiple clustering algorithms—K-means, DBSCAN, and Agglomerative Hierarchical Clustering—were applied to segment patients into meaningful groups. Statistical analysis and visualization techniques, using Python libraries like pandas, scikit-learn, matplotlib, and seaborn, supported the interpretation of cluster characteristics. Detailed reports and visualizations, such as dendrograms and cluster scatter plots, were created to communicate findings. The focus was on identifying high-risk profiles and common cancer trends to inform targeted healthcare interventions.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Analysis revealed distinct clusters highlighting high-risk patient profiles and demographics associated with advanced cancer stages. These insights enabled actionable recommendations for healthcare interventions, such as prioritizing resources for high-risk groups and tailoring preventive and treatment strategies. The project demonstrated the practical use of clustering in public health, providing evidence-based guidance for improving cancer care management and resource allocation.

CURRENT STATUS

Data Analyst · Working / employed



CAPSTONE Capstone Project

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FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The logistic regression model successfully predicted thyroid cancer recurrence with an accuracy of approximately 93.9% and an AUC score of 0.975, demonstrating excellent discriminatory ability between recurrence and non-recurrence cases. Chi-square statistical testing confirmed significant associations between recurrence and key clinical variables including gender, smoking status, adenopathy, focality, risk level, tumor size (T), metastasis (M), and disease stage. The trained model was serialized using joblib and deployed as an interactive Streamlit web application, enabling healthcare providers to input patient clinical features and demographics to receive real-time recurrence risk predictions. The project demonstrated the feasibility of applying machine learning to support clinical decision-making in oncology care, providing a data-driven tool to identify high-risk patients for enhanced monitoring and personalized treatment planning.

INTERNSHIP Internship Experience

Mt Kenya Sub-County Hospital — Non-Communicable Diseases (NCDs) risk factors, patient clustering, and health outcome analysis.

★ LED TO A JOB OFFER

Dataset Used: Mount Kenya Sub-County Hospital — patient records including age, blood pressure, chronic illness burden, and other demographic and clinical information. The datasets contained detailed clinical and demographic data, enabling the identification of patterns and risk factors among patients with NCDs. Information included age, blood pressure, comorbid conditions, and other relevant health metrics.

METHODOLOGY

Data cleaning and preprocessing were conducted to ensure accuracy and consistency. Exploratory data analysis (EDA) identified trends and patterns in patient health characteristics. Unsupervised clustering techniques, including K-means and hierarchical clustering, segmented patients into meaningful groups based on age, blood pressure, and chronic illness burden. Association rule mining was used to detect co-occurring conditions and relationships among risk factors. Visualizations and reports were produced to communicate findings effectively to healthcare professionals.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Three distinct patient clusters were identified, each with unique health profiles. Cluster 0 included younger patients with high blood pressure, Cluster 1 comprised patients with multiple chronic conditions requiring comprehensive care, and Cluster 2 consisted of older patients with well-controlled hypertension and stable health. These findings guided targeted interventions such as tailored health education, community-based support, and focused medical monitoring. Insights from the analysis informed strategies to improve NCD management and preventive care, highlighting the importance of data-driven decision-making in public health.

CURRENT STATUS

Junior Machine Learning Engineer · Working / employed

Sector relevance: Partially related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · Data visualization · ML modeling · Model evaluation · MLOps / Model deployment · Prompt engineering · Deep Learning · Generative AI / LLMs · Communication & teamwork

Confidence applying ENGAGE skills: **3 / 5**



CAPSTONE Capstone Project

Predicting Thyroid Cancer Recurrence Using Machine Learning for Clinical Decision Support

BACKGROUND & METHODOLOGY

This project developed a logistic-regression model to predict thyroid cancer recurrence using a clinical dataset of 383 patients with 17 features spanning demographics, habits, thyroid function, pathology, staging (T, N, M, Stage), and outcomes. Data exploration and preprocessing included confirming no missing values, encoding categorical variables (one-hot for binary fields; label encoding for ordinal fields), and visualizing distributions and correlations. Chi-square tests indicated significant associations between recurrence and several factors, including gender, smoking, adenopathy, focality, risk level, T, M, and stage. The data were split 70/30 into training and test sets (random_state=42). The trained logistic-regression model achieved ~93.9% test accuracy and an AUC of 0.975, with performance summarized via confusion matrix and classification report. The final model was saved to a .joblib file and deployed in a Streamlit application to enable interactive risk prediction from user-entered clinical inputs. The study recommended further validation, expanded data (e.g., treatment outcomes), and clinician collaboration prior to clinical integration.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The logistic regression model successfully predicted thyroid cancer recurrence with an accuracy of approximately 93.9% and an AUC score of 0.975, demonstrating excellent discriminatory ability between recurrence and non-recurrence cases. Chi-square statistical testing confirmed significant associations between recurrence and key clinical variables including gender, smoking status, adenopathy, focality, risk level, tumor size (T), metastasis (M), and disease stage. The trained model was serialized using joblib and deployed as an interactive Streamlit web application, enabling healthcare providers to input patient clinical features and demographics to receive real-time recurrence risk predictions. The project demonstrated the feasibility of applying machine learning to support clinical decision-making in oncology care, providing a data-driven tool to identify high-risk patients for enhanced monitoring and personalized treatment planning.

INTERNSHIP Internship Experience

Karatina Sub-county Hospital — HIV patient management, treatment adherence, and health risk stratification

Dataset Used: Comprehensive Care Centre (CCC), Karatina — patient records including treatment regimens, CD4 counts, BMI, blood pressure, age, and sex. The datasets captured detailed clinical and demographic information, enabling analysis of treatment outcomes, immune status, and health risks among HIV patients.

METHODOLOGY

Data cleaning and preprocessing were carried out to address missing values and standardize formats. Descriptive statistics and correlation analyses were applied to identify trends in patient health metrics. Unsupervised hierarchical clustering segmented patients into groups based on age, BMI, CD4 counts, blood pressure, and sex. Visualizations and detailed reports were generated to communicate findings to clinicians and inform intervention strategies.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Four distinct patient clusters were identified: young patients with stable health, middle-aged patients with low CD4 counts, elderly patients with high BMI, and mixed-risk groups. The findings informed targeted interventions such as personalized ART regimens and focused monitoring for high-risk patients. The analysis demonstrated the impact of data-driven approaches in improving treatment adherence, patient care, and allocation of healthcare resources.

CURRENT STATUS

Student · Not currently working

Sector relevance: Yes — Fully related

SKILLS IN USE

Python · Data visualization · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Predicting Individual Health Insurance Costs to Inform Risk-Adjusted Coverage

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This project develops and compares machine-learning models to predict individual medical insurance costs using demographic and modifiable risk factors (age, BMI, smoking, region, etc.). By improving performance from linear regression to Random Forest ($R \approx 0.86$), it enables fairer pricing, targeted risk counseling, and scenario testing in population health financing, with a simple Streamlit tool for frontline use.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The capstone project successfully developed and evaluated predictive models for insurance expense estimation. The initial multilinear regression model achieved an R-squared value of 0.7833, explaining 78.33% of the variance in insurance costs, with a Mean Absolute Error of 4,186.94. Model performance was significantly improved by implementing a Random Forest algorithm, which increased R^2 to 0.8629, reduced MSE to 21,274,322.98, and lowered MAE to 2,534.53, achieving an overall accuracy of approximately 86.29%. The trained model was serialized using joblib and successfully deployed as an interactive Streamlit web application, enabling users to input demographic and health characteristics to receive real-time insurance expense predictions. Key findings identified smoking status as the strongest predictor of insurance costs, followed by age and BMI, providing actionable insights for both risk assessment and personalized premium calculation in healthcare financing.

INTERNSHIP Internship Experience

Nyeri Town Health Center — HIV testing, ART adherence, and patient care outcomes in maternity and CCC patients.

Dataset Used: Nyeri Town Health Centre, Maternity section and Comprehensive Care Centre (CCC). Two datasets were accessed: the –MAT TST– dataset with HIV testing and care records, and the CCC dataset containing patient ART adherence and viral load data, including demographics and treatment outcomes.

METHODOLOGY

The work involved preprocessing, cleaning, and encoding the datasets, followed by exploratory data analysis. Hierarchical clustering was applied to the –MAT TST– dataset to identify patterns in HIV testing and linkage to care, while K-Means clustering segmented patients in the CCC dataset based on ART adherence and viral load outcomes. Visualization techniques such as dendrograms, scatter plots, histograms, box plots, and PCA were used to interpret clusters. Statistical analyses, including correlation testing and hypothesis evaluation, supported the validity of the findings. Python libraries like Pandas, Matplotlib, Seaborn, and Scikit-learn were extensively employed to implement these processes.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The analyses revealed meaningful patient clusters with distinct health profiles. In the –MAT TST– dataset, groups were identified, such as younger clients who had not been tested and patients successfully linked to care, offering insights for targeted HIV testing interventions. The CCC dataset analysis highlighted three clusters of ART patients: those with good adherence and low viral loads, patients with missed appointments and high viral loads, and intermediate cases, guiding recommendations for personalized follow-up and care. These outcomes provided actionable insights for improving patient treatment adherence, viral load management, and public health strategies in HIV care.

CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

Trends of Diarrhea Cases in Children Under Five in Kenya: A Predictive Analysis

BACKGROUND & METHODOLOGY

This project investigates the trends of diarrhea cases in children under five years old in Kenya, a significant public health concern due to its high rates of illness and death. The main objective is to analyze past cases, identify patterns, and develop models to predict future trends. By studying reported cases in different counties all over the country, we aimed to understand how these trends evolve over time. To achieve this, we used machine learning techniques, focusing on the linear regression model to create our predictive models. Our analysis uncovered seasonal peaks in diarrhea cases, which typically occur during the rainy season. We also identified specific regions in Kenya where the rates of diarrhea among young children are particularly high. Most of the cases, for example, were identified in Murang'a, Nakuru and Nairobi counties. The linear regression model we developed showed strong accuracy in predicting future trends, offering valuable insights for public health planning. These findings are crucial for health officials, as they provide a clearer understanding of when and where diarrhea outbreaks are likely to occur. This information can be used to improve resource allocation and prepare more effectively for future outbreaks, ultimately helping to reduce the impact of diarrhea on young children under the age of five in Kenya.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The capstone project successfully analyzed diarrhea trends in children under five years old in Kenya using a linear regression predictive model based on historical health facility data from 2018 to 2022. The analysis identified significant seasonal patterns, with diarrhea cases peaking during the rainy season (March-May), and revealed notable regional disparities, with Murang'a, Nakuru, and Nairobi counties recording the highest case volumes. The linear regression model demonstrated strong predictive performance, achieving a Mean Absolute Error (MAE) of 3,447 cases per month and a Root Mean Square Error (RMSE) of 58 cases per month, indicating reliable forecasting capability for public health planning purposes. Key findings confirmed that environmental factors, particularly rainfall patterns, significantly influence diarrhea incidence among young children, while geographic location emerged as a critical determinant of disease burden. The project successfully deployed an interactive Streamlit web application (<https://under-five-diarrhea-cases-prediction-n4u8dsa5g6q2pwjvlwwr3m.streamlit.app/>) enabling health officials to input regional and temporal parameters to generate real-time forecasts of expected diarrhea cases. These outcomes provide actionable evidence for optimizing resource allocation, strengthening seasonal preparedness protocols, and targeting high-burden counties with preventive interventions such as water sanitation improvements and community health education campaigns. The project demonstrated the practical utility of applying machine learning techniques to routine health surveillance data for enhancing early warning systems and supporting evidence-based decision-making in pediatric public health management in resource-limited settings.

INTERNSHIP Internship Experience

Chulaimbo County Hospital — Predictive modeling of patient outcomes for TB, HIV, outpatient diagnoses, and ART treatment.

Dataset Used: Health records from Chulaimbo County Hospital's CCC, Laboratory, and Outpatient departments. Datasets included TB clinic outcomes, TB lab HIV tests, outpatient visit records, and ART viral load suppression.

METHODOLOGY

Performed data collection, preprocessing, feature engineering, and balancing of datasets. Applied logistic regression, random forest, and other machine learning models for prediction. Trained, evaluated, and visualized models using Python libraries (Pandas, Scikit-learn, Matplotlib, Seaborn). Deployed models through Streamlit to create interactive web applications for healthcare teams.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Developed predictive tools for HIV diagnosis, TB treatment outcomes, outpatient disease likelihood, and viral load suppression. Delivered user-friendly web apps for real-time predictions, enhancing data-driven clinical decisions. Improved accuracy in patient outcome forecasting, empowered healthcare providers with actionable insights, and strengthened integration of predictive analytics into public health practice.

CURRENT STATUS

Maths Teacher · Working / employed



CAPSTONE Capstone Project

Trends of Diarrhea Cases in Children Under Five in Kenya: A Predictive Analysis

BACKGROUND & METHODOLOGY

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FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The capstone project successfully analyzed diarrhea trends in children under five years old in Kenya using a linear regression predictive model based on historical health facility data from 2018 to 2022. The analysis identified significant seasonal patterns, with diarrhea cases peaking during the rainy season (March-May), and revealed notable regional disparities, with Murang'a, Nakuru, and Nairobi counties recording the highest case volumes. The linear regression model demonstrated strong predictive performance, achieving a Mean Absolute Error (MAE) of 3,447 cases per month and a Root Mean Square Error (RMSE) of 58 cases per month, indicating reliable forecasting capability for public health planning purposes. Key findings confirmed that environmental factors, particularly rainfall patterns, significantly influence diarrhea incidence among young children, while geographic location emerged as a critical determinant of disease burden. The project successfully deployed an interactive Streamlit web application (<https://under-five-diarrhea-cases-prediction-n4u8dsa5g6q2pwjvlwwr3m.streamlit.app/>) enabling health officials to input regional and temporal parameters to generate real-time forecasts of expected diarrhea cases. These outcomes provide actionable evidence for optimizing resource allocation, strengthening seasonal preparedness protocols, and targeting high-burden counties with preventive interventions such as water sanitation improvements and community health education campaigns. The project demonstrated the practical utility of applying machine learning techniques to routine health surveillance data for enhancing early warning systems and supporting evidence-based decision-making in pediatric public health management in resource-limited settings.

INTERNSHIP Internship Experience

Kisumu Public Health Emergency Operation Center — EOC Referral data: Predicting patient outcomes, supporting clinical decision-making, and forecasting disease trends to strengthen public health planning.

Dataset Used: Emergency Operation Centre and patient health records from Kisumu County Referral Hospital. Data on referrals, lab test results, and malaria incidence analyzed using Python and Excel for cleaning, visualization, and predictive modeling to generate actionable public health insights.

METHODOLOGY

Cleaned and processed health records, performed exploratory data analysis, applied predictive modeling, and generated visualizations using Python and Excel.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Produced insights on referral patterns, lab test results, and malaria trends; supported public health planning; improved decision-making for patient care and resource allocation.

CURRENT STATUS

BIOINFORMATICS INTERN · Not currently working

Sector relevance: Yes – Fully related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · Data visualization · ML modeling · MLOps / Model deployment · Natural Language Processing (NLP) · Deep Learning · Generative AI / LLMs · Public health data analysis · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Diabetes Prediction from Nhanes Health Data

BACKGROUND & METHODOLOGY

This project utilizes the NHANES Dataset to investigate predictors for diabetes prevalence. The primary objective is to identify and analyze key variables that contribute to the likelihood of diabetes diagnosis. The data set includes various demographic and lifestyle factors such as age, weight, height, smoking status, socioeconomic conditions (poverty and housing). These variables are analyzed to determine their correlation with diabetes, aiming to identify significant relationships and patterns.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The capstone project successfully developed and evaluated a logistic regression model to predict diabetes risk based on demographic and symptom data from the NHANES dataset. The model achieved an overall accuracy of 75%, with precision of 95% for non-diabetic cases and 89% for diabetic cases, while recall values were 73% and 94% respectively, demonstrating balanced performance across both classes. The F1-scores of 0.89 for non-diabetic and 0.91 for diabetic cases confirmed the model's reliability in identifying individuals at risk. Key predictors identified through statistical testing and model coefficients included age, BMI, general health status, physical health, high blood pressure, and high cholesterol, all of which showed statistically significant associations with diabetes risk. The confusion matrix confirmed the model correctly classified 11,563 instances out of 14,804 total test cases. The trained model was serialized using joblib and successfully deployed as an interactive Streamlit web application, enabling users to input patient demographic and clinical features to receive real-time diabetes risk predictions. These findings demonstrate the feasibility of using routinely collected demographic and health data to support early diabetes screening and risk stratification in primary healthcare settings, enabling timely interventions and personalized preventive care strategies.

INTERNSHIP Internship Experience

Homabay County Teaching and Referral Hospital — HIV care and prevention, maternal and child health, and non-communicable diseases

Dataset Used: Centre for Health Solutions—Kenya (CHS), the Comprehensive Care Centre, DREAMS Safe Spaces, and health facilities, mainly through Kenya EMR, data warehouses, and patient registers. The datasets contained patient-level and program-level information such as ART treatment records, DREAMS enrollment and intervention data, and HIV testing reports, providing demographic, clinical, and behavioral indicators for analysis and predictive modeling.

METHODOLOGY

Collected, entered, and managed health data using Kenya EMR, CHS data warehouse, and physical registers. Cleaned, validated, and analyzed datasets with Python, Excel, and SQL, applying pivot tables, filtering, and visualization techniques. Developed predictive models to assess ART impact on viral load suppression, determine DREAMS program vulnerability, and predict HIV risk categories. Deployed models through Streamlit applications, created dashboards, and participated in multidisciplinary meetings, site visits, and program evaluations with PEPFAR and CDC teams.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Improved data accuracy and completeness through systematic quality checks and validation processes. Enhanced reporting efficiency and program monitoring by integrating registers into EMR systems and streamlining data handling workflows. Supported clinical and public health decision-making through predictive insights, treatment outcome tracking, and identification of vulnerable populations. Strengthened health information systems while ensuring ethical data use, confidentiality, and alignment with organizational goals.

CURRENT STATUS

Network Engineer · Working / employed



CAPSTONE Capstone Project

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FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The capstone project successfully analyzed diarrhea trends in children under five years old in Kenya using a linear regression predictive model based on historical health facility data from 2018 to 2022. The analysis identified significant seasonal patterns, with diarrhea cases peaking during the rainy season (March-May), and revealed notable regional disparities, with Murang'a, Nakuru, and Nairobi counties recording the highest case volumes. The linear regression model demonstrated strong predictive performance, achieving a Mean Absolute Error (MAE) of 3,447 cases per month and a Root Mean Square Error (RMSE) of 58 cases per month, indicating reliable forecasting capability for public health planning purposes. Key findings confirmed that environmental factors, particularly rainfall patterns, significantly influence diarrhea incidence among young children, while geographic location emerged as a critical determinant of disease burden. The project successfully deployed an interactive Streamlit web application (<https://under-five-diarrhea-cases-prediction-n4u8dsa5g6q2pwjvlwwr3m.streamlit.app/>) enabling health officials to input regional and temporal parameters to generate real-time forecasts of expected diarrhea cases. These outcomes provide actionable evidence for optimizing resource allocation, strengthening seasonal preparedness protocols, and targeting high-burden counties with preventive interventions such as water sanitation improvements and community health education campaigns. The project demonstrated the practical utility of applying machine learning techniques to routine health surveillance data for enhancing early warning systems and supporting evidence-based decision-making in pediatric public health management in resource-limited settings.

INTERNSHIP Internship Experience

Chulaimbo County Hospital — Predicting HIV prevalence among TB patients using demographic factors.

Dataset Used: TB laboratory diagnostic data from Chulaimbo County Hospital. The dataset included TB diagnostic results with variables such as HIV status, age, and gender, supporting analysis of TB—HIV co-infection patterns.

METHODOLOGY

Performed data collection, cleaning, and preprocessing; applied logistic regression modeling in Python using pandas, NumPy, and scikit-learn; incorporated demographic variables into the model; developed a Streamlit dashboard to visualize HIV prevalence patterns; and collaborated with supervisors and hospital staff to refine model accuracy.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Identified higher HIV prevalence among older age groups and females, enabling insights for targeted interventions; delivered an interactive dashboard for real-time analysis of TB—HIV co-infection trends; strengthened technical skills in predictive modeling and visualization; and provided public health teams with data-driven evidence to inform decision-making.

CURRENT STATUS

Teacher of maths, science kiswahili (primary) · Working / employed



CAPSTONE Capstone Project

Diabetes Prediction from Nhanes Health Data

BACKGROUND & METHODOLOGY

A logistic regression model was developed to estimate the likelihood of a child participating in the campaign, with the outcome variable being campaign participation (1 for participated, 2 for not participated). The model's predictors included factors that potentially influence participation, such as the child's health status, residence, and family's socioeconomic standing. 0 Additionally, younger children and those with more educated parents were more likely to be vaccinated, emphasizing the role of parental awareness and education in public health initiatives.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The capstone project successfully developed and evaluated a logistic regression model to predict diabetes risk based on demographic and symptom data from the NHANES dataset. The model achieved an overall accuracy of 75%, with precision of 95% for non-diabetic cases and 89% for diabetic cases, while recall values were 73% and 94% respectively, demonstrating balanced performance across both classes. The F1-scores of 0.89 for non-diabetic and 0.91 for diabetic cases confirmed the model's reliability in identifying individuals at risk. Key predictors identified through statistical testing and model coefficients included age, BMI, general health status, physical health, high blood pressure, and high cholesterol, all of which showed statistically significant associations with diabetes risk. The confusion matrix confirmed the model correctly classified 11,563 instances out of 14,804 total test cases. The trained model was serialized using joblib and successfully deployed as an interactive Streamlit web application, enabling users to input patient demographic and clinical features to receive real-time diabetes risk predictions. These findings demonstrate the feasibility of using routinely collected demographic and health data to support early diabetes screening and risk stratification in primary healthcare settings, enabling timely interventions and personalized preventive care strategies.

INTERNSHIP Internship Experience

Kakamega County General Teaching and Referral Hospital — Cancer data analysis for early detection and improved patient outcomes.

Dataset Used: Patient records from Kakamega County General Hospital Cancer Centre. Dataset included demographic details, cancer type, stage at diagnosis, HIV status, and residence, used to assess patterns in cancer incidence and diagnosis trends.

METHODOLOGY

Activities included managing cancer patient data by collecting, organizing, and cleaning records to address inconsistencies and missing values. Descriptive and statistical analyses were performed using Python and Excel, focusing on demographic distributions, cancer type prevalence, and stage at diagnosis. Logistic regression was applied to explore associations between age, HIV status, residence, and cancer stage. Visualizations were created using Matplotlib and Seaborn to present patterns of cancer incidence and diagnostic stages, supporting hospital stakeholders with clear insights for decision-making.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Outcomes showed that patients over 50 were more likely to be diagnosed at late stages, highlighting the urgency for targeted screening and awareness in older populations. Breast, cervical, oesophageal, and prostate cancers were found most prevalent, with many cases diagnosed at advanced stages. The project produced actionable recommendations for hospital interventions, including scaling up cancer awareness campaigns, improving access to screening in rural areas, and addressing socio-economic disparities in healthcare. These insights reinforced the role of data analytics in strengthening cancer prevention, early detection, and public health planning in Kakamega County.

CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

Diabetes Prediction from Nhanes Health Data

BACKGROUND & METHODOLOGY

A logistic regression model was developed to estimate the likelihood of a child participating in the campaign, with the outcome variable being campaign participation (1 for participated, 2 for not participated). The model's predictors included factors that potentially influence participation, such as the child's health status, residence, and family's socioeconomic standing. 0 Additionally, younger children and those with more educated parents were more likely to be vaccinated, emphasizing the role of parental awareness and education in public health initiatives.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The capstone project successfully developed and evaluated a logistic regression model to predict diabetes risk based on demographic and symptom data from the NHANES dataset. The model achieved an overall accuracy of 75%, with precision of 95% for non-diabetic cases and 89% for diabetic cases, while recall values were 73% and 94% respectively, demonstrating balanced performance across both classes. The F1-scores of 0.89 for non-diabetic and 0.91 for diabetic cases confirmed the model's reliability in identifying individuals at risk. Key predictors identified through statistical testing and model coefficients included age, BMI, general health status, physical health, high blood pressure, and high cholesterol, all of which showed statistically significant associations with diabetes risk. The confusion matrix confirmed the model correctly classified 11,563 instances out of 14,804 total test cases. The trained model was serialized using joblib and successfully deployed as an interactive Streamlit web application, enabling users to input patient demographic and clinical features to receive real-time diabetes risk predictions. These findings demonstrate the feasibility of using routinely collected demographic and health data to support early diabetes screening and risk stratification in primary healthcare settings, enabling timely interventions and personalized preventive care strategies.

INTERNSHIP Internship Experience

Kisumu County Referral Hospital — Predictive healthcare analytics for timely diagnosis and patient management

Dataset Used: Kisumu Public Health Investigation Records and referral calls data. Dataset included demographic and clinical details such as gender, age, location, clinical diagnoses, and investigation titles, used to develop predictive models and healthcare applications.

METHODOLOGY

The internship focused on developing three predictive applications: the Investigation Test Results Prediction App, the Malaria Prediction App, and the Readmission Prediction App. Tasks included collecting and preprocessing patient and investigation datasets across Kisumu County, applying supervised machine learning algorithms in Python with scikit-learn, performing statistical analyses to identify patterns, and building and validating classification models using metrics such as accuracy, recall, precision, and F1-score. User-friendly applications were developed to integrate predictive insights into clinical workflows.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The predictive tools enhanced clinical decision-making, improved patient care, and optimized diagnostic processes. The Investigation Test Results Prediction App enabled real-time insights into test likelihoods, supporting timely allocation of resources. Overall, the internship contributed to more efficient healthcare delivery, better patient outcomes across Kisumu County, and strengthened the link between data-driven insights and public health impact. The experience also reinforced technical, analytical, and problem-solving skills while demonstrating practical applications of predictive analytics in healthcare.

CURRENT STATUS

Cohort 2 Supervisor · Working / employed



CAPSTONE Capstone Project

Breathe Easy: AI-Driven Pneumonia Severity Checker

BACKGROUND & METHODOLOGY

The Breathe Easy project focuses on developing an AI-driven tool to accurately assess the severity of pneumonia among patients at Mirugi Kariuki Subcounty Hospital. Pneumonia remains a major health concern in the region due to poor living conditions, environmental factors, limited healthcare access, and low public awareness. The project leverages machine learning techniques—specifically Random Forest, Logistic Regression, and Decision Tree models—to analyze clinical and demographic variables such as age, temperature, blood pressure, SpO₂, and fever to predict disease severity. Random Forest emerged as the most accurate and efficient model. By identifying key risk factors and supporting resource allocation, the project aims to enhance personalized patient care and contribute to improved public health strategies. The system was deployed using Streamlit for ease of access and integration.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project successfully developed an AI-driven pneumonia severity checker using machine learning techniques to assess pneumonia risk based on patient symptoms and vital signs at Mirugi Kariuki Subcounty Hospital. The analysis evaluated multiple classification algorithms including Random Forest, Logistic Regression, and Decision Tree, with Random Forest identified as the most efficient and accurate model for predicting pneumonia severity. Key predictors influencing pneumonia outcomes included age, body temperature, pulse rate, SpO₂ levels, fever presence, weight, and blood pressure readings. The trained model was deployed as an interactive Streamlit web application to support real-time clinical decision-making for healthcare workers. The project demonstrated the feasibility of applying machine learning to routine clinical data for early pneumonia detection and severity stratification in resource-limited healthcare settings. Key challenges encountered included incomplete or inaccurate patient records, limited dataset accessibility, sample size constraints, and data bias contributing to potential model overfitting. Despite these limitations, the project established a practical foundation for data-driven pneumonia management, emphasized the critical importance of robust data collection and quality assurance systems, and highlighted the transformative potential of AI tools to enhance diagnostic accuracy, optimize resource allocation, and improve patient outcomes in endemic regions.

INTERNSHIP Internship Experience

Mirugi Kariuki Sub-county Hospital — Respiratory Disease (Pneumonia) and Mother-Child Health Data

Dataset Used: Hospital EMR system (CC patients, NHIF & Linda Mama, reception, records). Patient demographics, clinical presentations, treatment outcomes, maternal health services, and insurance claims

METHODOLOGY

The team cleaned and analyzed pneumonia data, observed seasonal trends, and examined resource availability. Work involved managing data across multiple departments, performing data retrieval, and developing predictive models for pneumonia prevalence.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Analysis revealed higher pneumonia incidence during colder months, especially among children under five and the elderly. Inadequate medical resources were identified, prompting recommendations for better resource allocation and targeted preventive interventions.

CURRENT STATUS

Client Relations Officer · Working / employed



CAPSTONE Capstone Project

Empowering Women's Health Through a Gynecological Care Chatbot

BACKGROUND & METHODOLOGY

The Gynecological Care Chatbot project focuses on improving women's health awareness and access to reliable medical information, particularly for women in remote and underserved areas. The project combines a mobile-friendly website and an interactive chatbot to provide educational resources on menstrual hygiene, pregnancy care, family planning, sexually transmitted infection (STI) prevention, and cancer awareness. Designed to overcome social, economic, and infrastructural barriers, the platform offers downloadable guides, offline access, and links to local healthcare centers. By integrating AI-driven conversational support, the chatbot enables women to ask questions and receive instant, personalized guidance. The initiative promotes informed decision-making, reduces stigma around women's health, and supports early interventions for better health outcomes. Future expansions include multilingual support and broader coverage of topics such as mental health and domestic violence, ensuring inclusivity and long-term impact on women's well-being.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The final outcome included a simple Flask web application where users can input tumor-related measurements and receive a prediction. This system can assist healthcare professionals by providing a quick second opinion, supporting earlier diagnosis, and helping improve treatment planning and patient outcomes.

INTERNSHIP Internship Experience

Soin Sub-county Hospital — Patient Records and Disease Prevalence Data

Dataset Used: Kenya Health Information System (KHIS) and Kenya EMR. Included patient demographics, diagnoses, services billed to insurance, and hospital visit information

METHODOLOGY

The team undertook extensive data cleaning and preprocessing to address inconsistencies such as missing entries and null values. They analyzed patient records to identify patterns in disease prevalence, focusing on common conditions like diarrhea, malaria, urinary tract infections (UTIs), and minor injuries. Advanced analysis included stratifying patients by age, gender, and diagnosis to detect trends and high-risk groups. Visualization was done using Python libraries (matplotlib and seaborn) to generate charts, graphs, and heatmaps that clearly communicated patterns to hospital stakeholders. The interns also evaluated insurance billing records to link services provided with specific patient conditions, aiming to identify resource utilization patterns.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The analysis highlighted diarrhea, malaria, UTIs, and minor injuries as the most prevalent health conditions at the facility, suggesting the need for targeted prevention and treatment programs. Patterns in age and gender distributions of patients were identified, which can guide future health interventions. The study also exposed gaps in data quality, with frequent null values and incomplete records, prompting recommendations for point-of-entry validation and improved data management practices. Furthermore, insights from the analysis were used to propose community outreach programs, resource prioritization for high-burden conditions, and improvements to the hospital's network and technical infrastructure to support more reliable data collection and reporting.

CURRENT STATUS

MBA & Data Analytics Student · Not currently working



CAPSTONE Capstone Project

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FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project successfully developed an accessible, mobile-optimized digital health platform featuring an interactive gynecological care chatbot and educational website designed to empower women in remote and underserved areas with reliable health information. The platform includes comprehensive sections on menstrual hygiene, pregnancy care, family planning, STI prevention, and cancer awareness, all presented in simple, accessible language. The chatbot provides real-time, personalized responses to common health questions and directs users to local healthcare resources, while downloadable PDF guides ensure offline accessibility for users with limited internet connectivity. Key outcomes include increased health literacy among target users, reduced barriers to accessing reproductive health information, and improved capacity for women to make informed decisions about their care. The project demonstrated the feasibility of leveraging lightweight, AI-driven tools to address geographic, social, and economic barriers to healthcare access in resource-limited settings. Future enhancements include expanding language support for greater inclusivity, incorporating more complex health queries into the chatbot's knowledge base, and integrating telehealth referral pathways to strengthen the link between information access and clinical service utilization.

INTERNSHIP Internship Experience

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CURRENT STATUS

Call centre Support · Working / employed



CAPSTONE Capstone Project

Empowering Women's Health Through a Gynecological Care Chatbot

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CAPSTONE Capstone Project

Breathe Easy: AI-Driven Pneumonia Severity Checker

BACKGROUND & METHODOLOGY

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FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project successfully developed an AI-driven pneumonia severity checker using machine learning techniques to assess pneumonia risk based on patient symptoms and vital signs at Mirugi Kariuki Subcounty Hospital. The analysis evaluated multiple classification algorithms including Random Forest, Logistic Regression, and Decision Tree, with Random Forest identified as the most efficient and accurate model for predicting pneumonia severity. Key predictors influencing pneumonia outcomes included age, body temperature, pulse rate, SpO₂ levels, fever presence, weight, and blood pressure readings. The trained model was deployed as an interactive Streamlit web application to support real-time clinical decision-making for healthcare workers. The project demonstrated the feasibility of applying machine learning to routine clinical data for early pneumonia detection and severity stratification in resource-limited healthcare settings. Key challenges encountered included incomplete or inaccurate patient records, limited dataset accessibility, sample size constraints, and data bias contributing to potential model overfitting. Despite these limitations, the project established a practical foundation for data-driven pneumonia management, emphasized the critical importance of robust data collection and quality assurance systems, and highlighted the transformative potential of AI tools to enhance diagnostic accuracy, optimize resource allocation, and improve patient outcomes in endemic regions.

INTERNSHIP Internship Experience

Mirugi Kariuki Sub-county Hospital — Respiratory Disease (Pneumonia) and Mother-Child Health Data

Dataset Used: Hospital EMR system (CC patients, NHIF & Linda Mama, reception, records). Patient demographics, clinical presentations, treatment outcomes, maternal health services, and insurance claims

METHODOLOGY

The team cleaned and analyzed pneumonia data, observed seasonal trends, and examined resource availability. Work involved managing data across multiple departments, performing data retrieval, and developing predictive models for pneumonia prevalence.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Analysis revealed higher pneumonia incidence during colder months, especially among children under five and the elderly. Inadequate medical resources were identified, prompting recommendations for better resource allocation and targeted preventive interventions.

CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

Breathe Easy: AI-Driven Pneumonia Severity Checker

BACKGROUND & METHODOLOGY

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INTERNSHIP Internship Experience

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METHODOLOGY

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FINDINGS, CONCLUSIONS & RECOMMENDATIONS

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CURRENT STATUS

IT Expert · Working / employed



CAPSTONE Capstone Project

Breast Cancer Detection System Using Machine Learning: Early Detection for Better Outcomes

BACKGROUND & METHODOLOGY

This project developed a machine learning-based system to support early breast cancer detection using clinical tumor data. The system used the Breast Cancer Wisconsin Diagnostic Dataset, which contains tumor features such as radius, texture, perimeter, and area. After data cleaning, preprocessing, and train-test splitting, a Random Forest Classifier was trained to predict whether a tumor is malignant or benign. The model achieved an accuracy of over 95%, showing strong potential in supporting faster and more informed screening decisions. The project also demonstrated how the trained model can be deployed through a Flask web interface, allowing users to enter patient-related tumor features and receive prediction results.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project successfully produced a machine learning model capable of predicting breast cancer status as either malignant or benign using clinical tumor characteristics. The Random Forest model achieved high accuracy of over 95%, indicating that it can support early detection and assist healthcare professionals with preliminary decision-making. A Flask-based web interface was also proposed/developed to make the model easier to use by allowing users to input tumor features and receive prediction results. Overall, the outcome shows that machine learning can improve early screening support, reduce delays in detection, and contribute to better treatment planning, while future improvements could include larger datasets, additional medical history features, and mobile deployment.

INTERNSHIP Internship Experience

LangaLanga Sub County Hospital — Outpatient, Maternal, Child, HIV, and Inventory Data

★ LED TO A JOB OFFER

Dataset Used: Hospital Records including KeMR System, ANC, CCC, Maternity, and Bin Card Inventories. Included patient demographics, disease codes, visit records, maternal and child health data, HIV treatment details (appointments, lab results), and medical stock levels

METHODOLOGY

The team conducted comprehensive data cleaning and preprocessing to handle incomplete records, missing values, and inconsistent coding across multiple datasets. They analyzed patient follow-ups, TCA appointment adherence, and viral load monitoring for HIV-positive patients. Child welfare clinic (CWC) and antenatal clinic (ANC) data were examined to track immunization coverage, maternal health, and child growth patterns. Inventory management data (bin cards) was cross-checked with physical stock to identify discrepancies in medical supplies. Statistical analyses, correlation checks, and data visualizations were performed using Python to uncover patterns in patient outcomes, service utilization, and supply management.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The analysis revealed significant gaps in patient follow-up, particularly among HIV-positive patients, which could affect continuity of care. Child health data highlighted cases of incomplete immunization and inconsistent growth monitoring. Inventory checks showed misalignment between physical stock and recorded data, indicating potential stock mismanagement. These findings provided actionable insights for improving patient tracking systems, streamlining medical supply management, and strengthening maternal and child health interventions. Additionally, the analysis emphasized the importance of anonymizing sensitive patient data and standardizing records to ensure reliable and ethical reporting. The outcomes support the hospital in prioritizing resources, improving continuity of care, and developing predictive models for patient follow-ups and inventory management.

CURRENT STATUS

Freelance · Not currently working

Sector relevance: Yes — Fully related

SKILLS IN USE

Python · Data cleaning & wrangling · ML modeling · MLOps / Model deployment · Natural Language Processing (NLP) · Prompt engineering · Generative AI / LLMs · Freelancing in data science / building a portfolio

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

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INTERNSHIP Internship Experience

LangaLanga Sub County Hospital — Outpatient, Maternal, Child, HIV, and Inventory Data

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METHODOLOGY

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CURRENT STATUS

Freelancing · Not currently working



CAPSTONE Capstone Project

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INTERNSHIP Internship Experience

LangaLanga Sub County Hospital — Outpatient, Maternal, Child, HIV, and Inventory Data

★ LED TO A JOB OFFER

Dataset Used: Hospital Records including KeMR System, ANC, CCC, Maternity, and Bin Card Inventories. Included patient demographics, disease codes, visit records, maternal and child health data, HIV treatment details (appointments, lab results), and medical stock levels

METHODOLOGY

The team conducted comprehensive data cleaning and preprocessing to handle incomplete records, missing values, and inconsistent coding across multiple datasets. They analyzed patient follow-ups, TCA appointment adherence, and viral load monitoring for HIV-positive patients. Child welfare clinic (CWC) and antenatal clinic (ANC) data were examined to track immunization coverage, maternal health, and child growth patterns. Inventory management data (bin cards) was cross-checked with physical stock to identify discrepancies in medical supplies. Statistical analyses, correlation checks, and data visualizations were performed using Python to uncover patterns in patient outcomes, service utilization, and supply management.

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CURRENT STATUS

Community Level Trainer · Working / employed

Sector relevance: Not related

SKILLS IN USE

Data collection · Communication & teamwork · Field survey

Confidence applying ENGAGE skills: 5 / 5



CAPSTONE Capstone Project

Data-Driven Hospital Admissions using Multilinear Regression Model

BACKGROUND & METHODOLOGY

Efficient hospital admission management is critical to optimizing patient care, yet many healthcare systems still struggle with delays caused by limited data-driven insights. This study applied a multiple linear regression model to predict hospital admissions using cycle threshold (CT_Blood) values alongside patient variables such as age, weight, and height. Data preprocessing involved handling missing values and duplicates, followed by statistical and graphical analyses of patient demographics and CT_Blood patterns. The model achieved exceptional performance with $R^2 = 1.0$, $MAE \approx 0$, and $MSE \approx 0$, indicating highly accurate predictions. An interactive application was deployed using Streamlit and Joblib to enable healthcare professionals to forecast admissions and manage resources proactively. Findings highlight significant associations between CT_Blood levels and patient characteristics, supporting risk identification and personalized interventions. This approach provides a scalable solution for reducing wait times, improving patient flow, and enhancing hospital operational efficiency.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The project developed a multiple linear regression machine learning model to predict hospital admission risk using cycle threshold blood values, with supporting variables such as age, gender, height, and weight. The study used hospital admissions data, applied data cleaning, missing value checks, duplicate checks, exploratory analysis, and a 70:30 training-testing split. The model was evaluated using Mean Squared Error, R-squared, and Mean Absolute Error, with results showing very strong predictive performance, including an R-squared value of 1.0 and very low error values. The outcome demonstrates that data-driven modelling can support hospital admission planning, help identify high-risk patients, improve patient flow, reduce waiting times, and support better resource allocation through an interactive Streamlit deployment.

INTERNSHIP Internship Experience

Mephi Hospital — Maternity Health Data

CURRENT STATUS

Not currently working

Sector relevance: Not related

SKILLS IN USE

Python · Data collection · Data visualization · Model evaluation

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

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INTERNSHIP Internship Experience

Mikindani Catholic Dispensary — Maternal health assessment and risk factors among pregnant women.

Dataset Used: ANC datasets from the Maternal and Child Health (MCH) department, including variables on blood group, hemoglobin level, weight, height, age, and HIV status. Analysis of maternal health indicators to identify patterns, risk factors, and correlations that could inform targeted interventions for improving pregnancy outcomes and maternal care.

METHODOLOGY

Collected, analyzed, visualized and modelled ANC data using excel and google Collab. The focus was on the relationship between maternal health indicators (blood group, hemoglobin level, weight, height, age, and HIV status) in relation to pregnant women. Performed data cleaning and analysis and visualized findings using scatter plots, bar graphs, histograms and box plots to create informative dashboard.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The analysis revealed critical associations between hemoglobin levels and factors such as weight, age, and blood group. Clustering highlighted three groups of women: one with higher age and weight but moderate hemoglobin, one with lower hemoglobin requiring extra monitoring, and one with lower age and weight but high hemoglobin. Blood group analysis showed mothers with negative rhesus factor could be at risk of having infants with sickle cell disease, indicating the need for targeted medical care. Overall, these findings provided evidence to guide interventions for maternal health, including anemia management, weight monitoring, and preventive care for high-risk pregnancies.

CURRENT STATUS

Not currently working



CAPSTONE Capstone Project

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INTERNSHIP Internship Experience

Pwani University Health Unit — Student health monitoring and service utilization analysis.

Dataset Used: Pwani University Health Unit dataset (September 2023 — April 2024), including patient demographics, treatment outcomes, and health service usage. Analysis of student health data to identify trends in disease prevalence, treatment outcomes, and resource allocation. The dataset includes variables such as age, gender, disease incidence (e.g., URTI, hyperacidity, eye conditions, candidiasis, malaria), and service utilization metrics.

METHODOLOGY

Gathered data on student visits, treatment outcomes, and health service utilization from the Pwani University health unit. Conducted exploratory data analysis (EDA) to identify trends and patterns in patient demographics, disease prevalence, and service usage. Created visual representations. Assisted in developing predictive models to forecast how Urinary Tract Infections (UTI) increased in every month.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Analysis highlighted the most prevalent health issues among students, including URTI, hyperacidity, eye conditions, candidiasis, and malaria. The predictive model provided insights into seasonal trends in URTI incidence, informing better allocation of medical supplies and staffing. Findings also revealed patterns in service utilization, guiding decisions for targeted health interventions and community health planning. These results can directly inform policy adjustments and improve healthcare delivery within the university setting.

CURRENT STATUS

Water Technician · Working / employed

Sector relevance: Not related

SKILLS IN USE

nothing

Confidence applying ENGAGE skills: **4 / 5**

App Development

Projects that culminated in a working, deployed application.

72

beneficiaries

DeKUT (10) · SEKU (12) · JOOUST (4) · Kabarak (16) · MUST (20) · Pwani (10)

**CAPSTONE** **Capstone Project***COVID-19 Death Percentage Prediction Using Multilinear Regression***BACKGROUND & METHODOLOGY**

This study develops a multilinear regression model to predict COVID-19 death percentage using global country-level data. The dataset includes variables such as population, total cases, total deaths, cases per million, and deaths per million across multiple continents. After data cleaning, encoding, and exploratory analysis, the model was trained and evaluated with regression metrics. Results show that total cases, population size, and deaths per million significantly influence COVID-19 death percentages. The model was deployed through a Streamlit interface, allowing users to input current country data and receive predicted death percentages. This tool can aid policymakers, healthcare managers, and researchers in anticipating health resource demands and guiding interventions.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a multilinear regression model for predicting COVID-19 death percentages with strong predictive performance. Key predictors identified were total cases, deaths per million, and population size, all showing significant positive relationships with death percentage. The model achieved meaningful R- and Adjusted R- values with RMSE and MAE within acceptable limits, indicating good generalization. An interactive Streamlit web interface was created enabling policymakers and health officials to input country-level data and receive real-time death percentage predictions without programming knowledge. The tool provides actionable estimates for healthcare resource planning, policy enforcement, and pandemic preparedness, while highlighting the need to incorporate additional variables such as vaccination rates and healthcare capacity for improved accuracy.

INTERNSHIP **Internship Experience***Karatina Sub County Hospital — Morbidity and Mortality Analysis and Patient Profiling | Hierarchical Clustering for Patient Segmentation | AI for Healthcare Quality Improvement*

Dataset Used: Patient episode dataset from Karatina Sub County Hospital (January—August 2025) containing demographic, clinical diagnoses, service utilization (ward, referring facility), and outcome variables (alive/mortality). Dataset containing patient episodes with variables including age, length of stay, ward, referring facility, diagnosis groups, number of diagnoses, and mortality outcomes, used to analyze morbidity and mortality patterns, identify patient subgroups through hierarchical agglomerative clustering, assess clinical segmentation by condition type, and support evidence-based resource allocation and quality improvement initiatives in hospital settings

METHODOLOGY

Data entry and validation of patient registers; Cleaning and standardizing health records by handling missing values, removing duplicates, and recoding categorical variables; Feature engineering (Length of Stay, Standardized Age, Diagnosis Group Flags, Mortality Flag, Month of Admission); Exploratory Data Analysis using descriptive statistics, frequency analysis, stratification by ward/diagnosis, and correlation testing (Spearman, Point-Biserial); Applied Hierarchical Agglomerative Clustering (Ward linkage) to Alive Morbidity cohort with feature scaling and one-hot encoding; Generated visualizations (histograms, bar plots, stacked charts, dendrograms, PCA plots) using Python; Conducted clinical segmentation based on Neonatal/Perinatal, Surgical, and External Injury flags; Performed risk profile analysis for defined patient groups

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully cleaned and preprocessed patient episode dataset with derived features for analysis; Identified statistically significant variation in Length of Stay across hospital wards; Established weak negative correlation between LOS and mortality suggesting shorter stays may associate with increased mortality; Hierarchical clustering identified four distinct patient subgroups within Alive Morbidity cohort differentiated by age and length of stay (older patients with longer LOS vs. younger patients with shorter stays); Clinical segmentation revealed key characteristics of Neonatal/Perinatal, Surgical Conditions, and External Injury patient groups including size, LOS, mortality rates, and top wards; Risk profile analysis identified potential predictive markers for elderly medical patients with long LOS and post-operative return cases; Highlighted data quality gaps in categorical fields and mortality-related variables requiring improved recording protocols; Provided actionable recommendations for targeted interventions including KMC strengthening for neonatal cases, 48-72h post-op checklists for surgical patients, trauma triage improvements, and comprehensive discharge planning for elderly patients; Demonstrated feasibility of applying unsupervised machine learning for patient stratification in resource-limited hospital settings

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Predictive Analysis of Insurance Expenses Based on Health and Demographic Characteristics

BACKGROUND & METHODOLOGY

This study analyzes a health insurance dataset of 1,338 individuals to identify the key demographic and health-related factors that influence medical expenses. The research examines how age, gender, body mass index (BMI), smoking status, number of children, and geographic region affect insurance-billed medical costs. The findings reveal that smoking status is the strongest predictor of medical costs, with smokers incurring expenses nearly four times higher than non-smokers. Age shows a strong positive correlation with costs, while BMI demonstrates a moderate positive relationship. The number of children has a non-linear effect, whereas gender and geographic region have relatively minor influence. These insights are valuable for insurers in designing risk-adjusted premium models, implementing targeted wellness programs—particularly smoking cessation and weight management initiatives—and optimizing cost-control strategies. Overall, the analysis underscores the significant financial impact of modifiable health behaviors on healthcare expenditures.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a predictive model for insurance expenses using Elastic Net regression with cross-validated hyperparameters. Key findings identified smoking status as the strongest predictor, with smokers incurring costs nearly four times higher than non-smokers (\$32,050 vs \$8,434). Age showed strong positive correlation (0.30) with expenses, while BMI demonstrated moderate positive correlation (0.20). The model was deployed via an interactive Streamlit web application enabling real-time insurance expense estimates based on individual health and demographic inputs. Results support risk-adjusted premium modeling, targeted wellness programs, and optimized cost control strategies for insurers.

INTERNSHIP Internship Experience

Mt. Kenya Subcounty Hospital — Predictive Analytics for Emergency Response Optimization | AI & Data Science for Public Health

Dataset Used: Ambulance dispatch data from Nyeri County (~2,000 incident records). Capstone project leveraging ambulance dispatch data from Nyeri County to improve emergency response decision-making through predictive analytics. Aimed to predict hospital admission likelihood and estimate total response time from call receipt to incident resolution using classification and regression models.

METHODOLOGY

Data Import and Cleaning: Parsed timestamps, handled missing data, standardized Age/Gender, managed outliers | Feature Engineering: Created Duration_min and Response_time_sec variables, encoded categorical features | EDA: Generated summary statistics, distributions, visualizations (bar charts, scatter plots) | Model Development: Built Logistic Regression & Random Forest Classifier for admission prediction; Random Forest Regressor for response time (80/20 train-test split) | Model Evaluation: Used ROC-AUC, Precision-Recall, F1-score, R², RMSE, MAE; feature importance & SHAP analysis

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Cleaned ~2,000 records, improving data completeness by ~20% | Classification model: ROC-AUC = 1.0, Precision-Recall AUC = 1.0 for hospital admission prediction | Regression model: R² = 0.72, RMSE = 6.8 min, MAE = 5.1 min for response time | Key predictors identified: Class (Referral/Rescue), Case Type, Priority, Gender | Evidence-based recommendations provided for ambulance operations | Demonstrated practical application of data science in public health emergency response

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Predictive Analysis of Depression Survey Data: Smoking Status Prediction from Health Examination Metrics

BACKGROUND & METHODOLOGY

This project aims to develop a machine learning model to predict an individual's smoking status based on their health examination metrics. Utilizing a dataset containing 27 attributes including physiological, biochemical, and oral examination data for 55,692 individuals, we performed extensive data pre-processing, exploratory data analysis (EDA), and feature engineering. A key step involved using K-Means clustering to identify inherent patterns and structures within the data, which could inform the predictive modeling process. The project outlines a complete data science pipeline, from data loading and cleaning to preprocessing and preliminary clustering analysis. The results from the EDA and clustering reveal significant differences in key health indicators (e.g., hemoglobin, GPT levels) between smokers and non-smokers. This analysis serves as a crucial foundation for building a subsequent classification model to accurately predict smoking behavior, which could be a valuable tool for public health initiatives and personalized healthcare interventions.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully completed data preprocessing and exploratory analysis on 55,692 health examination records with 27 features, establishing a robust foundation for smoking status prediction. Key findings from EDA revealed significant physiological differences between smokers and non-smokers, including elevated hemoglobin and GPT (?-GTP) levels in smokers. The dataset was standardized and encoded using an automated pipeline, with K-Means clustering implemented to uncover natural groupings correlated with smoking behavior. The target variable showed 37% smokers and 63% non-smokers, indicating class imbalance requiring SMOTE or class weighting for future classification. A conceptual Streamlit deployment interface was designed for real-time prediction input and probability scoring. The project provides a complete analytical framework ready for supervised model development using Logistic Regression, Random Forest, or XGBoost, with potential applications in public health screening, clinical practice, and personalized healthcare interventions.

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

COVID-19 Death Percentage Prediction Using Multilinear Regression

BACKGROUND & METHODOLOGY

This study develops a multilinear regression model to predict COVID-19 death percentage using global country-level data. The dataset includes variables such as population, total cases, total deaths, cases per million, and deaths per million across multiple continents. After data cleaning, encoding, and exploratory analysis, the model was trained and evaluated with regression metrics. Results show that total cases, population size, and deaths per million significantly influence COVID-19 death percentages. The model was deployed through a Streamlit interface, allowing users to input current country data and receive predicted death percentages. This tool can aid policymakers, healthcare managers, and researchers in anticipating health resource demands and guiding interventions.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a multilinear regression model for predicting COVID-19 death percentages with strong predictive performance. Key predictors identified were total cases, deaths per million, and population size, all showing significant positive relationships with death percentage. The model achieved meaningful R² and Adjusted R² values with RMSE and MAE within acceptable limits, indicating good generalization. An interactive Streamlit web interface was created enabling policymakers and health officials to input country-level data and receive real-time death percentage predictions without programming knowledge. The tool provides actionable estimates for healthcare resource planning, policy enforcement, and pandemic preparedness, while highlighting the need to incorporate additional variables such as vaccination rates and healthcare capacity for improved accuracy.

INTERNSHIP Internship Experience

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FINDINGS, CONCLUSIONS & RECOMMENDATIONS

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CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Predicting Thyroid Disease Recurrence Using Logistic Regression Model

BACKGROUND & METHODOLOGY

This project aimed to develop a logistic regression model to predict recurrence in thyroid patients using clinical and demographic data from the thyroid diff data set. The data set contained variables like gender, age, smoking history, pathology results, tumor stage and recurrence outcomes. The categorical variables were encoded and the ordinal variables treated as ordered features, and missing data were checked and replaced where necessary. Statistical tests such as chi-squared were used for feature selection for categorical variables. The model achieved an overall accuracy of 93% with 93% precision and 81% recall for predicting recurrence. These results showed that the model can be used as a decision support tool for early identification of high risk patients. The model was also deployed with a framework that can be used for clinical settings. This study contributes to improved patient management and optimized resource allocation in thyroid cancer care.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a logistic regression model achieving 93% accuracy, 93% precision, and 81% recall for predicting thyroid disease recurrence. Key predictors identified included Stage ($p < 0.01$) and Metastasis ($p < 0.01$) as strong recurrence indicators. The model correctly classified 107 out of 115 cases with only 8 misclassifications. A web-based deployment framework using Streamlit was designed for clinical integration, enabling clinicians to input patient data and receive recurrence probability scores. The model provides an interpretable decision-support tool for early identification of high-risk patients, personalized follow-up care, and optimized resource allocation in thyroid cancer management.

INTERNSHIP Internship Experience

Westlands Health Care — Development of Predictive and Analytical Dashboards for Healthcare Data Management | AI for Non-Communicable Disease Prediction | Digital Health Innovation

Dataset Used: Clinical patient health records from Westlands Health Care containing demographic data, vital parameters (blood sugar levels, blood pressure, BMI), and diagnostic information for diabetes and hypertension screening. Project developing four Streamlit-based digital health applications for clinical data management and disease prediction at Westlands Health Care. The applications include a Clinical Data Entry Dashboard for patient record management, a Diabetes Prediction App using machine learning models (Random Forest/Logistic Regression), a Hypertension Prediction App for risk assessment, and an integrated Analytics Dashboard for analyzing co-occurrence patterns between diabetes and hypertension. The system aims to streamline clinical workflows, enable early disease screening, and support evidence-based decision-making in resource-limited healthcare settings.

METHODOLOGY

Adopted project-based and agile methodology with iterative development and user-centered design | Developed Clinical Data Entry Form Dashboard with form-based data entry, real-time record display, and export functionality | Built Diabetes Prediction App using machine learning models trained on health data with interactive Streamlit interface | Created Hypertension Prediction App for vital health indicator-based risk assessment | Designed integrated Diabetes + Hypertension Analysis Dashboard with combined visualization and interactive filters | Utilized Python for backend development, Streamlit for web dashboards, Pandas & Plotly for data analysis and visualization, Scikit-learn for model training, and Excel/CSV for data storage | Conducted requirement gathering, system design, model training, dashboard integration, and deployment with user testing

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed four Streamlit-based applications using Streamlit Cloud | Improved healthcare data management through automated patient screening and enhanced accuracy in data entry | Introduced data-driven approach to preventive healthcare with early identification of high-risk patients for diabetes and hypertension | Demonstrated feasibility of integrating machine learning and digital dashboards in clinical settings | Enhanced technical proficiency in Python, data visualization, and machine learning integration | Provided practical insights into AI applications for improving healthcare outcomes in low-resource settings | Applications simplified clinical workflows, enhanced data quality, and improved disease monitoring capabilities for non-communicable disease management

CURRENT STATUS

Unemployed — Actively searching

Sector relevance: Not related

SKILLS IN USE

I am not currently engaged

Confidence applying ENGAGE skills: 5 / 5



CAPSTONE Capstone Project

Logistic Regression Model for Identifying Depression Risk Based on Depression Survey Data Set

BACKGROUND & METHODOLOGY

This project addresses the prediction of depression using depression survey data. Leveraging a dataset containing demographic, financial, and health-related features, the study employs Logistic Regression to build a predictive model. Key steps involved data cleaning, data visualization, exploratory data analysis to understand feature distributions and relationships, handling of class imbalance using SMOTE, and model training and evaluation. The results indicate that while the base Logistic Regression model struggled with the imbalanced dataset, applying SMOTE improved the model's ability to identify the minority class (depressed individuals), as reflected in better recall and F1-score for the positive class, albeit with a trade-off in overall accuracy and precision. The project culminates in a proposed Streamlit application for model deployment, allowing users to input specific features and receive a depression risk prediction.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and evaluated a Logistic Regression model for predicting depression risk using survey data containing demographic, financial, and health-related features. Applied SMOTE to address significant class imbalance, improving recall for the minority class (depressed individuals) from near-zero to 0.30 with an F1-score of 0.19. Key predictors identified included sex, age, marital status, number of children, and education level. A user-friendly Streamlit web interface was designed for real-time depression risk prediction, allowing users to input personal features and receive probability-based assessments. The project demonstrates the potential of machine learning for early mental health screening while highlighting challenges posed by class imbalance and limited predictive power of available features.

INTERNSHIP Internship Experience

Nyeri Town Health Centre — Analyzing Patient Characteristics and Health Patterns in Non-Communicable Diseases Using Association Rule Mining and Clustering | AI for Patient Risk Stratification | Data-Driven Healthcare Decision Support

Dataset Used: : Non-communicable diseases patient dataset from Mukurweini health facility containing demographic information, dates, diagnoses, insurance status, and number of visits. Dataset containing patient records with variables such as age, sex, comorbidities, insurance status, diagnoses, and number of visits, used to apply association rule mining (FP-Growth algorithm) and K-Means clustering to identify relationships between patient demographics, diagnoses, and insurance status, and to segment patients into distinct groups based on their health profiles for targeted interventions in hypertension and diabetes management

METHODOLOGY

Data cleaning and preprocessing: handled missing values, standardized variables, transformed data for association rule mining | Feature engineering: created new features like Age Group, Gender, Has Insurance, Diagnoses, and Complications | Applied association rule mining using FP-Growth algorithm (mlxtend library) to find frequent itemsets and generate association rules evaluated using support, confidence, and lift metrics | Performed K-Means clustering on encoded data to group patients into distinct clusters using elbow method for optimal cluster selection | Conducted targeted analysis on specific patient clusters (uninsured group) to analyze demographic and diagnostic characteristics | Generated visualizations (histograms, bar plots, box plots, heatmaps, network graphs) using Python with pandas, scikit-learn, mlxtend, matplotlib, and seaborn

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Generated 102 association rules with top rules revealing key relationships among demographic and health variables | Identified that individuals aged 50-59 years had high likelihood of being female and insured with strong association to hypertension | Established female patients formed majority (over 500) in hypertension records with older adults (70+) most prominently represented | Insurance status was widespread with strong association between hypertension and being insured (confidence=0.98) | Clustering analysis (k=3) identified three distinct patient groups with Cluster 1 (498 patients) showing high hypertension prevalence (~90%) | High-confidence associations discovered: "Hypertension ? Insured" and "Female ? Hypertension" | Network graph and heatmap visualizations revealed interconnected risk factors for targeted interventions | Demonstrated feasibility of applying association rule mining and clustering to identify high-risk patient groups for targeted hypertension management | Enhanced practical skills in health data management, unsupervised machine learning, and result interpretation for public health applications | Provided actionable recommendations for integrating predictive analytics into clinical workflows and strengthening data quality in NCD management

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

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BACKGROUND & METHODOLOGY

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INTERNSHIP Internship Experience

Nyeri Town Health Centre — Analyzing Patient Characteristics and Health Patterns in Non-Communicable Diseases Using Association Rule Mining and Clustering | AI for Patient Risk Stratification | Data-Driven Healthcare Decision Support

Dataset Used: : Non-communicable diseases patient dataset from Mukurweini health facility containing demographic information, dates, diagnoses, insurance status, and number of visits. Dataset containing patient records with variables such as age, sex, comorbidities, insurance status, diagnoses, and number of visits, used to apply association rule mining (FP-Growth algorithm) and K-Means clustering to identify relationships between patient demographics, diagnoses, and insurance status, and to segment patients into distinct groups based on their health profiles for targeted interventions in hypertension and diabetes management

METHODOLOGY

Data cleaning and preprocessing: handled missing values, standardized variables, transformed data for association rule mining | Feature engineering: created new features like Age Group, Gender, Has Insurance, Diagnoses, and Complications | Applied association rule mining using FP-Growth algorithm (mlxtend library) to find frequent itemsets and generate association rules evaluated using support, confidence, and lift metrics | Performed K-Means clustering on encoded data to group patients into distinct clusters using elbow method for optimal cluster selection | Conducted targeted analysis on specific patient clusters (uninsured group) to analyze demographic and diagnostic characteristics | Generated visualizations (histograms, bar plots, box plots, heatmaps, network graphs) using Python with pandas, scikit-learn, mlxtend, matplotlib, and seaborn

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

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CURRENT STATUS

Not yet reported

**CAPSTONE** **Capstone Project***Predicting Thyroid Disease Recurrence Using Logistic Regression Model***BACKGROUND & METHODOLOGY**

This project aimed to develop a logistic regression model to predict recurrence in thyroid patients using clinical and demographic data from the thyroid diff data set. The data set contained variables like gender, age, smoking history, pathology results, tumor stage and recurrence outcomes. The categorical variables were encoded and the ordinal variables treated as ordered features, and missing data were checked and replaced where necessary. Statistical tests such as chi-squared were used for feature selection for categorical variables. The model achieved an overall accuracy of 93% with 93% precision and 81% recall for predicting recurrence. These results showed that the model can be used as a decision support tool for early identification of high risk patients. The model was also deployed with a framework that can be used for clinical settings. This study contributes to improved patient management and optimized resource allocation in thyroid cancer care.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a logistic regression model achieving 93% accuracy, 93% precision, and 81% recall for predicting thyroid disease recurrence. Key predictors identified included Stage ($p < 0.01$) and Metastasis ($p < 0.01$) as strong recurrence indicators. The model correctly classified 107 out of 115 cases with only 8 misclassifications. A web-based deployment framework using Streamlit was designed for clinical integration, enabling clinicians to input patient data and receive recurrence probability scores. The model provides an interpretable decision-support tool for early identification of high-risk patients, personalized follow-up care, and optimized resource allocation in thyroid cancer management.

INTERNSHIP **Internship Experience***Nyeri County Referral Hospital — Clustering and Survival Analysis of Cervical Cancer Patients | Machine Learning for Patient Stratification in Oncology Care*

Dataset Used: Cervical cancer patient records from Nyeri County Referral Hospital cancer registry (148 validated records containing demographic and clinical variables from 2018–2025). Dataset containing cervical cancer patient records with variables including age, marital status, stage at diagnosis, and treatment outcomes, used to explore associations between demographic/clinical factors and patient survival outcomes, and to apply hierarchical clustering techniques to uncover hidden patient subgroups with similar survival patterns for targeted oncology interventions

METHODOLOGY

Internship Centre Work: Prepared summaries for cancer case reporting and outcomes tracking; Conducted data quality checks to ensure completeness and accuracy of registry entries | Capstone Project: Loaded and cleaned cervical cancer patient dataset from hospital records; Ran descriptive statistics on demographic (age, marital status) and clinical (stage, outcome) variables; Applied Chi-square (χ^2) tests to assess associations between categorical variables; Conducted Hierarchical Clustering using Ward linkage on standardized variables (Age, Marital Status, Stage, Outcome); Generated dendrograms,

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Established that 75% of patients were diagnosed at late stages (III–IV), confirming late presentation as a major barrier to survival | Chi-square testing revealed stage at diagnosis as the only significant predictor of survival outcomes ($\chi^2 = 30.56$, $p < 0.001$), while age and marital status showed no significant associations | Hierarchical clustering identified three distinct patient subgroups with unexpected survival patterns, including 100% mortality in some early-stage clusters suggesting potential gaps in treatment access or follow-up | Enhanced practical skills in oncology data management, statistical testing, and unsupervised machine learning techniques | Provided actionable recommendations for strengthening cervical cancer screening programs, improving registry data quality, and integrating clustering analytics into routine cancer surveillance for evidence-based patient stratification

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Predictive Analysis of Insurance Expenses Based on Health and Demographic Characteristics

BACKGROUND & METHODOLOGY

This study analyzes a health insurance dataset of 1,338 individuals to identify the key demographic and health-related factors that influence medical expenses. The research examines how age, gender, body mass index (BMI), smoking status, number of children, and geographic region affect insurance-billed medical costs. The findings reveal that smoking status is the strongest predictor of medical costs, with smokers incurring expenses nearly four times higher than non-smokers. Age shows a strong positive correlation with costs, while BMI demonstrates a moderate positive relationship. The number of children has a non-linear effect, whereas gender and geographic region have relatively minor influence. These insights are valuable for insurers in designing risk-adjusted premium models, implementing targeted wellness programs—particularly smoking cessation and weight management initiatives—and optimizing cost-control strategies. Overall, the analysis underscores the significant financial impact of modifiable health behaviors on healthcare expenditures.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a predictive model for insurance expenses using Elastic Net regression with cross-validated hyperparameters. Key findings identified smoking status as the strongest predictor, with smokers incurring costs nearly four times higher than non-smokers (\$32,050 vs \$8,434). Age showed strong positive correlation (0.30) with expenses, while BMI demonstrated moderate positive correlation (0.20). The model was deployed via an interactive Streamlit web application enabling real-time insurance expense estimates based on individual health and demographic inputs. Results support risk-adjusted premium modeling, targeted wellness programs, and optimized cost control strategies for insurers.

INTERNSHIP Internship Experience

Othaya Sub County Hospital — Statistical Analysis and Unsupervised Association Rule Mining for Morbidity Pattern Detection | AI for Comorbidity Identification | Data-Driven Health Planning

★ LED TO A JOB OFFER

Dataset Used: Othaya Sub County Hospital morbidity dataset (January 2022–December 2024, 2,817 validated records containing age, sex, ICD codes/diagnosis text, sub-county, date of death, and mode of discharge). Project applying statistical analysis and Association Rule Mining (Apriori algorithm) to facility morbidity data to identify frequent diagnoses, comorbidity bundles, and demographic-outcome patterns. Dataset includes variables such as patient age, sex, diagnostic categories (Hypertension, Diabetes, HIV, TB, Malaria, Pneumonia, Asthma, Pregnancy), and discharge outcomes, used to uncover hidden relationships between demographic factors and clinical conditions for targeted public health interventions in a Level 4 rural-periurban facility.

METHODOLOGY

Data Cleaning and Preparation: Standardized age to years (converted days/months), imputed missing values where possible, dropped records with >50% missing essential fields, mapped ICD codes and free-text diagnoses into standardized diagnostic groups, created binary flags for diagnoses, one-hot encoded demographics for ARM | Statistical Analysis: Computed descriptive statistics (mean/median/range for age, frequencies for sex/diagnoses/outcomes), performed correlation analysis (age category — primary diagnosis), conducted Chi-square tests for categorical associations (sex — diagnosis, diagnosis — mode of discharge) at $p < 0.05$ significance | Association Rule Mining: Applied Apriori algorithm via mlxtend with thresholds (support ≥ 0.02 , confidence ≥ 0.6 , lift > 1), pruned rules for clinical relevance, generated visualizations (scatter plots, heatmaps, network graphs) | Clustering: Performed K-means clustering to segment patients into distinct clinical profiles based on demographic and diagnostic characteristics

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Identified significant association between patient sex and top 10 diagnoses ($\chi^2(18) = 184.87$, $p < 0.0001$), confirming gendered patterns in disease presentation (e.g., pregnancy-related conditions concentrated in females, hypertension more frequent in males) | Established significant association between primary diagnosis and mode of discharge ($\chi^2(9) = 35.44$, $p < 0.0001$), indicating outcome patterns aligned with disease severity | Discovered strong age-diagnosis links: hypertension overwhelmingly concentrated in older patients (lift nearly 3—), two-thirds of diabetes cases in patients aged 51+ confirming age-related NCD risk | Generated clinically meaningful association rules: Age 19-35 strongly associated with female sex (maternal health burden), pediatric pneumonia frequent but rarely fatal (positive outcome insight), elderly males with diabetes showing favorable survival | K-means clustering identified four distinct patient profiles: Cluster 0 (young females with acute infections, high survival), Cluster 1 (reproductive-age females with pregnancy complications), Cluster 2 (older patients with chronic conditions, slightly higher mortality), Cluster 3 (young males with acute infections) | Provided actionable recommendations: establish integrated NCD clinics for Hypertension+Diabetes, strengthen under-five illness triage for malaria/pneumonia during seasonal peaks, improve maternal health pathways for ages 15-49, integrate dashboards into hospital Quality Improvement meetings | Demonstrated practical application of unsupervised learning and statistical testing in routine health data for evidence-based planning in resource-limited settings

CURRENT STATUS

Regulatory pharmacist-Digital marketing · Employed (full-time)

Sector relevance: Yes — Fully related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · Data visualization · ML modeling · Model evaluation · Generative AI / LLMs · Ethics & bias in AI · Public health data analysis · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Predictive Analysis of Depression Survey Data: Smoking Status Prediction from Health Examination Metrics

BACKGROUND & METHODOLOGY

This project aims to develop a machine learning model to predict an individual's smoking status based on their health examination metrics. Utilizing a dataset containing 27 attributes including physiological, biochemical, and oral examination data for 55,692 individuals, we performed extensive data pre-processing, exploratory data analysis (EDA), and feature engineering. A key step involved using K-Means clustering to identify inherent patterns and structures within the data, which could inform the predictive modeling process. The project outlines a complete data science pipeline, from data loading and cleaning to preprocessing and preliminary clustering analysis. The results from the EDA and clustering reveal significant differences in key health indicators (e.g., hemoglobin, GPT levels) between smokers and non-smokers. This analysis serves as a crucial foundation for building a subsequent classification model to accurately predict smoking behavior, which could be a valuable tool for public health initiatives and personalized healthcare interventions.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully completed data preprocessing and exploratory analysis on 55,692 health examination records with 27 features, establishing a robust foundation for smoking status prediction. Key findings from EDA revealed significant physiological differences between smokers and non-smokers, including elevated hemoglobin and GPT (?-GTP) levels in smokers. The dataset was standardized and encoded using an automated pipeline, with K-Means clustering implemented to uncover natural groupings correlated with smoking behavior. The target variable showed 37% smokers and 63% non-smokers, indicating class imbalance requiring SMOTE or class weighting for future classification. A conceptual Streamlit deployment interface was designed for real-time prediction input and probability scoring. The project provides a complete analytical framework ready for supervised model development using Logistic Regression, Random Forest, or XGBoost, with potential applications in public health screening, clinical practice, and personalized healthcare interventions.

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Predicting Malaria Using Risk Factors in Machine Learning

BACKGROUND & METHODOLOGY

Malaria remains a major public health challenge, particularly in tropical regions, where conventional diagnostic methods are often slow, costly, and limited in accessibility. This study explores the application of machine learning to predict malaria risk using key environmental, climatic, socio-economic, and health-related factors, including antimalarial history, vaccination status, bed net use, indoor spraying, nutritional status, and previous malaria test results. Secondary data from Kaggle was analyzed, and a logistic regression model was developed to classify malaria risk. The model achieved an overall accuracy of 70%, with precision of 0.81 and recall of 0.70 for non-infected cases, and precision of 0.56 and recall of 0.70 for infected cases. The ROC-AUC of 0.766 demonstrates good discriminatory ability, while the PR-AUC of 0.655 indicates moderate performance in identifying infections. The results highlight the potential of machine learning to enhance early detection, support timely interventions, and guide resource allocation in malaria prevention and control.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a logistic regression model to predict malaria infection risk using key categorical risk factors including antimalarial history, vaccination status, bed net use, indoor spraying history, nutritional status, and malaria RDT results, sourced from Kaggle. The model achieved an overall accuracy of 70%, a ROC-AUC of 0.766, and a PR-AUC of 0.655, demonstrating good discriminatory ability in classifying malaria risk. The model was deployed as a secure API application, providing a practical, data-driven tool to support early detection, targeted interventions, and resource allocation in malaria prevention and control.

INTERNSHIP Internship Experience

Jordan Hospital — Predicting Mode of Delivery Using Logistic Regression Model | Machine Learning for Maternal Health Analytics

Dataset Used: Maternal health records from Jordan Hospital (3,106 validated records containing age, parity, gravidae, marital status, ANC visits, and vaginal examination results). Dataset containing maternal health variables used to predict mode of delivery (normal spontaneous vaginal delivery vs Caesarean Section) using logistic regression. The project aimed to identify key clinical and demographic factors influencing delivery outcomes to support early risk identification and evidence-based obstetric care planning in resource-limited settings.

METHODOLOGY

Internship Centre Work: Data entry and updating maternity health records; Cleaning and validating maternal datasets for accuracy and completeness; Performing exploratory data analysis and visualization to identify trends; Training and evaluating predictive models using machine learning techniques; Preparing summary reports and visual dashboards for review | Capstone Project: Extracted 3,106 maternal health records; Cleaned and recoded variables (age, parity, gravidae, marital status, ANC visits, vaginal exam results); Performed descriptive analysis and logistic regression modeling in Python; Generated visualizations (pie charts, bar graphs, distribution plots) to illustrate maternal characteristics and delivery patterns; Evaluated model performance using accuracy metrics

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Logistic regression model achieved 77% accuracy in predicting mode of delivery, demonstrating reliable predictive performance for clinical decision support | Identified parity, number of ANC visits, and pregnancy history as key predictors: mothers with higher parity more likely to have normal spontaneous vaginal delivery (SVD), while first-time mothers with limited ANC attendance (<4 visits) showed higher likelihood of Caesarean Section | Established that regular prenatal monitoring (4+ ANC visits) strongly associated with safe SVD outcomes, reinforcing importance of antenatal care adherence | Enhanced practical skills in data cleaning, statistical analysis, machine learning modeling, and health data interpretation | Demonstrated feasibility of using routine maternal health data for predictive analytics in hospital settings | Provided evidence-based recommendations for integrating predictive tools into maternal health workflows to support early risk stratification and targeted clinical interventions

CURRENT STATUS

I recently completed my internship at Jordan, where I gained practical experience and strengthened my skills. I am currently actively searching for a job opportunity that aligns with my background and interests. I kindly request the Engage team to consider me for any available roles that match my qualifications. · Unemployed — Actively searching

Sector relevance: Not related

SKILLS IN USE

Give a job right away · so that I'll put into practice the skills I gained

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Predicting Malaria Using Risk Factors in Machine Learning

BACKGROUND & METHODOLOGY

Malaria remains a major public health challenge, particularly in tropical regions, where conventional diagnostic methods are often slow, costly, and limited in accessibility. This study explores the application of machine learning to predict malaria risk using key environmental, climatic, socio-economic, and health-related factors, including antimalarial history, vaccination status, bed net use, indoor spraying, nutritional status, and previous malaria test results. Secondary data from Kaggle was analyzed, and a logistic regression model was developed to classify malaria risk. The model achieved an overall accuracy of 70%, with precision of 0.81 and recall of 0.70 for non-infected cases, and precision of 0.56 and recall of 0.70 for infected cases. The ROC-AUC of 0.766 demonstrates good discriminatory ability, while the PR-AUC of 0.655 indicates moderate performance in identifying infections. The results highlight the potential of machine learning to enhance early detection, support timely interventions, and guide resource allocation in malaria prevention and control.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a logistic regression model to predict malaria infection risk using key categorical risk factors including antimalarial history, vaccination status, bed net use, indoor spraying history, nutritional status, and malaria RDT results, sourced from Kaggle. The model achieved an overall accuracy of 70%, a ROC-AUC of 0.766, and a PR-AUC of 0.655, demonstrating good discriminatory ability in classifying malaria risk. The model was deployed as a secure API application, providing a practical, data-driven tool to support early detection, targeted interventions, and resource allocation in malaria prevention and control.

INTERNSHIP Internship Experience

Neema Hospital Limited — Predicting Urinary Tract Infection (UTI) Diagnosis Using Clinical Data | Machine Learning for Clinical Decision Support

Dataset Used: Patient clinical records from Neema Hospital Limited (8,949-9,949 records) containing variables: age, gender, nitrites, erythrocytes, glucose, ketones, leukocytes, and diagnosis outcome (UTI positive/negative). Dataset containing routine patient clinical and urine test data used to develop and evaluate classification models for predicting Urinary Tract Infection (UTI) diagnosis. The project aimed to support rapid clinical screening by identifying key laboratory indicators associated with UTI and building predictive tools for early detection in resource-limited healthcare settings.

METHODOLOGY

Internship Centre Work: Entered and validated over 10,000 patient records into the EMR system; assisted in compiling weekly laboratory and outpatient department reports; conducted data quality checks by cross-verifying registers with electronic records | Capstone Project: Extracted and cleaned UTI dataset; dropped irrelevant columns (Unnamed: 8-14) and standardized variable formats; performed descriptive statistics (age distribution, gender proportions, test outcomes); trained multiple classification models (Logistic Regression, Random Forest, Support Vector Machine) to predict UTI diagnosis; evaluated models using accuracy, precision, recall, and F1-score; generated visualizations for data distribution and model performance

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Support Vector Machine model achieved best performance: Accuracy = 94%, Precision = 92%, Recall = 91%, F1-score = 91%; identified leukocytes, nitrites, and erythrocytes as strongest predictors of UTI; established female patients comprised 70% of cases with higher UTI prevalence, aligning with medical literature; demonstrated feasibility of using routine clinical data for rapid UTI screening; highlighted challenges with missing values and class imbalance affecting model performance; enhanced practical skills in data cleaning, statistical analysis, and machine learning model deployment for healthcare applications

CURRENT STATUS

Not yet reported

**CAPSTONE** **Capstone Project***Detecting Key Predictors for Poor Uptake of Measles 2 Vaccine Among Under-Five Children in Kenya Using Machine Learning***BACKGROUND & METHODOLOGY**

Measles remains a major public health challenge in Kenya, particularly due to the suboptimal uptake of the second dose of the Measles-Containing Vaccine (MCV2) among children under five. This study aims to identify key socio-demographic, socioeconomic, and health-related predictors contributing to poor MR2 uptake and develop a machine learning model to support targeted public health interventions. Utilizing a dataset from electronic health records accessed via Kaggle, the study applied data cleaning, exploratory analysis, and a Random Forest algorithm to predict MR2 vaccination status. The model achieved a moderate overall accuracy of 61%, with strong performance in predicting children who missed the MR2 vaccine (recall = 78%, F1-score = 0.71), but weaker performance in identifying those who completed the vaccine (recall = 36%). Feature importance analysis revealed that child's age, caregiver's occupation, transport mode, household income, and distance to health facility were the most influential predictors of MR2 uptake. Based on these findings, the study recommends targeted health education for caregivers, improved healthcare accessibility, enhanced reminder systems, and integrated socioeconomic support to improve MR2 coverage and reduce measles-related morbidity and mortality in Kenya.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Random Forest classification model to detect key predictors of poor Measles 2 (MR2) vaccine uptake among under-five children in Kenya, using electronic health records sourced from Kaggle. The model achieved an overall accuracy of 61%, with a recall of 78% and F1-score of 0.71 for identifying children who missed the vaccine. Feature importance analysis highlighted child's age, caregiver's occupation, transport mode, household income, and distance to health facility as the most influential predictors. The model was deployed as a web application via Streamlit and GitHub, providing a practical tool to support health authorities in designing targeted, data-driven interventions to improve MR2 vaccination coverage.

INTERNSHIP **Internship Experience***Westlands Health Centre — Predictive Model for Hypertension Risk Assessment | Machine Learning for Clinical Decision Support*

Dataset Used: Aggregated patient entries from manual diabetes clinic register at Westlands Health Centre (643 validated records containing demographics, physiological measurements including BMI, blood glucose, cholesterol, and clinical observations). Capstone project developing a machine learning model to predict individual patient risk of hypertension based on comprehensive physiological and demographic health data. The tool aims to assist public health officials and clinicians in identifying high-risk patients early, enabling timely intervention strategies and transitioning hypertension risk identification from reactive screening to proactive, data-driven prevention.

METHODOLOGY

Internship Centre Work: Performed daily data entry and validation of maternal and child health records into electronic systems (Taifa Care, Tiberbu, Afya Yangu); assisted in reconciling paper registers with electronic databases; supported preparation of weekly/monthly health claims and program reports; participated in data review meetings to identify trends and defaulters | Capstone Project: Cleaned and preprocessed diabetes clinic data using Python (handled missing values via mode/mean/median imputation, recoded categorical variables using one-hot encoding, managed outliers via IQR method); performed descriptive statistics and exploratory visualization using Pandas, Matplotlib, and Seaborn; trained and compared classification models (Support Vector Machine, Random Forest) using Scikit-learn; applied K-fold cross-validation and hyperparameter optimization for model tuning; evaluated performance using accuracy, precision, recall, and F1-score; deployed final model as interactive Streamlit web application for clinical risk stratification

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Random Forest Classifier achieved exceptional balanced performance: Accuracy = 94.1%, Precision = 94.1%, Recall = 94.1%, F1-score = 94.1 | Identified key predictors of hypertension: Age (most dominant non-modifiable factor), BMI (highly significant), Mean Blood Glucose Level, and initial Systolic Blood Pressure reading | Established hypertension prevalence of 42.5% within diabetes clinic population with mean patient age 55.3 — 10.1 years and mean BMI 30.8 — 4.5 indicating high obesity burden | Successfully deployed Streamlit web application providing clinicians with real-time risk stratification into low/medium/high categories for targeted intervention | Enhanced practical skills in end-to-end data science pipeline from data cleaning and EDA to model deployment and interpretation | Demonstrated feasibility of using routine clinical data for non-invasive hypertension risk screening in resource-limited settings | Provided actionable recommendations for pilot integration of predictive tool, transition to electronic data capture, expansion of data collection parameters, and longitudinal model validation

CURRENT STATUS**Unemployed — Actively searching**

Sector relevance: Yes — Fully related

SKILLS IN USE

Python · Data cleaning & wrangling · Data visualization · ML modeling · Model evaluation · MLOps / Model deployment · Natural Language Processing (NLP) · Generative AI / LLMs · Ethics & bias in AI · Public health data analysis

Confidence applying ENGAGE skills: **4 / 5**



CAPSTONE Capstone Project

Detecting Key Predictors for Poor Uptake of Measles 2 Vaccine Among Under-Five Children in Kenya Using Machine Learning

BACKGROUND & METHODOLOGY

Measles remains a major public health challenge in Kenya, particularly due to the suboptimal uptake of the second dose of the Measles-Containing Vaccine (MCV2) among children under five. This study aims to identify key socio-demographic, socioeconomic, and health-related predictors contributing to poor MR2 uptake and develop a machine learning model to support targeted public health interventions. Utilizing a dataset from electronic health records accessed via Kaggle, the study applied data cleaning, exploratory analysis, and a Random Forest algorithm to predict MR2 vaccination status. The model achieved a moderate overall accuracy of 61%, with strong performance in predicting children who missed the MR2 vaccine (recall = 78%, F1-score = 0.71), but weaker performance in identifying those who completed the vaccine (recall = 36%). Feature importance analysis revealed that child's age, caregiver's occupation, transport mode, household income, and distance to health facility were the most influential predictors of MR2 uptake. Based on these findings, the study recommends targeted health education for caregivers, improved healthcare accessibility, enhanced reminder systems, and integrated socioeconomic support to improve MR2 coverage and reduce measles-related morbidity and mortality in Kenya.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Random Forest classification model to detect key predictors of poor Measles 2 (MR2) vaccine uptake among under-five children in Kenya, using electronic health records sourced from Kaggle. The model achieved an overall accuracy of 61%, with a recall of 78% and F1-score of 0.71 for identifying children who missed the vaccine. Feature importance analysis highlighted child's age, caregiver's occupation, transport mode, household income, and distance to health facility as the most influential predictors. The model was deployed as a web application via Streamlit and GitHub, providing a practical tool to support health authorities in designing targeted, data-driven interventions to improve MR2 vaccination coverage.

INTERNSHIP Internship Experience

Equity Afya Medical Centre — Kitui Branch — Predicting T2D Complication Risk Using Machine Learning | AI for Clinical Decision Support in Diabetes Care | Multi-Layer Perceptron for Patient Risk Stratification

Dataset Used: 7,000 anonymized patient records from Equity Afya Medical Centre (Kitui) containing demographic, clinical, and treatment variables including age, gender, diet quality, BMI, LDL, HDL, blood glucose, and complication status. Capstone project applying machine learning to predict likelihood of complication occurrence among Type 2 Diabetes patients. Dataset includes clinical measurements, lifestyle factors, and treatment history used to train an MLP classifier for early risk detection and targeted intervention planning in resource-limited healthcare settings.

METHODOLOGY

Internship Centre Work: Data entry and updating digital patient registers; Cleaning and validating patient records; Preparing visual summaries of complication trends; Supporting reporting and data review sessions with health staff | Capstone Project: Data preprocessing (handling missing values, encoding categorical variables, scaling numerical features); Descriptive analysis and visualization of glucose, BMI, and age distributions using Python (Pandas, Matplotlib); Training and testing Multi-Layer Perceptron (MLP) classifier using Scikit-learn; Model evaluation using accuracy, precision, recall, F1-score, and ROC-AUC metrics; Developing Streamlit web application for real-time clinical risk prediction and decision support

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully entered and validated 7,000 patient records, improving data quality for facility reporting | Generated interactive dashboards and charts on patient demographics and complication occurrence for health worker review | MLP classifier achieved exceptional performance: Accuracy = 96.7%, Precision = 96.0%, Recall = 96.9%, F1-score = 96.5%, ROC-AUC = 0.996 | Identified key predictors of complication risk: elevated blood glucose, high BMI, older age, and longer treatment duration | Model demonstrated strong balance between detecting true complication cases and minimizing false negatives, critical for clinical utility | Successfully deployed Streamlit web app providing clinicians with real-time risk stratification for proactive patient management | Enhanced practical skills in end-to-end data science pipeline from data cleaning to model deployment in public health settings | Demonstrated feasibility of using routine clinical data for predictive analytics in diabetes care | Provided actionable recommendations for integrating AI-driven tools into clinical workflows to support early intervention and personalized care strategies

CURRENT STATUS

Not yet reported



CAPSTONE **Capstone Project**

Predicting Maternal Mortality Rate in Kenya

BACKGROUND & METHODOLOGY

Maternal mortality refers to deaths due to complications from pregnancy, childbirth, or within a year of pregnancy resolution, and remains a significant public health concern disproportionately affecting low- and middle-income countries. This study explores the prediction of maternal mortality rates in Kenya by identifying key contributing factors including postpartum hemorrhage, hypertensive disorders, sepsis, obstructed labour, and indirect causes such as malaria, HIV/AIDS, and anemia. Using secondary data from the Kenya Demographic and Health Surveys (DHS) and the DHIS2 platform, a logistic regression model was developed with maternal death as the dependent variable. The model achieved an overall accuracy of 56%, with a ROC-AUC of 0.55 and a PR-AUC of 0.23, reflecting challenges posed by significant class imbalance in the dataset. While the model performed well in predicting survival cases (precision: 82%), it struggled to reliably identify maternal death cases (precision: 23%, recall: 51%). The study underscores the potential of machine learning to support evidence-based strategies for reducing preventable maternal deaths and advancing SDG 3.1

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a logistic regression model to predict maternal mortality in Kenya using health facility data from DHS and DHIS2, incorporating variables related to underlying conditions, hospital location, and obstetric risk factors. The model achieved an overall accuracy of 56%, with strong precision for survival prediction (82%) but limited performance for mortality detection (precision: 23%, F1-score: 0.32, ROC-AUC: 0.55), largely attributable to class imbalance between death and survival cases. The model was designed for deployment via Flask or Streamlit, with recommendations to improve predictive reliability through SMOTE-based data balancing, advanced algorithms, and enriched feature sets to better support maternal health interventions and policy planning in Kenya.

INTERNSHIP **Internship Experience**

Provide International Hospital — Analysis of Maternal Data Using K-Means Clustering | Machine Learning for Patient Segmentation in Maternal Health

Dataset Used: Manually recorded birth records from Provide International Hospital (302 validated records containing maternal demographics, clinical measurements, and delivery outcomes). Dataset containing maternal health variables including Duration of Labour, Mode of Delivery, History of Caesarean Section, Mode of Presentation, Maternal Age, Marital Status, Education Level, BMI, Gravida, Para, and Fetal Weight, used to identify distinct patient segments through unsupervised machine learning clustering techniques to support targeted maternal health interventions and resource planning

METHODOLOGY

Internship Centre Work: Entered and validated 302 manually recorded birth records in Excel | Assisted in preparing monthly MCH reports including immunization data | Supported facility health app (Banda Health) implementation | Conducted data quality checks by comparing available registers with electronic records | Capstone Project: Extracted 302 available manual birth records | Cleaned and recoded variables (History of Cs, Fetal/Mother weight, Mode of Delivery, Mode of Presentation, Maternal education, Para/Gravida, Duration of Labour, Maternal Age) | Ran descriptive and categorical analysis in Python | Generated tables, graphs, and clusters using K-Means Clustering algorithm | Visualized relationships between variables through histograms, scatter plots, and box plots

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully identified three distinct patient clusters using K-Means clustering: Cluster 0 (older mothers ~31.8 years, higher BMI ~76.8, higher Gravida ~3.6, slightly higher fetal weight ~3.3kg, mostly married, mix of SVD/C-S delivery) | Cluster 1 (younger mothers ~23.9 years, lower BMI ~73.1, lower Gravida ~1.5, slightly lower fetal weight ~3.0kg, primarily SVD delivery, more single mothers) | Cluster 2 (single outlier case with data quality concerns) | Established key variable relationships: older individuals tend to have higher gravida; higher gravida more frequent among married individuals and those with secondary/college education; SVD births tend to have longer labour durations than C-S births; fetal weight shows weak positive correlation with maternal BMI; Cephalic presentation most common | Enhanced practical data management and analytical skills | Provided actionable insights for strengthening maternal and child health programs through data-driven patient segmentation and targeted intervention strategies

CURRENT STATUS

No employment · Unemployed — Actively searching

Sector relevance: Yes — Fully related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · Data visualization · ML modeling · Model evaluation · Natural Language Processing (NLP) · Ethics & bias in AI · Public health data analysis · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Logistic Regression Model to Predict Hypertension

BACKGROUND & METHODOLOGY

Hypertension is one of the leading causes of cardiovascular disease, kidney damage, and premature death worldwide. In Kenya, cases are increasing due to lifestyle changes, poor diet, and low awareness. Early identification of high-risk individuals is vital for prevention. This study developed a logistic regression model to predict hypertension using demographic, lifestyle, and clinical variables including age, sex, BMI, smoking status, family history of diabetes, alcohol intake, physical activity, cardiovascular disease, cholesterol levels, haemoglobin count, and systolic and diastolic blood pressure. The model achieved an accuracy of about 50%, with sensitivity ~48% and specificity ~52%, indicating limited predictive power. While results show the model is not reliable for diagnosis, it demonstrates how data-driven methods can be applied to public health and highlights the importance of including additional predictors for stronger models.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a logistic regression model to predict hypertension risk among adults in Kenya using demographic, lifestyle, and clinical variables from secondary health survey data. The model achieved an overall accuracy of approximately 50%, with sensitivity of ~48%, specificity of ~52%, and a ROC-AUC of 0.55, identifying systolic and diastolic blood pressure as the strongest predictors, followed by age, BMI, and cholesterol levels. While the model's predictive power remains limited due to class imbalance and dataset constraints, it was designed for deployment via Flask or Streamlit as an early risk screening tool, with recommendations to expand clinical variables, apply SMOTE for class balancing, and explore advanced algorithms such as Random Forest and Gradient Boosting to improve reliability.

INTERNSHIP Internship Experience

Mulutu Catholic Dispensary — Visualization of Hypertension Prescription Patterns and Predictive Modeling for Patient Risk Stratification | Random Forest Classifier for Hypertension Outcome Prediction

Dataset Used: Hypertension medication and patient records from Mulutu Catholic Dispensary (2,205 validated patient records containing demographic, clinical, and prescription variables). Dataset containing hypertension patient records with variables including age, sex, residency, blood pressure readings, and prescribed medications, used to visualize prescription patterns, identify demographic trends, and develop a Random Forest classifier to predict hypertension-related outcomes for targeted clinical interventions and resource planning

METHODOLOGY

Data cleaning and management using pandas for hypertension medication records | Created visualizations with matplotlib and seaborn including prescription trend charts, demographic distribution plots, and residency-based prevalence maps | Applied feature engineering and train-test splitting for model preparation | Trained and evaluated Random Forest classifier using scikit-learn with accuracy, precision, recall, and F1-score metrics | Generated confusion matrix and feature importance analysis to interpret model performance and key predictors

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Random Forest classifier achieved 73% overall accuracy with balanced performance across classes (Yes class: Precision=0.76, Recall=0.81, F1=0.79; No class: Precision=0.69, Recall=0.62, F1=0.65) | Visual analysis revealed older adults (40+) receive majority of hypertension prescriptions and female patients show slightly higher prescription counts | Identified top 5 residencies with highest hypertension prevalence for targeted outreach | Feature importance analysis highlighted key predictors including age, BMI, and diabetic status | Provided evidence-based recommendations for early screening protocols, targeted health education programs, integrated data-driven clinical decision support, strengthened hypertension management workflows, and geographic-specific interventions for high-prevalence areas

CURRENT STATUS

Unemployed — Actively searching

Sector relevance: Not related

SKILLS IN USE

Python · Ethics & bias in AI

Confidence applying ENGAGE skills: **4 / 5**



CAPSTONE Capstone Project

Machine Learning Model for Detecting Suicidal Behavior Among Youth in Kenya

BACKGROUND & METHODOLOGY

This study explores the application of machine learning, specifically the XGBoost algorithm, to predict causes of death using socio-demographic, behavioral, and clinical data collected from morgue records across various counties in Kenya. The dataset comprises anonymized records with features such as age, gender, marital status, employment status, substance use, mental health history, and socioeconomic indicators. After preprocessing and feature engineering, an XGBoost classification model was trained to predict the primary cause of death. The model achieved an accuracy of 61%, indicating moderate predictive performance. While the results suggest that socio-behavioral factors are informative in understanding mortality patterns, the relatively modest accuracy highlights the complexity and multifactorial nature of causes of death, as well as potential limitations in data quality, class imbalance, and feature representation. This work underscores the potential of machine learning in public health surveillance while emphasizing the need for richer, longitudinal, and more standardized data collection to improve predictive modeling in forensic and epidemiological contexts.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed an XGBoost classification model trained on anonymized morgue records from multiple Kenyan counties to predict cause of death, achieving an accuracy of 61.5%. The model identified socio-behavioral patterns linked to suicidal behavior and was deployed via a Django web application. Findings highlight the need for improved data quality, class balancing, and exploration of stronger models such as Random Forest and LightGBM to enhance predictive reliability for real-world suicide prevention applications.

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Predicting Maternal Mortality Rate in Kenya

BACKGROUND & METHODOLOGY

Maternal mortality refers to deaths due to complications from pregnancy, childbirth, or within a year of pregnancy resolution, and remains a significant public health concern disproportionately affecting low- and middle-income countries. This study explores the prediction of maternal mortality rates in Kenya by identifying key contributing factors including postpartum hemorrhage, hypertensive disorders, sepsis, obstructed labour, and indirect causes such as malaria, HIV/AIDS, and anemia. Using secondary data from the Kenya Demographic and Health Surveys (DHS) and the DHIS2 platform, a logistic regression model was developed with maternal death as the dependent variable. The model achieved an overall accuracy of 56%, with a ROC-AUC of 0.55 and a PR-AUC of 0.23, reflecting challenges posed by significant class imbalance in the dataset. While the model performed well in predicting survival cases (precision: 82%), it struggled to reliably identify maternal death cases (precision: 23%, recall: 51%). The study underscores the potential of machine learning to support evidence-based strategies for reducing preventable maternal deaths and advancing SDG 3.1

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a logistic regression model to predict maternal mortality in Kenya using health facility data from DHS and DHIS2, incorporating variables related to underlying conditions, hospital location, and obstetric risk factors. The model achieved an overall accuracy of 56%, with strong precision for survival prediction (82%) but limited performance for mortality detection (precision: 23%, F1-score: 0.32, ROC-AUC: 0.55), largely attributable to class imbalance between death and survival cases. The model was designed for deployment via Flask or Streamlit, with recommendations to improve predictive reliability through SMOTE-based data balancing, advanced algorithms, and enriched feature sets to better support maternal health interventions and policy planning in Kenya.

INTERNSHIP Internship Experience

Mulutu Catholic Dispensary — Visualization of Hypertension Prescription Patterns and Predictive Modeling for Patient Risk Stratification | Random Forest Classifier for Hypertension Outcome Prediction

Dataset Used: Hypertension medication and patient records from Mulutu Catholic Dispensary (2,205 validated patient records containing demographic, clinical, and prescription variables). Dataset containing hypertension patient records with variables including age, sex, residency, blood pressure readings, and prescribed medications, used to visualize prescription patterns, identify demographic trends, and develop a Random Forest classifier to predict hypertension-related outcomes for targeted clinical interventions and resource planning

METHODOLOGY

Data cleaning and management using pandas for hypertension medication records | Created visualizations with matplotlib and seaborn including prescription trend charts, demographic distribution plots, and residency-based prevalence maps | Applied feature engineering and train-test splitting for model preparation | Trained and evaluated Random Forest classifier using scikit-learn with accuracy, precision, recall, and F1-score metrics | Generated confusion matrix and feature importance analysis to interpret model performance and key predictors

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Random Forest classifier achieved 73% overall accuracy with balanced performance across classes (Yes class: Precision=0.76, Recall=0.81, F1=0.79; No class: Precision=0.69, Recall=0.62, F1=0.65) | Visual analysis revealed older adults (40+) receive majority of hypertension prescriptions and female patients show slightly higher prescription counts | Identified top 5 residencies with highest hypertension prevalence for targeted outreach | Feature importance analysis highlighted key predictors including age, BMI, and diabetic status | Provided evidence-based recommendations for early screening protocols, targeted health education programs, integrated data-driven clinical decision support, strengthened hypertension management workflows, and geographic-specific interventions for high-prevalence areas

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Prediction of Tuberculosis Prevalence Using Categorical Naïve Bayes Model

BACKGROUND & METHODOLOGY

Tuberculosis (TB) is a major public health challenge in Kenya, ranking the country among the 30 highest TB burden nations globally. Traditional prevalence estimation methods such as surveys and routine reporting are costly, slow, and often miss undiagnosed cases. This study applied a machine learning approach to predict TB prevalence using demographic, socio-economic, environmental, and health-related factors obtained from electronic health records, DHS, and Kaggle datasets. A Categorical Naïve Bayes model was trained and evaluated using accuracy, precision, recall, and F1-score. The model achieved 67.1% accuracy, 67.1% recall, 45.0% precision, and an F1-score of 53.9%. Results show the model is effective at detecting true TB cases but limited by low precision, leading to false positives. Despite these limitations, the study demonstrates the potential of machine learning to strengthen TB surveillance, guide targeted interventions, and support resource allocation, provided that data quality and modeling approaches are improved.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Categorical Naïve Bayes model to predict tuberculosis prevalence in Kenya using patient and demographic variables including age, sex, HIV status, BCG vaccination, and smoking history. The model achieved an accuracy of 67.1%, a recall of 67.1%, a precision of 45.0%, and an F1-score of 53.9%, demonstrating reliable detection of true TB cases. The model was deployed via a Streamlit web application, providing a practical tool to support TB surveillance, guide resource allocation, and inform early intervention strategies, with recommendations for further improvement through advanced algorithms and enriched datasets.

INTERNSHIP Internship Experience

Neema Hospital Limited — Predicting Urinary Tract Infection (UTI) Diagnosis Using Clinical Data | Machine Learning for Clinical Decision Support

Dataset Used: Patient clinical records from Neema Hospital Limited (8,949-9,949 records) containing variables: age, gender, nitrites, erythrocytes, glucose, ketones, leukocytes, and diagnosis outcome (UTI positive/negative). Dataset containing routine patient clinical and urine test data used to develop and evaluate classification models for predicting Urinary Tract Infection (UTI) diagnosis. The project aimed to support rapid clinical screening by identifying key laboratory indicators associated with UTI and building predictive tools for early detection in resource-limited healthcare settings.

METHODOLOGY

Internship Centre Work: Entered and validated over 10,000 patient records into the EMR system; assisted in compiling weekly laboratory and outpatient department reports; conducted data quality checks by cross-verifying registers with electronic records | Capstone Project: Extracted and cleaned UTI dataset; dropped irrelevant columns (Unnamed: 8-14) and standardized variable formats; performed descriptive statistics (age distribution, gender proportions, test outcomes); trained multiple classification models (Logistic Regression, Random Forest, Support Vector Machine) to predict UTI diagnosis; evaluated models using accuracy, precision, recall, and F1-score; generated visualizations for data distribution and model performance

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Support Vector Machine model achieved best performance: Accuracy = 94%, Precision = 92%, Recall = 91%, F1-score = 91%; identified leukocytes, nitrites, and erythrocytes as strongest predictors of UTI; established female patients comprised 70% of cases with higher UTI prevalence, aligning with medical literature; demonstrated feasibility of using routine clinical data for rapid UTI screening; highlighted challenges with missing values and class imbalance affecting model performance; enhanced practical skills in data cleaning, statistical analysis, and machine learning model deployment for healthcare applications

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Prediction of Tuberculosis Prevalence Using Categorical Naïve Bayes Model

BACKGROUND & METHODOLOGY

Tuberculosis (TB) is a major public health challenge in Kenya, ranking the country among the 30 highest TB burden nations globally. Traditional prevalence estimation methods such as surveys and routine reporting are costly, slow, and often miss undiagnosed cases. This study applied a machine learning approach to predict TB prevalence using demographic, socio-economic, environmental, and health-related factors obtained from electronic health records, DHS, and Kaggle datasets. A Categorical Naïve Bayes model was trained and evaluated using accuracy, precision, recall, and F1-score. The model achieved 67.1% accuracy, 67.1% recall, 45.0% precision, and an F1-score of 53.9%. Results show the model is effective at detecting true TB cases but limited by low precision, leading to false positives. Despite these limitations, the study demonstrates the potential of machine learning to strengthen TB surveillance, guide targeted interventions, and support resource allocation, provided that data quality and modeling approaches are improved.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Categorical Naïve Bayes model to predict tuberculosis prevalence in Kenya using patient and demographic variables including age, sex, HIV status, BCG vaccination, and smoking history. The model achieved an accuracy of 67.1%, a recall of 67.1%, a precision of 45.0%, and an F1-score of 53.9%, demonstrating reliable detection of true TB cases. The model was deployed via a Streamlit web application, providing a practical tool to support TB surveillance, guide resource allocation, and inform early intervention strategies, with recommendations for further improvement through advanced algorithms and enriched datasets.

INTERNSHIP Internship Experience

SEKU University Health Clinic — Predicting Patient Sex Using Age and Five Common Diagnoses | Gaussian Naïve Bayes for Clinical Classification | AI for Demographic Pattern Analysis

Dataset Used: Manual outpatient clinic records from South Eastern Kenya University Health Clinic (779 validated outpatient cases with age, sex, and five most common diagnosis variables). Study applying machine learning to predict patient sex using age and five common outpatient diagnoses at the SEKU University Clinic. The project aimed to assess whether simple demographic and diagnostic information could serve as effective predictors in clinical classification tasks, demonstrating the application of AI and data science in healthcare training and internship clinic settings.

METHODOLOGY

Converted manual clinic records from books into Excel format; cleaned data by handling missing values and removing unnecessary columns; performed exploratory data analysis with visualizations (pie charts, histograms) to understand data distribution; applied Gaussian Naïve Bayes classifier using Python and Jupyter Notebook; trained and tested model with standard train-test split; evaluated performance using accuracy, precision, recall, and F1-score metrics; generated classification reports and confusion matrix for model interpretation

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Gaussian Naïve Bayes model achieved 61.7% accuracy, 61.8% precision, 61.7% recall, and 61.2% F1-score for predicting patient sex; Class 0 (n=408) showed stronger performance with 72% recall and 0.66 F1-score versus Class 1 (n=371) with 50% recall and 0.56 F1-score, indicating class imbalance in predictive performance; demonstrated moderate but meaningful predictive power beyond random guessing; enhanced practical skills in health data management, AI model application, and result interpretation; identified key challenges including limited dataset size, few predictive features, and class imbalance; provided foundation for future studies to expand features and test alternative models for improved clinical classification

CURRENT STATUS

No current job · Unemployed — Actively searching

Sector relevance: Not related

SKILLS IN USE

Data collection

Confidence applying ENGAGE skills: **5 / 5**

**CAPSTONE** **Capstone Project***Machine Learning Model for Detecting Suicidal Behavior Among Youth in Kenya***BACKGROUND & METHODOLOGY**

This study explores the application of machine learning, specifically the XGBoost algorithm, to predict causes of death using socio-demographic, behavioral, and clinical data collected from morgue records across various counties in Kenya. The dataset comprises anonymized records with features such as age, gender, marital status, employment status, substance use, mental health history, and socioeconomic indicators. After preprocessing and feature engineering, an XGBoost classification model was trained to predict the primary cause of death. The model achieved an accuracy of 61%, indicating moderate predictive performance. While the results suggest that socio-behavioral factors are informative in understanding mortality patterns, the relatively modest accuracy highlights the complexity and multifactorial nature of causes of death, as well as potential limitations in data quality, class imbalance, and feature representation. This work underscores the potential of machine learning in public health surveillance while emphasizing the need for richer, longitudinal, and more standardized data collection to improve predictive modeling in forensic and epidemiological contexts.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed an XGBoost classification model trained on anonymized morgue records from multiple Kenyan counties to predict cause of death, achieving an accuracy of 61.5%. The model identified socio-behavioral patterns linked to suicidal behavior and was deployed via a Django web application. Findings highlight the need for improved data quality, class balancing, and exploration of stronger models such as Random Forest and LightGBM to enhance predictive reliability for real-world suicide prevention applications.

INTERNSHIP **Internship Experience***Jordan Hospital — Monitoring Child Growth Using Random Forest Classifier | Machine Learning for Malnutrition Risk Prediction | AI for Pediatric Health Analytics*

Dataset Used: Child health records from Jordan Hospital, Kitui County (2,616 records collected between 2018–2025 from routine growth monitoring registers). Dataset containing child growth monitoring records with variables including age, gender, height, weight, height categories, and weight categories, used to classify children into nutritional status groups (underweight, normal, overweight, obese) using Random Forest Classifier to support early detection of malnutrition risks and improve clinical decision-making in maternal and child health services

METHODOLOGY

Internship Centre Work: Entered and validated 2,616 child health records into the DHIS2 system; conducted data quality checks by comparing paper registers with electronic records | Capstone Project: Extracted child growth monitoring records from 2018–2025 registers; cleaned and recoded variables (Age, Gender, Height, Weight, Height categories, Weight categories); performed descriptive analysis using Python; trained Random Forest Classifier to predict nutritional status categories; generated visualizations (distribution charts, feature importance bar graphs); evaluated model performance using accuracy metrics

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Random Forest Classifier achieved 84% overall accuracy in predicting child weight categories; model performed well in distinguishing between normal weight and underweight children but showed moderate performance for overweight/obese categories due to class imbalance; identified height, weight, and their categorized versions as key predictors of nutritional status; revealed that nearly 80% of children in dataset were classified as normal weight; demonstrated feasibility of using routinely collected clinical measurements for predictive analytics in hospital settings; enhanced practical skills in data cleaning, machine learning modeling, and health data management; provided evidence-based recommendations for integrating predictive tools into routine child growth monitoring to support early intervention for at-risk children

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Logistic Regression Model to Predict Hypertension

BACKGROUND & METHODOLOGY

Hypertension is one of the leading causes of cardiovascular disease, kidney damage, and premature death worldwide. In Kenya, cases are increasing due to lifestyle changes, poor diet, and low awareness. Early identification of high-risk individuals is vital for prevention. This study developed a logistic regression model to predict hypertension using demographic, lifestyle, and clinical variables including age, sex, BMI, smoking status, family history of diabetes, alcohol intake, physical activity, cardiovascular disease, cholesterol levels, haemoglobin count, and systolic and diastolic blood pressure. The model achieved an accuracy of about 50%, with sensitivity ~48% and specificity ~52%, indicating limited predictive power. While results show the model is not reliable for diagnosis, it demonstrates how data-driven methods can be applied to public health and highlights the importance of including additional predictors for stronger models.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a logistic regression model to predict hypertension risk among adults in Kenya using demographic, lifestyle, and clinical variables from secondary health survey data. The model achieved an overall accuracy of approximately 50%, with sensitivity of ~48%, specificity of ~52%, and a ROC-AUC of 0.55, identifying systolic and diastolic blood pressure as the strongest predictors, followed by age, BMI, and cholesterol levels. While the model's predictive power remains limited due to class imbalance and dataset constraints, it was designed for deployment via Flask or Streamlit as an early risk screening tool, with recommendations to expand clinical variables, apply SMOTE for class balancing, and explore advanced algorithms such as Random Forest and Gradient Boosting to improve reliability.

INTERNSHIP Internship Experience

Jordan Hospital — Predicting Mode of Delivery Using Logistic Regression Model | Machine Learning for Maternal Health Analytics

Dataset Used: Maternal health records from Jordan Hospital (3,106 validated records containing age, parity, gravidae, marital status, ANC visits, and vaginal examination results). Dataset containing maternal health variables used to predict mode of delivery (normal spontaneous vaginal delivery vs Caesarean Section) using logistic regression. The project aimed to identify key clinical and demographic factors influencing delivery outcomes to support early risk identification and evidence-based obstetric care planning in resource-limited settings.

METHODOLOGY

Internship Centre Work: Data entry and updating maternity health records; Cleaning and validating maternal datasets for accuracy and completeness; Performing exploratory data analysis and visualization to identify trends; Training and evaluating predictive models using machine learning techniques; Preparing summary reports and visual dashboards for review | Capstone Project: Extracted 3,106 maternal health records; Cleaned and recoded variables (age, parity, gravidae, marital status, ANC visits, vaginal exam results); Performed descriptive analysis and logistic regression modeling in Python; Generated visualizations (pie charts, bar graphs, distribution plots) to illustrate maternal characteristics and delivery patterns; Evaluated model performance using accuracy metrics

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Logistic regression model achieved 77% accuracy in predicting mode of delivery, demonstrating reliable predictive performance for clinical decision support | Identified parity, number of ANC visits, and pregnancy history as key predictors: mothers with higher parity more likely to have normal spontaneous vaginal delivery (SVD), while first-time mothers with limited ANC attendance (<4 visits) showed higher likelihood of Caesarean Section | Established that regular prenatal monitoring (4+ ANC visits) strongly associated with safe SVD outcomes, reinforcing importance of antenatal care adherence | Enhanced practical skills in data cleaning, statistical analysis, machine learning modeling, and health data interpretation | Demonstrated feasibility of using routine maternal health data for predictive analytics in hospital settings | Provided evidence-based recommendations for integrating predictive tools into maternal health workflows to support early risk stratification and targeted clinical interventions

CURRENT STATUS

Social work Internship · Internship

Sector relevance: Partially related

SKILLS IN USE

Data collection · Generative AI / LLMs · Ethics & bias in AI · Communication & teamwork

Confidence applying ENGAGE skills: **4 / 5**



CAPSTONE Capstone Project

Personalized Drug Recommendation System Based on Patient Records

BACKGROUND & METHODOLOGY

This project explores the application of Artificial Intelligence (AI) and Machine Learning (ML) in public health through the development of a Personalized Drug Recommendation System. The aim was to create an intelligent model that recommends the most appropriate drug for a patient based on their medical profile, specifically utilizing attributes such as Age, Sex, Blood Pressure (BP), Cholesterol, and Sodium-to-Potassium ratio (Na_to_K). The study employed the Drug200 dataset and trained a Random Forest Classifier within a pipeline architecture. The model achieved an overall accuracy of 97.5%, with Top-2 and Top-3 accuracies of 100%. Feature importance analysis revealed that Na_to_K and Age were the most influential predictors. An interactive prototype was deployed using Gradio, demonstrating that ML can provide highly accurate, data-driven support for drug prescription.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The study successfully developed a robust predictive framework where the Random Forest model emerged as the superior performer, achieving a high accuracy of 97.5%. In comparison, other models such as SVM (94%), Decision Tree (91%), and KNN (89%) provided strong but lower baseline performances. The analysis identified the Sodium-to-Potassium ratio (Na_to_K) as the most critical determinant in the recommendation process, followed by Age and Blood Pressure. To ensure practical clinical utility, the system was implemented as an interactive Gradio prototype, allowing for real-time drug suggestions based on patient data. Future work will focus on integrating more complex variables like genetics and comorbidities, exploring deep learning architectures, and expanding deployment across multiple healthcare centers to assess generalizability.

INTERNSHIP Internship Experience

St. Paul Methodist Health Centre — Mother-to-Child Transmission (MTCT) Risk Prediction Using Machine Learning for PMTCT Program Enhancement

★ LED TO A JOB OFFER

Dataset Used: Retrospective ART program electronic health records, PMTCT registers, laboratory records, and pharmacy records at St. Paul Methodist Health Centre (September 2023 — August 2024, 818 patient records). Dataset of 818 HIV-positive pregnant women and mothers enrolled in the ART/PMTCT program, containing demographic variables (age, sex), clinical variables (viral load, CD4 count, ART start date, current regimen, BMI, blood pressure), PMTCT program variables (enrollment status, pregnancy status, ANC visits, infant feeding practices, TPT outcome), and an engineered binary target variable (MTCT risk: High vs. Low) derived from viral load levels, ART duration, and PMTCT participation status.

METHODOLOGY

Supported PMTCT program activities at the health centre including patient registration and data entry into clinical registers; extracted and compiled retrospective data from ART registers, laboratory records, and pharmacy records; performed extensive data cleaning including handling missing values (62.3% missing ART duration), outlier detection, and data standardization; engineered features including ART duration in months, viral load suppression status, and risk categorization; conducted exploratory data analysis with correlation heatmaps and distribution visualizations; applied SMOTE to address significant class imbalance (96% low risk vs. 3.2% high risk); trained, tuned via GridSearchCV, and evaluated Logistic Regression and Random Forest classifiers using 5-fold cross-validation; and deployed the best-performing model as a Streamlit web application with real-time risk assessment and clinical decision support.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a Random Forest Classifier to predict MTCT risk of HIV among pregnant women enrolled in the ART program, achieving perfect performance with 100% accuracy, precision, recall, F1-score, and AUC-ROC of 1.00 on the test dataset, confirmed across 5-fold cross-validation (mean accuracy 99.6%); the model significantly outperformed the Logistic Regression baseline (98% accuracy, limited high-risk recall of 0.80); feature importance analysis identified last viral load (0.45), ART duration in months (0.28), and viral load suppression status (0.15) as the top three predictors, with elevated viral load (>1000 copies/mL) associated with 5.2 times higher MTCT risk; SMOTE improved overall F1-score from 0.81 to 1.00; the model was deployed as a live Streamlit web application enabling frontline healthcare workers to input patient clinical data and receive real-time MTCT risk classifications with probability scores and evidence-based recommendations, with recommendations to validate externally across multiple facilities, integrate into national PMTCT guidelines, and expand features with socioeconomic and behavioral variables.

CURRENT STATUS

Data scientist,data analyst · Unemployed — Actively searching

Sector relevance: Yes — Fully related

SKILLS IN USE

Python · Data collection · Data visualization · ML modeling · Model evaluation · Natural Language Processing (NLP) · Deep Learning · Ethics & bias in AI · Freelancing in data science / building a portfolio · Public health data analysis · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Predicting Cholera Outcomes in Kenya Through Environmental and Socio-Economic Factors with Machine Learning

BACKGROUND & METHODOLOGY

Cholera remains a major public health challenge in Kenya, where fragile water, sanitation, and hygiene (WASH) systems contribute to recurrent outbreaks. This study applied machine learning (ML) techniques to predict cholera outcomes using Kenyan surveillance data, including demographic details, water source, sanitation status, income level, and settlement type. The research developed two classification models—Logistic Regression and Random Forest—to provide a predictive, data-driven approach to outbreak management and resource allocation. By identifying key socio-economic and environmental determinants, the study aims to create decision-support tools that enhance epidemic preparedness and reduce the burden of the disease.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The study successfully developed a high-performing predictive framework, with the Random Forest model emerging as the superior performer, achieving an exceptional accuracy of 97.7%. In comparison, the Logistic Regression model provided a baseline accuracy of 72%. The analysis identified sanitation, water source, settlement type, sex, and age as the primary predictors of patient survival outcomes. To facilitate practical use, the Random Forest model was deployed via an interactive Streamlit web application, allowing healthcare workers to receive real-time risk predictions. Future work will focus on validating the model with larger datasets, integrating clinical and behavioral variables, and utilizing Explainable AI (XAI) to improve clinical transparency.

INTERNSHIP Internship Experience

JOOUST Bondo Medical Clinic — Application of Machine Learning in Predicting Malaria Infection Based on Symptoms and Vital Signs | AI for Clinical Decision Support in Malaria Diagnosis | Data-Driven Diagnostic Innovation

Dataset Used: Patient health records from JOOUST Bondo Medical Clinic, Siaya County (345-718 validated records containing demographic variables, vital signs including temperature/blood pressure/pulse rate, and clinical symptoms such as fever/headache/joint pain alongside malaria diagnosis outcomes). Project applying supervised machine learning techniques to predict malaria infection status using routinely collected clinical data in a resource-limited rural healthcare setting. Dataset includes demographic factors (age, sex), physiological measurements (temperature, blood pressure, pulse rate, weight), and patient-reported symptoms (fever, headache, joint pain) used to train Logistic Regression and Random Forest classifiers for binary classification of malaria-positive versus malaria-negative cases, supporting early diagnostic decision-making where rapid diagnostic tests may be unavailable or stock-limited

METHODOLOGY

Internship Centre Work: Data entry and patient registration; Organizing patient records in Excel; Supporting billing through SHA system; Preparing monthly Tuberculosis (TB) reports for facility reporting | Capstone Project: Data preprocessing including handling missing values, encoding categorical symptom variables into binary format (1=present, 0=absent), and normalizing continuous vital sign variables | Applied SMOTE (Synthetic Minority Oversampling Technique) to address class imbalance in malaria-positive cases | Trained and evaluated Logistic Regression and Random Forest classifiers using 80/20 stratified train-test split | Model evaluation using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix | Generated visualizations including correlation heatmaps, stacked bar plots of malaria diagnosis by temperature category, and scatter plots of diagnosis distribution by symptoms | Developed and deployed Streamlit web application for real-time clinical risk prediction accessible to healthcare workers

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Random Forest classifier achieved superior performance: Accuracy = 97.7%, Precision = 0.95-1.00, Recall = 0.96-1.00, F1-score = 0.98, with zero false negatives critical for medical screening applications | Logistic Regression also performed strongly: Accuracy = 96.9%, Precision = 0.94-0.99, Recall = 0.95-0.99, F1-score = 0.97 | Identified clinical symptoms (fever, headache, joint pain) as strongest predictors of malaria infection with correlation coefficients $r = 0.93-0.99$, significantly outperforming demographic and physiological variables | Temperature showed moderate correlation with diagnosis ($r = 0.27$), confirming fever as common but non-specific indicator | Successfully deployed Streamlit web application demonstrating feasibility of integrating predictive analytics into routine clinical workflows in rural Kenyan health facilities | Enhanced practical skills in health data management, machine learning model development, class imbalance handling, and web application deployment for public health applications | Provided actionable recommendations for implementing AI-assisted diagnostic support tools in resource-constrained settings to reduce misdiagnosis, optimize treatment allocation, and improve patient outcomes in malaria-endemic regions

CURRENT STATUS

I do online projects but they don't pay. Just to gain more experience · Unemployed — Actively searching

Sector relevance: Yes — Fully related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · Data visualization · Model evaluation · MLOps / Model deployment · Natural Language Processing (NLP)

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Predicting Cholera Outcomes in Kenya Through Environmental and Socio-Economic Factors with Machine Learning

BACKGROUND & METHODOLOGY

Cholera remains a major public health challenge in Kenya, where fragile water, sanitation, and hygiene (WASH) systems contribute to recurrent outbreaks. This study applied machine learning (ML) techniques to predict cholera outcomes using Kenyan surveillance data, including demographic details, water source, sanitation status, income level, and settlement type. The research developed two classification models—Logistic Regression and Random Forest—to provide a predictive, data-driven approach to outbreak management and resource allocation. By identifying key socio-economic and environmental determinants, the study aims to create decision-support tools that enhance epidemic preparedness and reduce the burden of the disease.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The study successfully developed a high-performing predictive framework, with the Random Forest model emerging as the superior performer, achieving an exceptional accuracy of 97.7%. In comparison, the Logistic Regression model provided a baseline accuracy of 72%. The analysis identified sanitation, water source, settlement type, sex, and age as the primary predictors of patient survival outcomes. To facilitate practical use, the Random Forest model was deployed via an interactive Streamlit web application, allowing healthcare workers to receive real-time risk predictions. Future work will focus on validating the model with larger datasets, integrating clinical and behavioral variables, and utilizing Explainable AI (XAI) to improve clinical transparency.

INTERNSHIP Internship Experience

St. Paul Methodist Health Centre, Ugunja Town, Siaya County — Predicting Missed ART Appointments and Assessing TB/NCD Risks Using AI Models at St. Paul Methodist Health Centre

Dataset Used: Active on ART Patients Linelist (August 2025) from facility electronic medical records containing demographic, clinical, and treatment-related variables for patients on antiretroviral therapy. Project applying predictive analytics to identify patients at risk of missing ART appointments and assess individual risk profiles for tuberculosis (TB) and non-communicable diseases (NCDs) among HIV-positive patients. Dataset included variables such as patient ID, age, sex, visit dates, ART regimen, comorbidities, TB screening status, and NCD indicators (blood pressure, glucose levels), used to train machine learning models for real-time clinical risk prediction and proactive patient management in resource-limited settings

METHODOLOGY

Internship Centre Work: Updated ART registers and validated patient records for completeness, consistency, and accuracy | Supported preparation of monthly HIV and TB reports submitted to sub-county health office | Participated in MCH data reviews and assisted in updating immunization and antenatal care records | Conducted spot checks on ART files to identify missing data and support corrective actions | Capstone Project Work: Data cleaning and preprocessing using Python in Google Colab (handling missing values, recoding categorical variables, feature engineering, normalizing continuous variables) | Trained and evaluated three supervised learning algorithms (Logistic Regression, Random Forest, Gradient Boosting) for binary classification of missed appointments and TB/NCD risk stratification | Model evaluation using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix | Cross-validation and hyperparameter tuning using GridSearchCV | Deployed final models as interactive Streamlit web applications via GitHub integration for real-time clinical decision support

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Validated over 1,200 patient records, improving data completeness and consistency across paper-based and electronic systems | Improved monthly reporting accuracy by approximately 15% through reconciliation of ART, TB, and MCH indicators | Random Forest model achieved 87% accuracy, 84% precision, 89% recall, 86% F1-score, and 0.91 ROC-AUC for predicting missed ART appointments | Gradient Boosting models for TB and NCD risk assessment achieved ROC-AUC scores of 0.88 and 0.90 respectively | Key predictors identified: age (18-35 years at higher risk), visit frequency, comorbidities, BMI, blood pressure, ART adherence | Low BMI and poor ART adherence identified as strong predictors of TB risk; elevated blood pressure and smoking history as key indicators for NCD vulnerability | Successfully deployed two Streamlit web applications (ART Appointment Risk Predictor and TB/NCD Risk Stratification Tool) accessible via secure links for real-time risk prediction by clinicians and data officers | Facility staff reported tools were intuitive and helpful for identifying patients needing additional support | Demonstrated feasibility of integrating AI-driven predictive analytics into routine HIV program workflows in rural Kenyan health facilities

CURRENT STATUS

Public Health Officer · Unemployed — Actively searching

Sector relevance: Yes – Fully related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · Data visualization · ML modeling · Model evaluation · MLOps / Model deployment · Public health data analysis · Communication & teamwork

Confidence applying ENGAGE skills: 5 / 5



CAPSTONE Capstone Project

Personalized Drug Recommendation System Based on Patient Records

BACKGROUND & METHODOLOGY

This project explores the application of Artificial Intelligence (AI) and Machine Learning (ML) in public health through the development of a Personalized Drug Recommendation System. The aim was to create an intelligent model that recommends the most appropriate drug for a patient based on their medical profile, specifically utilizing attributes such as Age, Sex, Blood Pressure (BP), Cholesterol, and Sodium-to-Potassium ratio (Na_to_K). The study employed the Drug200 dataset and trained a Random Forest Classifier within a pipeline architecture. The model achieved an overall accuracy of 97.5%, with Top-2 and Top-3 accuracies of 100%. Feature importance analysis revealed that Na_to_K and Age were the most influential predictors. An interactive prototype was deployed using Gradio, demonstrating that ML can provide highly accurate, data-driven support for drug prescription.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

The study successfully developed a robust predictive framework where the Random Forest model emerged as the superior performer, achieving a high accuracy of 97.5%. In comparison, other models such as SVM (94%), Decision Tree (91%), and KNN (89%) provided strong but lower baseline performances. The analysis identified the Sodium-to-Potassium ratio (Na_to_K) as the most critical determinant in the recommendation process, followed by Age and Blood Pressure. To ensure practical clinical utility, the system was implemented as an interactive Gradio prototype, allowing for real-time drug suggestions based on patient data. Future work will focus on integrating more complex variables like genetics and comorbidities, exploring deep learning architectures, and expanding deployment across multiple healthcare centers to assess generalizability.

INTERNSHIP Internship Experience

Kisumu County Referral Hospital — An AI-Powered Platform for Integrated Referral, Ambulance Management, and Clinical Decision Support

★ LED TO A JOB OFFER

Dataset Used: Synthetic patient records (1,847), county health facilities database (34 facilities), simulated ambulance fleet (20 vehicles), referral logs, clinical encounter data, laboratory results, and operational cost data (fuel prices, distance metrics). Enterprise-grade integrated health system designed to transform clinical workflows and emergency medical services across Kisumu County, Kenya. The platform addresses diagnostic delays, fragmented patient data, inefficient resource allocation, and systemic failures in patient transfer logistics through two synergistic cores: (1) Clinical Hub featuring AI-powered diagnostic assistant, HL7 FHIR-compliant EHR, predictive analytics for 30-day readmission risk, and automated revenue cycle management; (2) Emergency Logistics & Referral System with real-time ambulance tracking via PyDeck, automated notification engine, and cost-optimization service using geolocation and fuel price data for trip expense modeling.

METHODOLOGY

Internship Centre Work: Data entry and validation in DHIS2 system, cross-checking patient records for accuracy, correcting inconsistencies and missing values | Capstone Project Work: System architecture design using modular client-server model with Streamlit frontend and SQLite backend | Database development using SQLAlchemy and custom EHRDatabase class for clinical/logistical operations | Implementation of Role-Based Access Control (RBAC) with HealthcareSecurity class for authentication and audit logging | Curation of structured disease knowledge base for 25+ locally relevant conditions | Development of synthetic data generation scripts for realistic patient records and referral requests | AI diagnostic assistant development using deterministic rule-based matching algorithm for symptom-to-diagnosis confidence scoring | Predictive model for 30-day readmission risk using heuristic clinical feature analysis | Spatial analysis and intelligent dispatch algorithm for nearest-ambulance assignment using Haversine formula | Cost-calculation service modeling trip expenses using distance and fuel consumption data | Interactive dashboard development using Plotly and PyDeck for clinical KPIs, public health trends, real-time ambulance locations, and cost-saving projections | Full-stack development using Python, Streamlit, SQLAlchemy, Pandas, NumPy, and Transformers library

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully engineered and deployed unified web application consolidating 8 fragmented workflows (Patient Registration, Clinical Review, Lab Testing, Pharmacy, Billing, Referral, Ambulance Dispatch, Analytics) into single integrated platform | AI diagnostic assistant achieved high clinical accuracy: Malaria (85% confidence), Heart Disease (92% confidence), Tuberculosis (78% confidence) | Intelligent dispatch algorithm reduced average estimated response distance by 32% compared to non-optimized methods | Predictive analytics stratified patient readmission risk (High: 12%, Medium: 28%, Low: 60%) with key risk factors identified: critical lab values at discharge (OR: 4.7), multiple comorbidities (OR: 3.5) | Proactive interventions for high-risk cohort projected to achieve KES 2.8 million cost avoidance per quarter | Cost analysis revealed fuel constitutes 58% of ambulance trip expenses (avg. KES 2,500/trip) with 15-20% potential savings through optimal fleet management | Real-time revenue cycle analytics provided departmental performance insights using localized Kenyan Shilling pricing | Platform operationalized with comprehensive database managing synthetic data for demonstration and testing without protected health information | Demonstrated scalability model for strengthening health systems through improved diagnostic accuracy, proactive care via predictive risk stratification, and optimized emergency medical service efficiency | Enhanced technical proficiency in full-stack Python/Streamlit development, relational database design, AI logic integration, and public health informatics | Reinforced importance of systems thinking, user-centric design, and continuous clinical/operational validation in healthcare technology development

CURRENT STATUS

Finance Intern · Internship

Sector relevance: Partially related

SKILLS IN USE

Data collection · Communication & teamwork

Confidence applying ENGAGE skills: **4 / 5**

**CAPSTONE** Capstone Project*Patient-Centered PCOS Risk Prediction Using a Logistic Regression Model Based on Clinical Features***BACKGROUND & METHODOLOGY**

Polycystic Ovary Syndrome (PCOS) represents the most common endocrine disorder among women of reproductive age, with a global prevalence estimated between 6% and 13%. This burden is particularly evident in resource-constrained settings like Kenya, where the condition is significantly underdiagnosed, with up to 70% of cases remaining unidentified, leading to severe and often preventable long-term complications including infertility, type 2 diabetes mellitus, cardiovascular morbidity, and considerable psychological distress. Traditional diagnostic approaches are often subjective, inconsistent, and heavily reliant on clinician experience, creating a critical need for data-driven and patient-centered tools that support early prediction, empower women to track their symptoms, and improve health outcomes. This study aims to develop, evaluate, and deploy a logistic regression model designed to predict an individual's risk of developing PCOS based on a patient's clinical features. The methodological approach entailed a secondary analysis of the publicly available PCOS Extended Dataset extracted from Kaggle, comprising data from 2000 women of reproductive age (18-49 years) as the target population. Using Python programming, we cleaned the data, handled missing values, and prepared it for analysis. Logistic regression was used to identify the relationships between clinical features and PCOS outcomes. The final model was operationalized into an accessible, patient-friendly web application using Streamlit.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a patient-centered PCOS risk prediction tool using logistic regression on 2,000 women from the PCOS Extended Dataset. The model achieved 85% overall accuracy with high precision and recall for non-PCOS cases (0.89-0.90) and good performance for PCOS cases (0.73-0.77). Odds ratio analysis identified key risk predictors: hair growth (OR 1.84), skin darkening (OR 1.74), irregular menstrual cycles (OR 1.70), and weight gain (OR 1.66) significantly increased PCOS risk, while older age (OR 0.62) and longer periods (OR 0.81) showed protective effects. A tiered risk stratification system was implemented based on probability thresholds: low risk (<0.2), moderate risk (0.2-0.38), and high risk (>0.38), enabling personalized clinical pathways. The model was deployed via an interactive Streamlit web application allowing women to input symptoms and receive immediate, interpretable risk assessments with clear explanations. This approach addresses the 70% underdiagnosis rate in Kenya by empowering community health workers and patients with an affordable, accessible early screening tool that transforms subjective symptoms into objective, evidence-based risk scores, facilitating timely referral and proactive health management.

INTERNSHIP Internship Experience*Bahati Sub-County Hospital, Bahati Sub-County, Nakuru County — Child health; pneumonia early detection; under-five clinical risk prediction; Community Health Promoter screening tool*

Dataset Used: Secondary inpatient and outpatient hospital records from Bahati Sub-County Hospital / healthcare facility records. Dataset of 800 under-five patient records with 18 variables, including clinical signs and demographic attributes related to pneumonia such as cough, fever/hotness of body, fast respiratory rate, chest indrawing, age, and pneumonia outcome

METHODOLOGY

Extracted and compiled inpatient and outpatient records; cleaned and preprocessed data; handled duplicates and consistency issues; conducted exploratory data analysis using Python; visualized distributions and correlations using Matplotlib and Seaborn; trained and tuned Logistic Regression, Random Forest, and XGBoost models; compared models using accuracy, recall, precision, and F1-score; deployed the final model using Streamlit

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Developed a clinical feature-based pneumonia prediction model for children under five; identified cough, fever, fast respiratory rate, and chest indrawing as key predictors; found Random Forest and XGBoost to perform better than Logistic Regression, with Random Forest recommended for integration into a mobile/web tool; produced a user-friendly screening tool to help Community Health Promoters identify high-risk children early and refer them for timely care

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Predictive Modeling for Prediabetes Risk Using Machine Learning Approaches

BACKGROUND & METHODOLOGY

Prediabetes is a transitional stage between normal glucose regulation and type 2 diabetes mellitus, and its early detection is vital for preventing disease progression. This study develops a predictive model using machine learning algorithms to assess individual risk of prediabetes based on demographic and lifestyle features such as age, body mass index, diet, physical activity, alcohol consumption, family history of diabetes, and healthcare access. Data preprocessing, algorithm selection, and model development were carried out to ensure robust predictions, while performance was validated using accuracy, precision, recall, and ROC analysis. The results demonstrate that machine learning models can stratify individuals into risk categories, providing insights into modifiable risk factors. Clinical relevance lies in preventive recommendations such as weight management, lifestyle adjustments, and healthcare engagement. Limitations include data availability and generalizability, but future directions point toward integrating larger datasets and advanced deep learning methods. The study concludes that predictive modeling is a valuable tool for early detection and prevention of prediabetes, with strong potential for public health applications.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and validated machine learning models for prediabetes risk prediction using demographic and lifestyle features on a dataset of 500 records. Random Forest emerged as the best-performing model achieving 100% recall and perfect identification of all true positive prediabetes cases. Gradient Boosting achieved the highest overall predictive accuracy with ROC-AUC above 0.85, indicating strong discriminative power. Key predictors identified through feature importance analysis were BMI, age, and diet patterns as most significant, followed by family history of diabetes and alcohol consumption. The model was refined to seven optimal features (Age, BMI, Physical Activity, Diet Pattern, Family History, Healthcare Access) to improve clinical feasibility. A conceptual web application interface was designed for healthcare professionals to input patient data and receive individualized risk assessments with personalized preventive recommendations. The study demonstrates strong potential for integrating predictive modeling into community healthcare programs as a cost-effective, scalable decision-support tool for early intervention and lifestyle modification in resource-limited settings.

INTERNSHIP Internship Experience

Lanet Health Center — Maternal health; antenatal care completion risk prediction; ANC visit adherence

Dataset Used: ANC_Dataset_with_Risk.xlsx; antenatal care records from Lanet Health Center. ANC dataset containing maternal demographic, socio-economic, clinical, and service utilization variables, including age, parity, gestational age, previous ANC visits, education level, marital status, income source, distance to facility, previous pregnancy complications, phone ownership, and ANC completion risk category

METHODOLOGY

Cleaned and preprocessed ANC data; handled missing values; encoded categorical variables; scaled numerical variables; conducted exploratory data analysis; analyzed risk patterns by education, marital status, income, distance to facility, pregnancy complications, and phone ownership; trained and evaluated multiple ML models including Logistic Regression, Decision Tree, Naive Bayes, KNN, SVM, Random Forest, XGBoost, and LightGBM; developed a Streamlit deployment pipeline

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Developed an ANC completion risk prediction system to classify pregnant women as high or low risk of missing recommended ANC visits; identified key risk factors such as low income, long distance to facility, lack of phone ownership, lower education, and limited social support; selected strong-performing models, especially XGBoost/LightGBM, for deployment; created a Streamlit-based decision-support tool to help health workers identify mothers needing early follow-up and targeted intervention

CURRENT STATUS

No job yet, still searching · Unemployed — Actively searching

Sector relevance: Not related

SKILLS IN USE

Communication & teamwork

Confidence applying ENGAGE skills: **1 / 5**

**CAPSTONE** **Capstone Project***Coronary Heart Disease Prediction Model and Recommender System***BACKGROUND & METHODOLOGY**

Cardiovascular diseases (CVDs) are considered to be one of the major causes of mortality in the world, killing millions of people annually. The most important approaches to reducing this burden and improving the patient outcomes are preventive measures and early diagnoses. The project uses machine learning to identify the chances of cardiovascular disease through the examination of significant health variables related to age, blood pressure, cholesterol level, and other lifestyle considerations. Besides prediction, the system also has a custom health recommender system, which recommends specific lifestyle changes, including dietary changes, physical exercises, and check-ups with doctors and related medical services, depending on the risk profile of the individual. This two-fold strategy is able not only to pinpoint the people involved but also give some realistic and evidence-based guidelines on ways to prevent and deal with such individuals. Integrating predictive analytics and individualized advice, the project shows how artificial intelligence can assist healthcare providers, improve patient education, and encourage early treatment-eventually lessening hospitalization and saving lives.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a Coronary Heart Disease prediction system combining machine learning with personalized lifestyle recommendations. Logistic Regression was selected as the final model from five algorithms tested (Random Forest, Gradient Boosting, XGBoost, LightGBM) due to superior interpretability and strong performance metrics. Key predictors identified included age over 50, high cholesterol, and elevated systolic blood pressure as strongest risk factors. SMOTE was applied to address class imbalance, ensuring balanced learning across CHD-positive and CHD-negative cases. The system was deployed via an interactive Streamlit web application enabling real-time risk assessment and tailored recommendations including diet, exercise, and medical check-ups based on individual risk profiles. The model provides transparent coefficient-based explanations for clinical trust, with potential to democratize preventive healthcare access, particularly in underserved regions. Future enhancements include wearable device integration, expanded demographic diversity, and NLP-based conversational recommendations.

CURRENT STATUS

Not yet reported

**CAPSTONE** **Capstone Project***Coronary Heart Disease Prediction Model and Recommender System***BACKGROUND & METHODOLOGY**

Cardiovascular diseases (CVDs) are considered to be one of the major causes of mortality in the world, killing millions of people annually. The most important approaches to reducing this burden and improving the patient outcomes are preventive measures and early diagnoses. The project uses machine learning to identify the chances of cardiovascular disease through the examination of significant health variables related to age, blood pressure, cholesterol level, and other lifestyle considerations. Besides prediction, the system also has a custom health recommender system, which recommends specific lifestyle changes, including dietary changes, physical exercises, and check-ups with doctors and related medical services, depending on the risk profile of the individual. This two-fold strategy is able not only to pinpoint the people involved but also give some realistic and evidence-based guidelines on ways to prevent and deal with such individuals. Integrating predictive analytics and individualized advice, the project shows how artificial intelligence can assist healthcare providers, improve patient education, and encourage early treatment-eventually lessening hospitalization and saving lives.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a Coronary Heart Disease prediction system combining machine learning with personalized lifestyle recommendations. Logistic Regression was selected as the final model from five algorithms tested (Random Forest, Gradient Boosting, XGBoost, LightGBM) due to superior interpretability and strong performance metrics. Key predictors identified included age over 50, high cholesterol, and elevated systolic blood pressure as strongest risk factors. SMOTE was applied to address class imbalance, ensuring balanced learning across CHD-positive and CHD-negative cases. The system was deployed via an interactive Streamlit web application enabling real-time risk assessment and tailored recommendations including diet, exercise, and medical check-ups based on individual risk profiles. The model provides transparent coefficient-based explanations for clinical trust, with potential to democratize preventive healthcare access, particularly in underserved regions. Future enhancements include wearable device integration, expanded demographic diversity, and NLP-based conversational recommendations.

CURRENT STATUS**Mechanical engineering trainee · Internship**

Sector relevance: Partially related

SKILLS IN USE

Prompt engineering

Confidence applying ENGAGE skills: **5 / 5**

**CAPSTONE** **Capstone Project***Development of a Machine Learning-Powered Pre-Vitals Triage Tool for Enhanced Patient Prioritization in Resource-Constrained Healthcare Settings***BACKGROUND & METHODOLOGY**

Overcrowding in emergency departments and triage areas is a critical global health challenge, leading to prolonged waiting times, delayed interventions, and preventable mortality. This problem is exacerbated in resource-limited settings by a shortage of trained triage staff and the reliance on vital signs for decision-making, which is time consuming and could lead to preventable deaths. This project proposes the development and deployment of a Machine Learning pre-vitals triage tool to address this gap. Leveraging the World Health Organization's Emergency Triage, Assessment and Treatment (ETAT) guidelines, a synthetic dataset of over 1500 patient records was generated, incorporating demographics, observable signs, reported symptoms, and underlying conditions. A Random Forest Classifier machine learning model was trained on this data to predict patient triage category (Emergency/Red, Urgent/Yellow, Routine/Green) without the initial need for vital signs. The model achieved a high predictive accuracy of 97%. The model was operated through a user-friendly, web-based application built with Streamlit, allowing healthcare workers to input patient data via checkboxes and dropdown menus and receive an immediate, color-coded triage recommendation. This tool standardizes decision-making, empowers frontline health workers, and facilitates the rapid identification of critical cases, thereby potentially reducing waiting times, improving patient flow, and preventing adverse outcomes in overcrowded healthcare facilities. The study concludes that AI and Machine Learning driven pre-vitals triage is a viable and promising strategy for enhancing emergency care efficiency and equity.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a machine learning-powered pre-vitals triage tool achieving 97% accuracy using Random Forest Classifier on a synthetic dataset of 1500+ patient records based on WHO ETAT guidelines. The model predicts triage category (Red/Emergency, Yellow/Urgent, Green/Routine) using 40+ binary indicators for observable signs, reported symptoms, and underlying conditions without requiring vital signs. Key clinical patterns validated include strong correlations between obstructed airway, severe respiratory distress, and unconsciousness with Emergency classification. An interactive Streamlit web application was deployed featuring checkbox inputs, dropdown menus, color-coded output banners, confidence bar charts, and session distribution summaries for situational awareness. The tool addresses critical healthcare gaps by reducing triage delays from minutes to seconds, augmenting less experienced staff with standardized evidence-based recommendations, improving consistency regardless of fatigue or workload, and aligning with Universal Health Coverage goals for equitable emergency care access.

INTERNSHIP **Internship Experience***Equity Afia Section 58, Nakuru — Healthcare; patient follow-up default prediction; continuity of care*

Dataset Used: Equity Afia Section 58 internal facility records; Microsoft Excel spreadsheets maintained by the data clerk; supplementary patient registers from reception. Patient discharge and follow-up records from September to October 2025, including age, sex, occupation, diagnosis/condition type, number of medications, insurance status, reason for follow-up, follow-up gap, and follow-up/default risk indicators

METHODOLOGY

Conducted staff consultations with clinicians, receptionist, and data clerk; collected and cleaned patient follow-up data; performed exploratory data analysis; trained and compared Logistic Regression, Random Forest, and Gradient Boosting models; selected Gradient Boosting; built and deployed a Streamlit dashboard for follow-up risk prediction and monitoring

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Developed a machine-learning model that classifies patients as low, medium, or high risk of missing follow-up appointments; created a dashboard showing overdue/due patients, top diagnoses, follow-up schedules, and risk categories; helped clinic staff prioritize patient follow-up calls and shift from manual/reactive tracking to proactive patient monitoring

CURRENT STATUS**Volunteering as a PHO · Unemployed — Actively searching**

Sector relevance: Yes – Fully related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · Data visualization · ML modeling · Model evaluation · MLOps / Model deployment · Natural Language Processing (NLP) · Ethics & bias in AI · Public health data analysis · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Predictive Modeling for Prediabetes Risk Using Machine Learning Approaches

BACKGROUND & METHODOLOGY

Prediabetes is a transitional stage between normal glucose regulation and type 2 diabetes mellitus, and its early detection is vital for preventing disease progression. This study develops a predictive model using machine learning algorithms to assess individual risk of prediabetes based on demographic and lifestyle features such as age, body mass index, diet, physical activity, alcohol consumption, family history of diabetes, and healthcare access. Data preprocessing, algorithm selection, and model development were carried out to ensure robust predictions, while performance was validated using accuracy, precision, recall, and ROC analysis. The results demonstrate that machine learning models can stratify individuals into risk categories, providing insights into modifiable risk factors. Clinical relevance lies in preventive recommendations such as weight management, lifestyle adjustments, and healthcare engagement. Limitations include data availability and generalizability, but future directions point toward integrating larger datasets and advanced deep learning methods. The study concludes that predictive modeling is a valuable tool for early detection and prevention of prediabetes, with strong potential for public health applications.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and validated machine learning models for prediabetes risk prediction using demographic and lifestyle features on a dataset of 500 records. Random Forest emerged as the best-performing model achieving 100% recall and perfect identification of all true positive prediabetes cases. Gradient Boosting achieved the highest overall predictive accuracy with ROC-AUC above 0.85, indicating strong discriminative power. Key predictors identified through feature importance analysis were BMI, age, and diet patterns as most significant, followed by family history of diabetes and alcohol consumption. The model was refined to seven optimal features (Age, BMI, Physical Activity, Diet Pattern, Family History, Healthcare Access) to improve clinical feasibility. A conceptual web application interface was designed for healthcare professionals to input patient data and receive individualized risk assessments with personalized preventive recommendations. The study demonstrates strong potential for integrating predictive modeling into community healthcare programs as a cost-effective, scalable decision-support tool for early intervention and lifestyle modification in resource-limited settings.

INTERNSHIP Internship Experience

Lanet Health Center — Maternal health; antenatal care completion risk prediction; ANC visit adherence

★ LED TO A JOB OFFER

Dataset Used: ANC_Dataset_with_Risk.xlsx; antenatal care records from Lanet Health Center. ANC dataset containing maternal demographic, socio-economic, clinical, and service utilization variables, including age, parity, gestational age, previous ANC visits, education level, marital status, income source, distance to facility, previous pregnancy complications, phone ownership, and ANC completion risk category

METHODOLOGY

Cleaned and preprocessed ANC data; handled missing values; encoded categorical variables; scaled numerical variables; conducted exploratory data analysis; analyzed risk patterns by education, marital status, income, distance to facility, pregnancy complications, and phone ownership; trained and evaluated multiple ML models including Logistic Regression, Decision Tree, Naive Bayes, KNN, SVM, Random Forest, XGBoost, and LightGBM; developed a Streamlit deployment pipeline

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Developed an ANC completion risk prediction system to classify pregnant women as high or low risk of missing recommended ANC visits; identified key risk factors such as low income, long distance to facility, lack of phone ownership, lower education, and limited social support; selected strong-performing models, especially XGBoost/LightGBM, for deployment; created a Streamlit-based decision-support tool to help health workers identify mothers needing early follow-up and targeted intervention

CURRENT STATUS

Data Analyst/Data Scientist · Self-employed / Freelance

Sector relevance: Yes – Fully related

SKILLS IN USE

Python · Data cleaning & wrangling · Data visualization · Prompt engineering · Ethics & bias in AI · Freelancing in data science / building a portfolio

Confidence applying ENGAGE skills: **3 / 5**



CAPSTONE Capstone Project

Development of a Machine Learning Screening Model for Endometriosis in Women of Reproductive Age

BACKGROUND & METHODOLOGY

Endometriosis is a chronic gynecological disorder affecting approximately 190 million women worldwide. Despite its significant health and socioeconomic burden, there is a substantial delay in diagnosis, particularly in low- and middle-income countries (LMICs) like Kenya. This delay is attributed to the normalization of symptoms, social stigma, and the invasive and costly nature of the current diagnostic gold standard, laparoscopy. This report outlines the development of a non-invasive machine learning-based screening model designed to estimate the likelihood of endometriosis. By utilizing demographic, reproductive, and clinical data, the model aims to serve as a predictive tool to enable earlier detection and referral. The study identified key predictors, constructed and trained machine learning models (Logistic Regression and Naïve Bayes), and evaluated their performance. The models achieved an impressive accuracy of 86%, demonstrating their potential as a cost-effective and accessible screening alternative for resource-limited settings.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a non-invasive machine learning screening model for endometriosis achieving 86% accuracy using Logistic Regression and Naïve Bayes algorithms. Key predictors identified included dysmenorrhea score, pelvic pain score, dyspareunia score, dyschezia score, duration of periods, age of menarche, family history, absenteeism days per month, and menstrual flow characteristics. Logistic Regression achieved 86% accuracy with precision 0.85-0.87, recall 0.84-0.88, and F1-Score 0.86 across both classes. Naïve Bayes achieved 86.67% accuracy with AUC-ROC of 0.9431, precision 0.85-0.89, recall 0.84-0.89, and F1-Score 0.86-0.87. The model addresses the 7-10 year diagnostic delay in LMICs by providing an accessible alternative to invasive laparoscopy, enabling healthcare workers to identify high-risk women using readily available clinical data. A mobile application interface was designed for deployment in resource-constrained settings, facilitating early referral, reducing patient suffering, and aligning with Sustainable Development Goal 3 for women's health and well-being.

INTERNSHIP Internship Experience

Bondeni Hospital / Bondeni Sub-County Hospital, Nakuru — Maternal health; anemia risk screening in pregnant women; antenatal care risk prediction

Dataset Used: Antenatal care records from Bondeni Hospital covering 2024–2025. ANC dataset containing health records for pregnant women, including hemoglobin levels and anemia-related predictors such as parity, gravidity, HIV status, blood pressure, malaria infection, BMI, previous history, and hemoglobin concentration. The model classified women into low-risk and high-risk anemia categories based mainly on hemoglobin thresholds

METHODOLOGY

Cleaned and preprocessed ANC data; explored anemia patterns using visualizations; trained and compared machine learning models including Logistic Regression, Random Forest, and XGBoost; classified anemia risk into low-risk and high-risk groups; deployed the trained model for clinical/community screening use

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Developed a machine-learning screening model for early anemia risk detection among pregnant women; identified a high proportion of women at risk of anemia; Random Forest and XGBoost were reported as the best-performing models; created an AI-powered screening tool to support healthcare providers and CHPs in identifying high-risk mothers early and enabling timely intervention

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Machine Learning-Based Community Screening Model for Predicting Malnutrition Risk in Children Under Five

BACKGROUND & METHODOLOGY

Malnutrition is a leading cause of morbidity and mortality among children under five, contributing to over 3 million deaths annually worldwide. In Kenya, stunting, wasting, and underweight remain prevalent, posing a significant public health burden. This project developed a machine learning--based community screening model to predict malnutrition risk early, enabling intervention before severe cases occur. Using synthetic data (3,000 records, 10 variables), Logistic Regression, Random Forest, and XGBoost models were trained and evaluated. XGBoost achieved the best performance with an AUC score of 0.91. The model was deployed with Python and Streamlit, demonstrating its potential for community-based health screening.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a machine learning-based community screening model for predicting malnutrition risk in children under five using synthetic data of 3,000 records with 10 demographic and socio-economic variables. XGBoost emerged as the best-performing model achieving an AUC score of 0.9113, MSE of 0.0889, and R² of 0.6437, outperforming Logistic Regression and Random Forest. Key predictors identified included parental education, income level, access to clean water, age, and family size. The model was deployed via an interactive offline Streamlit interface for community health practitioners, enabling household-level risk assessment without requiring advanced clinical measurements. The study demonstrates a scalable, data-driven approach to early malnutrition detection, supporting timely intervention, reducing child mortality, and aligning with Kenya's commitment to SDG 2 (Zero Hunger) and SDG 3 (Good Health and Well-being).

INTERNSHIP Internship Experience

Bahati Sub-County Hospital, Bahati Sub-County, Nakuru County — Child health; pneumonia early detection; under-five clinical risk prediction; Community Health Promoter screening tool

Dataset Used: Secondary inpatient and outpatient hospital records from Bahati Sub-County Hospital / healthcare facility records. Dataset of 800 under-five patient records with 18 variables, including clinical signs and demographic attributes related to pneumonia such as cough, fever/hotness of body, fast respiratory rate, chest indrawing, age, and pneumonia outcome

METHODOLOGY

Extracted and compiled inpatient and outpatient records; cleaned and preprocessed data; handled duplicates and consistency issues; conducted exploratory data analysis using Python; visualized distributions and correlations using Matplotlib and Seaborn; trained and tuned Logistic Regression, Random Forest, and XGBoost models; compared models using accuracy, recall, precision, and F1-score; deployed the final model using Streamlit

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Developed a clinical feature-based pneumonia prediction model for children under five; identified cough, fever, fast respiratory rate, and chest indrawing as key predictors; found Random Forest and XGBoost to perform better than Logistic Regression, with Random Forest recommended for integration into a mobile/web tool; produced a user-friendly screening tool to help Community Health Promoters identify high-risk children early and refer them for timely care

CURRENT STATUS

Volunteering at Deaf Empowerment Kenya · Unemployed — Actively searching

Sector relevance: Not related

SKILLS IN USE

Ethics & bias in AI · Public health data analysis · Communication & teamwork

Confidence applying ENGAGE skills: **4 / 5**



CAPSTONE Capstone Project

Preventing Missed Vaccinations: Predicting the Risk of Vaccination Defaulting in Nakuru Using Machine Learning

BACKGROUND & METHODOLOGY

Childhood immunization is a significant health promotion campaign, but defaulting is one of the most significant concerns, particularly in low-resource contexts. This is an initiative to develop a predictive model to recognize at-risk children when it comes to defaulting on their vaccines in Nakuru County, Kenya. We performed a large-scale exploratory data analysis (EDA) in our study to investigate the critical socioeconomic, geographic, health system, knowledge-based, and social and cultural variables to understand how they can impact the vaccination adherence. Several machine learning models were trained and tested, they include Logistic Regression, Random Forest and XGBoost. The logistic regression model was observed to give a more optimal performance with 65 percent accuracy and AUC of 0.68 of good predictive performance with stronger predictive capacity to identify non-defaulters. The most important predictive factors included traditional barriers (+1.24), religious barriers (+1.17) and the lack of an immunization card. There was creation of a risk scoring system to classify children as Low, Medium and High-risk. The model was deployed as an interactive web-based application in Streamlit, according to which healthcare workers and CHVs were able to input patient data and receive real-time risk forecasts. The predicted risk of defaulting can be visualized to visualize areas that are highly affected therefore enhance target interventions. The tool can facilitate the shift of reactive methods to public health toward targeted interventions and, by extension, improve the coverage of vaccination and the incidence of vaccine-preventable diseases. The study points out how effective data-based solutions are to address complicated healthcare problems.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a predictive system for vaccination defaulting risk in Nakuru County using machine learning on a synthetic dataset of 2,000 records with 19 features. Logistic Regression was selected as the final model achieving 64% accuracy and 0.68 ROC-AUC, prioritizing interpretability over ensemble methods for healthcare transparency. Key risk predictors identified were traditional barriers (+1.24 coefficient), religious barriers (+1.17), and lack of immunization card (-0.83 protective factor). A rule-based risk scoring system was created classifying children into Low, Medium, and High-risk categories with high-risk group showing 45% default rate. An interactive Streamlit web application was deployed with three tabs: Risk Prediction (real-time default probability), Risk Map (Folium-based geographical visualization with color-coded priority areas), and Interventions (targeted evidence-based strategies). Geographical analysis identified Rongai, Molo, and Naivasha as priority sub-counties requiring targeted outreach. The system enables Community Health Volunteers to shift from reactive defaulter tracing to proactive risk-based interventions, improving vaccination coverage and reducing vaccine-preventable disease burden.

INTERNSHIP Internship Experience

Lanet Health Center — Maternal health; antenatal care completion risk prediction; ANC visit adherence

Dataset Used: ANC_Dataset_with_Risk.xlsx; antenatal care records from Lanet Health Center. ANC dataset containing maternal demographic, socio-economic, clinical, and service utilization variables, including age, parity, gestational age, previous ANC visits, education level, marital status, income source, distance to facility, previous pregnancy complications, phone ownership, and ANC completion risk category

METHODOLOGY

Cleaned and preprocessed ANC data; handled missing values; encoded categorical variables; scaled numerical variables; conducted exploratory data analysis; analyzed risk patterns by education, marital status, income, distance to facility, pregnancy complications, and phone ownership; trained and evaluated multiple ML models including Logistic Regression, Decision Tree, Naive Bayes, KNN, SVM, Random Forest, XGBoost, and LightGBM; developed a Streamlit deployment pipeline

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Developed an ANC completion risk prediction system to classify pregnant women as high or low risk of missing recommended ANC visits; identified key risk factors such as low income, long distance to facility, lack of phone ownership, lower education, and limited social support; selected strong-performing models, especially XGBoost/LightGBM, for deployment; created a Streamlit-based decision-support tool to help health workers identify mothers needing early follow-up and targeted intervention

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Preventing Missed Vaccinations: Predicting the Risk of Vaccination Defaulting in Nakuru Using Machine Learning

BACKGROUND & METHODOLOGY

Childhood immunization is a significant health promotion campaign, but defaulting is one of the most significant concerns, particularly in low-resource contexts. This is an initiative to develop a predictive model to recognize at-risk children when it comes to defaulting on their vaccines in Nakuru County, Kenya. We performed a large-scale exploratory data analysis (EDA) in our study to investigate the critical socioeconomic, geographic, health system, knowledge-based, and social and cultural variables to understand how they can impact the vaccination adherence. Several machine learning models were trained and tested, they include Logistic Regression, Random Forest and XGBoost. The logistic regression model was observed to give a more optimal performance with 65 percent accuracy and AUC of 0.68 of good predictive performance with stronger predictive capacity to identify non-defaulters. The most important predictive factors included traditional barriers (+1.24), religious barriers (+1.17) and the lack of an immunization card. There was creation of a risk scoring system to classify children as Low, Medium and High-risk. The model was deployed as an interactive web-based application in Streamlit, according to which healthcare workers and CHVs were able to input patient data and receive real-time risk forecasts. The predicted risk of defaulting can be visualized to visualize areas that are highly affected therefore enhance target interventions. The tool can facilitate the shift of reactive methods to public health toward targeted interventions and, by extension, improve the coverage of vaccination and the incidence of vaccine-preventable diseases. The study points out how effective data-based solutions are to address complicated healthcare problems.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a predictive system for vaccination defaulting risk in Nakuru County using machine learning on a synthetic dataset of 2,000 records with 19 features. Logistic Regression was selected as the final model achieving 64% accuracy and 0.68 ROC-AUC, prioritizing interpretability over ensemble methods for healthcare transparency. Key risk predictors identified were traditional barriers (+1.24 coefficient), religious barriers (+1.17), and lack of immunization card (-0.83 protective factor). A rule-based risk scoring system was created classifying children into Low, Medium, and High-risk categories with high-risk group showing 45% default rate. An interactive Streamlit web application was deployed with three tabs: Risk Prediction (real-time default probability), Risk Map (Folium-based geographical visualization with color-coded priority areas), and Interventions (targeted evidence-based strategies). Geographical analysis identified Rongai, Molo, and Naivasha as priority sub-counties requiring targeted outreach. The system enables Community Health Volunteers to shift from reactive defaulter tracing to proactive risk-based interventions, improving vaccination coverage and reducing vaccine-preventable disease burden.

INTERNSHIP Internship Experience

Lanet Health Center — Maternal health; antenatal care completion risk prediction; ANC visit adherence

Dataset Used: ANC_Dataset_with_Risk.xlsx; antenatal care records from Lanet Health Center. ANC dataset containing maternal demographic, socio-economic, clinical, and service utilization variables, including age, parity, gestational age, previous ANC visits, education level, marital status, income source, distance to facility, previous pregnancy complications, phone ownership, and ANC completion risk category

METHODOLOGY

Cleaned and preprocessed ANC data; handled missing values; encoded categorical variables; scaled numerical variables; conducted exploratory data analysis; analyzed risk patterns by education, marital status, income, distance to facility, pregnancy complications, and phone ownership; trained and evaluated multiple ML models including Logistic Regression, Decision Tree, Naive Bayes, KNN, SVM, Random Forest, XGBoost, and LightGBM; developed a Streamlit deployment pipeline

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Developed an ANC completion risk prediction system to classify pregnant women as high or low risk of missing recommended ANC visits; identified key risk factors such as low income, long distance to facility, lack of phone ownership, lower education, and limited social support; selected strong-performing models, especially XGBoost/LightGBM, for deployment; created a Streamlit-based decision-support tool to help health workers identify mothers needing early follow-up and targeted intervention

CURRENT STATUS

I have no job yet · Unemployed — Actively searching

Sector relevance: Partially related

SKILLS IN USE

Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Development of a Machine Learning-Powered Pre-Vitals Triage Tool for Enhanced Patient Prioritization in Resource-Constrained Healthcare Settings

BACKGROUND & METHODOLOGY

Overcrowding in emergency departments and triage areas is a critical global health challenge, leading to prolonged waiting times, delayed interventions, and preventable mortality. This problem is exacerbated in resource-limited settings by a shortage of trained triage staff and the reliance on vital signs for decision-making, which is time consuming and could lead to preventable deaths. This project proposes the development and deployment of a Machine Learning pre-vitals triage tool to address this gap. Leveraging the World Health Organization's Emergency Triage, Assessment and Treatment (ETAT) guidelines, a synthetic dataset of over 1500 patient records was generated, incorporating demographics, observable signs, reported symptoms, and underlying conditions. A Random Forest Classifier machine learning model was trained on this data to predict patient triage category (Emergency/Red, Urgent/Yellow, Routine/Green) without the initial need for vital signs. The model achieved a high predictive accuracy of 97%. The model was operated through a user-friendly, web-based application built with Streamlit, allowing healthcare workers to input patient data via checkboxes and dropdown menus and receive an immediate, color-coded triage recommendation. This tool standardizes decision-making, empowers frontline health workers, and facilitates the rapid identification of critical cases, thereby potentially reducing waiting times, improving patient flow, and preventing adverse outcomes in overcrowded healthcare facilities. The study concludes that AI and Machine Learning driven pre-vitals triage is a viable and promising strategy for enhancing emergency care efficiency and equity.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a machine learning-powered pre-vitals triage tool achieving 97% accuracy using Random Forest Classifier on a synthetic dataset of 1500+ patient records based on WHO ETAT guidelines. The model predicts triage category (Red/Emergency, Yellow/Urgent, Green/Routine) using 40+ binary indicators for observable signs, reported symptoms, and underlying conditions without requiring vital signs. Key clinical patterns validated include strong correlations between obstructed airway, severe respiratory distress, and unconsciousness with Emergency classification. An interactive Streamlit web application was deployed featuring checkbox inputs, dropdown menus, color-coded output banners, confidence bar charts, and session distribution summaries for situational awareness. The tool addresses critical healthcare gaps by reducing triage delays from minutes to seconds, augmenting less experienced staff with standardized evidence-based recommendations, improving consistency regardless of fatigue or workload, and aligning with Universal Health Coverage goals for equitable emergency care access.

INTERNSHIP Internship Experience

Equity Afia Section 58, Nakuru — Healthcare; patient follow-up default prediction; continuity of care

Dataset Used: Equity Afia Section 58 internal facility records; Microsoft Excel spreadsheets maintained by the data clerk; supplementary patient registers from reception. Patient discharge and follow-up records from September to October 2025, including age, sex, occupation, diagnosis/condition type, number of medications, insurance status, reason for follow-up, follow-up gap, and follow-up/default risk indicators

METHODOLOGY

Conducted staff consultations with clinicians, receptionist, and data clerk; collected and cleaned patient follow-up data; performed exploratory data analysis; trained and compared Logistic Regression, Random Forest, and Gradient Boosting models; selected Gradient Boosting; built and deployed a Streamlit dashboard for follow-up risk prediction and monitoring

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Developed a machine-learning model that classifies patients as low, medium, or high risk of missing follow-up appointments; created a dashboard showing overdue/due patients, top diagnoses, follow-up schedules, and risk categories; helped clinic staff prioritize patient follow-up calls and shift from manual/reactive tracking to proactive patient monitoring

CURRENT STATUS

Self employed · Unemployed — Actively searching

Sector relevance: Yes – Fully related

SKILLS IN USE

Data collection · Data cleaning & wrangling · Data visualization · ML modeling · Model evaluation · MLOps / Model deployment · Natural Language Processing (NLP) · Prompt engineering · Generative AI / LLMs · Ethics & bias in AI · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**

**CAPSTONE** **Capstone Project***Predictive Model for Maternal Mortality Using Machine Learning Approaches***BACKGROUND & METHODOLOGY**

Maternal mortality remains a major public health challenge, particularly in developing countries, where preventable causes account for most maternal deaths. The aim of this study was to design and implement a predictive model for maternal mortality using health-related risk factors such as hemorrhage, eclampsia, sepsis, obstructed labor, skilled birth attendance, antenatal care visits, and place of delivery. The data used for this project is a synthetic data. A machine learning model was trained using logistic regression and random forest classifiers to predict maternal outcomes (survived or died). The study found that hemorrhage, eclampsia, and lack of skilled birth attendance were the most significant predictors. The developed model was deployed on a user-friendly interface for healthcare providers. Findings suggest that predictive modeling can enhance early detection of high-risk cases, improve decision-making, and reduce maternal mortality.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a predictive model for maternal mortality using machine learning on synthetic data from Nakuru County, Kenya. Random Forest was selected as the final model achieving 76% accuracy with an exceptional ROC-AUC score of 0.91, outperforming Logistic Regression (78% accuracy) and SVM. Key predictors identified through feature importance ranking were hemorrhage as the strongest predictor, followed by skilled birth attendance, eclampsia, and place of delivery. Exploratory analysis confirmed that mothers with no skilled birth attendance, fewer than 4 ANC visits, and presence of complications (hemorrhage or eclampsia) were highly likely to experience maternal death. The model was deployed through a user-friendly interface enabling healthcare providers to input patient data and receive instant high-risk or low-risk classifications. The study contributes to Sustainable Development Goal 3 by providing a data-driven solution for early risk detection, supporting timely interventions, strengthening referral systems, and reducing preventable maternal deaths in resource-limited settings.

INTERNSHIP **Internship Experience***Kabarak University / ANC maternal health project — Maternal health; antenatal risk prediction; high-risk pregnancy screening*

Dataset Used: Simulated and hospital-like antenatal profile dataset based on routine ANC records. Dataset of 500 balanced antenatal profile records containing maternal age, hemoglobin level, systolic and diastolic blood pressure, number of ANC visits, HIV status, syphilis test, urine protein, blood sugar, malaria test, BMI, and maternal risk level classified as high risk or low risk

METHODOLOGY

Prepared antenatal profile data for modelling; cleaned and encoded categorical variables; normalized numeric variables; analyzed relationships between ANC indicators and maternal risk; trained and evaluated a Logistic Regression model; identified influential predictors of high maternal risk

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Developed a predictive model for classifying expectant mothers into high-risk and low-risk categories; achieved over 80% reported accuracy; identified hemoglobin level, blood pressure, urine protein, BMI, blood sugar, malaria status, and ANC attendance as important risk indicators; proposed use of the model as a decision-support tool for early maternal risk screening and timely clinical intervention

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Machine Learning-Based Community Screening Model for Predicting Malnutrition Risk in Children Under Five

BACKGROUND & METHODOLOGY

Malnutrition is a leading cause of morbidity and mortality among children under five, contributing to over 3 million deaths annually worldwide. In Kenya, stunting, wasting, and underweight remain prevalent, posing a significant public health burden. This project developed a machine learning--based community screening model to predict malnutrition risk early, enabling intervention before severe cases occur. Using synthetic data (3,000 records, 10 variables), Logistic Regression, Random Forest, and XGBoost models were trained and evaluated. XGBoost achieved the best performance with an AUC score of 0.91. The model was deployed with Python and Streamlit, demonstrating its potential for community-based health screening.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a machine learning-based community screening model for predicting malnutrition risk in children under five using synthetic data of 3,000 records with 10 demographic and socio-economic variables. XGBoost emerged as the best-performing model achieving an AUC score of 0.9113, MSE of 0.0889, and R- of 0.6437, outperforming Logistic Regression and Random Forest. Key predictors identified included parental education, income level, access to clean water, age, and family size. The model was deployed via an interactive offline Streamlit interface for community health practitioners, enabling household-level risk assessment without requiring advanced clinical measurements. The study demonstrates a scalable, data-driven approach to early malnutrition detection, supporting timely intervention, reducing child mortality, and aligning with Kenya's commitment to SDG 2 (Zero Hunger) and SDG 3 (Good Health and Well-being).

INTERNSHIP Internship Experience

Bahati Sub-County Hospital, Bahati Sub-County, Nakuru County — Child health; pneumonia early detection; under-five clinical risk prediction; Community Health Promoter screening tool

Dataset Used: Secondary inpatient and outpatient hospital records from Bahati Sub-County Hospital / healthcare facility records. Dataset of 800 under-five patient records with 18 variables, including clinical signs and demographic attributes related to pneumonia such as cough, fever/hotness of body, fast respiratory rate, chest indrawing, age, and pneumonia outcome

METHODOLOGY

Extracted and compiled inpatient and outpatient records; cleaned and preprocessed data; handled duplicates and consistency issues; conducted exploratory data analysis using Python; visualized distributions and correlations using Matplotlib and Seaborn; trained and tuned Logistic Regression, Random Forest, and XGBoost models; compared models using accuracy, recall, precision, and F1-score; deployed the final model using Streamlit

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Developed a clinical feature-based pneumonia prediction model for children under five; identified cough, fever, fast respiratory rate, and chest indrawing as key predictors; found Random Forest and XGBoost to perform better than Logistic Regression, with Random Forest recommended for integration into a mobile/web tool; produced a user-friendly screening tool to help Community Health Promoters identify high-risk children early and refer them for timely care

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Development of a Machine Learning Screening Model for Endometriosis in Women of Reproductive Age

BACKGROUND & METHODOLOGY

Endometriosis is a chronic gynecological disorder affecting approximately 190 million women worldwide. Despite its significant health and socioeconomic burden, there is a substantial delay in diagnosis, particularly in low- and middle-income countries (LMICs) like Kenya. This delay is attributed to the normalization of symptoms, social stigma, and the invasive and costly nature of the current diagnostic gold standard, laparoscopy. This report outlines the development of a non-invasive machine learning-based screening model designed to estimate the likelihood of endometriosis. By utilizing demographic, reproductive, and clinical data, the model aims to serve as a predictive tool to enable earlier detection and referral. The study identified key predictors, constructed and trained machine learning models (Logistic Regression and Naïve Bayes), and evaluated their performance. The models achieved an impressive accuracy of 86%, demonstrating their potential as a cost-effective and accessible screening alternative for resource-limited settings.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a non-invasive machine learning screening model for endometriosis achieving 86% accuracy using Logistic Regression and Naïve Bayes algorithms. Key predictors identified included dysmenorrhea score, pelvic pain score, dyspareunia score, dyschezia score, duration of periods, age of menarche, family history, absenteeism days per month, and menstrual flow characteristics. Logistic Regression achieved 86% accuracy with precision 0.85-0.87, recall 0.84-0.88, and F1-Score 0.86 across both classes. Naïve Bayes achieved 86.67% accuracy with AUC-ROC of 0.9431, precision 0.85-0.89, recall 0.84-0.89, and F1-Score 0.86-0.87. The model addresses the 7-10 year diagnostic delay in LMICs by providing an accessible alternative to invasive laparoscopy, enabling healthcare workers to identify high-risk women using readily available clinical data. A mobile application interface was designed for deployment in resource-constrained settings, facilitating early referral, reducing patient suffering, and aligning with Sustainable Development Goal 3 for women's health and well-being.

INTERNSHIP Internship Experience

Bondeni Hospital / Bondeni Sub-County Hospital, Nakuru — Maternal health; anemia risk screening in pregnant women; antenatal care risk prediction

Dataset Used: Antenatal care records from Bondeni Hospital covering 2024–2025. ANC dataset containing health records for pregnant women, including hemoglobin levels and anemia-related predictors such as parity, gravidity, HIV status, blood pressure, malaria infection, BMI, previous history, and hemoglobin concentration. The model classified women into low-risk and high-risk anemia categories based mainly on hemoglobin thresholds

METHODOLOGY

Cleaned and preprocessed ANC data; explored anemia patterns using visualizations; trained and compared machine learning models including Logistic Regression, Random Forest, and XGBoost; classified anemia risk into low-risk and high-risk groups; deployed the trained model for clinical/community screening use

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Developed a machine-learning screening model for early anemia risk detection among pregnant women; identified a high proportion of women at risk of anemia; Random Forest and XGBoost were reported as the best-performing models; created an AI-powered screening tool to support healthcare providers and CHPs in identifying high-risk mothers early and enabling timely intervention

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Predictive Model for Maternal Mortality Using Machine Learning Approaches

BACKGROUND & METHODOLOGY

Maternal mortality remains a major public health challenge, particularly in developing countries, where preventable causes account for most maternal deaths. The aim of this study was to design and implement a predictive model for maternal mortality using health-related risk factors such as hemorrhage, eclampsia, sepsis, obstructed labor, skilled birth attendance, antenatal care visits, and place of delivery. The data used for this project is a synthetic data. A machine learning model was trained using logistic regression and random forest classifiers to predict maternal outcomes (survived or died). The study found that hemorrhage, eclampsia, and lack of skilled birth attendance were the most significant predictors. The developed model was deployed on a user-friendly interface for healthcare providers. Findings suggest that predictive modeling can enhance early detection of high-risk cases, improve decision-making, and reduce maternal mortality.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a predictive model for maternal mortality using machine learning on synthetic data from Nakuru County, Kenya. Random Forest was selected as the final model achieving 76% accuracy with an exceptional ROC-AUC score of 0.91, outperforming Logistic Regression (78% accuracy) and SVM. Key predictors identified through feature importance ranking were hemorrhage as the strongest predictor, followed by skilled birth attendance, eclampsia, and place of delivery. Exploratory analysis confirmed that mothers with no skilled birth attendance, fewer than 4 ANC visits, and presence of complications (hemorrhage or eclampsia) were highly likely to experience maternal death. The model was deployed through a user-friendly interface enabling healthcare providers to input patient data and receive instant high-risk or low-risk classifications. The study contributes to Sustainable Development Goal 3 by providing a data-driven solution for early risk detection, supporting timely interventions, strengthening referral systems, and reducing preventable maternal deaths in resource-limited settings.

INTERNSHIP Internship Experience

Kabarak University / ANC maternal health project — Maternal health; antenatal risk prediction; high-risk pregnancy screening

Dataset Used: Simulated and hospital-like antenatal profile dataset based on routine ANC records. Dataset of 500 balanced antenatal profile records containing maternal age, hemoglobin level, systolic and diastolic blood pressure, number of ANC visits, HIV status, syphilis test, urine protein, blood sugar, malaria test, BMI, and maternal risk level classified as high risk or low risk

METHODOLOGY

Prepared antenatal profile data for modelling; cleaned and encoded categorical variables; normalized numeric variables; analyzed relationships between ANC indicators and maternal risk; trained and evaluated a Logistic Regression model; identified influential predictors of high maternal risk

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Developed a predictive model for classifying expectant mothers into high-risk and low-risk categories; achieved over 80% reported accuracy; identified hemoglobin level, blood pressure, urine protein, BMI, blood sugar, malaria status, and ANC attendance as important risk indicators; proposed use of the model as a decision-support tool for early maternal risk screening and timely clinical intervention

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Patient-Centered PCOS Risk Prediction Using a Logistic Regression Model Based on Clinical Features

BACKGROUND & METHODOLOGY

Polycystic Ovary Syndrome (PCOS) represents the most common endocrine disorder among women of reproductive age, with a global prevalence estimated between 6% and 13%. This burden is particularly evident in resource-constrained settings like Kenya, where the condition is significantly underdiagnosed, with up to 70% of cases remaining unidentified, leading to severe and often preventable long-term complications including infertility, type 2 diabetes mellitus, cardiovascular morbidity, and considerable psychological distress. Traditional diagnostic approaches are often subjective, inconsistent, and heavily reliant on clinician experience, creating a critical need for data-driven and patient-centered tools that support early prediction, empower women to track their symptoms, and improve health outcomes. This study aims to develop, evaluate, and deploy a logistic regression model designed to predict an individual's risk of developing PCOS based on a patient's clinical features. The methodological approach entailed a secondary analysis of the publicly available PCOS Extended Dataset extracted from Kaggle, comprising data from 2000 women of reproductive age (18-49 years) as the target population. Using Python programming, we cleaned the data, handled missing values, and prepared it for analysis. Logistic regression was used to identify the relationships between clinical features and PCOS outcomes. The final model was operationalized into an accessible, patient-friendly web application using Streamlit.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a patient-centered PCOS risk prediction tool using logistic regression on 2,000 women from the PCOS Extended Dataset. The model achieved 85% overall accuracy with high precision and recall for non-PCOS cases (0.89-0.90) and good performance for PCOS cases (0.73-0.77). Odds ratio analysis identified key risk predictors: hair growth (OR 1.84), skin darkening (OR 1.74), irregular menstrual cycles (OR 1.70), and weight gain (OR 1.66) significantly increased PCOS risk, while older age (OR 0.62) and longer periods (OR 0.81) showed protective effects. A tiered risk stratification system was implemented based on probability thresholds: low risk (<0.2), moderate risk (0.2-0.38), and high risk (>0.38), enabling personalized clinical pathways. The model was deployed via an interactive Streamlit web application allowing women to input symptoms and receive immediate, interpretable risk assessments with clear explanations. This approach addresses the 70% underdiagnosis rate in Kenya by empowering community health workers and patients with an affordable, accessible early screening tool that transforms subjective symptoms into objective, evidence-based risk scores, facilitating timely referral and proactive health management.

INTERNSHIP Internship Experience

Bahati Sub-County Hospital, Bahati Sub-County, Nakuru County — Child health; pneumonia early detection; under-five clinical risk prediction; Community Health Promoter screening tool

Dataset Used: Secondary inpatient and outpatient hospital records from Bahati Sub-County Hospital / healthcare facility records. Dataset of 800 under-five patient records with 18 variables, including clinical signs and demographic attributes related to pneumonia such as cough, fever/hotness of body, fast respiratory rate, chest indrawing, age, and pneumonia outcome

METHODOLOGY

Extracted and compiled inpatient and outpatient records; cleaned and preprocessed data; handled duplicates and consistency issues; conducted exploratory data analysis using Python; visualized distributions and correlations using Matplotlib and Seaborn; trained and tuned Logistic Regression, Random Forest, and XGBoost models; compared models using accuracy, recall, precision, and F1-score; deployed the final model using Streamlit

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Developed a clinical feature-based pneumonia prediction model for children under five; identified cough, fever, fast respiratory rate, and chest indrawing as key predictors; found Random Forest and XGBoost to perform better than Logistic Regression, with Random Forest recommended for integration into a mobile/web tool; produced a user-friendly screening tool to help Community Health Promoters identify high-risk children early and refer them for timely care

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Analyzing and Predicting the Role of Antenatal Care in Maternal Mortality and Complications

BACKGROUND & METHODOLOGY

Maternal mortality remains high especially in low- and middle-income countries, with many deaths preventable through timely and effective antenatal care (ANC). This study analyzes a hospital-based dataset of 387 mothers to evaluate the impact of ANC visits on maternal mortality and complications, using exploratory data analysis and logistic regression modeling. Variables include ANC visits, complications, socio-demographic factors such as age, education, marital status, parity, and gravida, as well as clinical factors including booking status, mode of delivery, and hospital stay duration. Key findings reveal that women with fewer or no ANC visits had significantly higher rates of maternal death and complications, while increased ANC attendance improved survival outcomes. The logistic regression model achieved an accuracy of 71%, correctly identifying 32 true negatives and 23 true positives, with a Mean Absolute Error of 0.29. The study concludes that early, consistent, and high-quality ANC reduces maternal mortality, with quality and timing of care being just as critical as the number of visits.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a logistic regression model to predict maternal death risk using a hospital-based dataset of 387 mothers, incorporating ANC visits, complications, booking status, education, marital status, parity, gravida, mode of delivery, and hospital stay days as predictors. The model achieved an accuracy of 71%, an MSE of 0.29, and an MAE of 0.29, correctly classifying 32 survivors and 23 maternal deaths while identifying 12 dangerous missed cases. Analysis confirmed a strong protective association between higher ANC attendance and maternal survival, with women having 0–1 visits showing the highest mortality rates and those with 5+ visits showing significantly better outcomes. The model was deployed as a Streamlit web application providing real-time maternal death risk predictions — classified as Low, Medium, or High — with confidence scores, supporting evidence-based clinical decision-making and early intervention planning.

INTERNSHIP Internship Experience

Beacon of Hope Hospital — Predicting Abnormal Urinalysis Results Using Routine Clinical Data | Machine Learning for Clinical Triage Support | AI for Frontline Decision-Making

Dataset Used: Urinalysis patient records from Beacon of Hope Hospital (June—September 2025, 600+ validated records containing age, sex, diagnosis, village, urine color, HPF ranges, and result flags '+', '++', '+++'). Dataset containing routine urinalysis results with variables including patient demographics, clinical diagnosis, village location, urine color, high power field (HPF) counts, and keyword-based abnormality flags, used to identify patterns of abnormal findings across age, sex, and diagnosis categories and to develop simple flagging logic to support early triage and clinical follow-up in resource-constrained primary care settings

METHODOLOGY

Internship Centre Work: Cleaned over 600 urinalysis records from June to September 2025 | Standardized result terminology (e.g., "NIL", "TRACE", "TURBID", "++") | Flagged abnormal results using keyword logic and HPF ranges | Encoded categorical variables (sex, diagnosis, village) | Supported health workers in interpreting results and identifying follow-up cases | Capstone Project: Selected predictors (age, sex, diagnosis, village) | Grouped and cleaned diagnosis labels to reduce duplication | Applied keyword-based flagging logic for abnormal results | Created age groups and stratified analysis by sex and village | Generated visualizations (bar charts, histograms, grouped summaries) using Python (Pandas, Matplotlib) | Produced structured summaries for health worker use

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Identified that most abnormalities were mild ('+', '++') but clinically important, often signaling early infection or inflammation in UTI and antenatal cases | Female patients and reproductive-age groups (20–49 years) showed highest rates of abnormal results | UTI and antenatal diagnoses were most frequently linked to abnormal urinalysis flags, confirming clinical relevance for targeted follow-up | RONGAI village contributed highest sample volume, enabling geographic prioritization for outreach | Flagging logic successfully surfaced subtle cases that might otherwise be missed, supporting early triage | Visual patterns by age, sex, and village guided CHW follow-up and resource allocation | Enhanced practical skills in health data cleaning, stratified analysis, and result interpretation | Provided actionable recommendations for improving diagnosis recording, training health workers on flag interpretation, and implementing structured reporting templates with color-coded abnormal result highlights

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Prevalence of Hearing Loss, Hearing Difficulties and Access to Healthcare

BACKGROUND & METHODOLOGY

Hearing loss is a prevalent but often under-recognized public health issue, particularly among individuals aged 18–55 exposed to modern lifestyle risk factors such as prolonged earphone use, noisy environments, and untreated ear infections. This cross-sectional survey study assessed the prevalence of hearing difficulties, awareness of hearing health, exposure to risk factors, and access to hearing healthcare among 150 adults aged 18–55 using structured questionnaires. Descriptive statistics were applied to analyze prevalence and trends, while associations between risk factors and hearing challenges were identified. Key findings revealed that over 40% of participants showed signs of mild to moderate hearing impairment, with early detection rates remaining low as most cases were identified only after significant communication challenges emerged. Fewer than 30% of respondents had ever undergone a hearing test, and awareness of preventive measures was notably low. Common barriers to accessing care included lack of awareness, cost of healthcare, and limited screening opportunities. The study highlights significant gaps in hearing wellness awareness and preventive practices, underscoring the need for targeted public health interventions including community-based screenings, safe listening education, and improved access to affordable hearing care.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully conducted a cross-sectional survey study among 150 participants aged 18–55 to assess hearing loss prevalence, risk factors, and healthcare access, with data analyzed using Python-powered descriptive statistics and a logistic regression model incorporating demographics, clinical information, and self-reported hearing status as predictors. Key findings established that over 40% of participants showed mild to moderate hearing impairment, noise-induced hearing loss was most prevalent among younger adults in industrial or agricultural settings, and a strong positive association was identified between belief in early hearing care and willingness to invest in hearing solutions. Age, percentage of correct test responses, and average reaction time emerged as the most important predictive features. The model was deployed as a web-based tool accessible via QR code, with recommendations for public awareness campaigns on early signs of hearing loss, workplace and community screening programs, and education on safe listening practices to reduce the burden of untreated hearing loss.

INTERNSHIP Internship Experience

Meru Hospice — Data-Driven Segmentation of Cancer Patients Receiving Palliative Care Using K-Means Clustering | Unsupervised Machine Learning for Patient Stratification in Oncology Care

Dataset Used: Palliative care patient records from Meru Hospice cancer registry containing demographic, clinical, and psychosocial variables (age, gender, diagnosis type, cancer site, HIV status, pain score, mental status). Study applying unsupervised machine learning (K-Means clustering) to segment cancer patients based on demographic, clinical, and psychosocial variables to identify distinct patient subgroups with unique clinical and symptom profiles ranging from elderly patients with advanced disease and high pain levels to younger patients with stable mental health and lower symptom burden, supporting personalized palliative care delivery and evidence-based resource allocation

METHODOLOGY

Internship Centre Work: Data entry and updating of cancer patient registers; Validating and cleaning patient health data by checking for missing values, duplicates, and inconsistencies within electronic health records; Compiling and preparing monthly cancer surveillance and treatment reports for hospital management and county health department; Supporting data review meetings to discuss trends in cancer cases, treatment outcomes, and palliative care needs | Capstone Project: Data collection, cleaning, encoding, and normalization of palliative care patient data; Applied K-Means clustering algorithm using Python to group patients based on shared characteristics; Performed descriptive statistics and data visualization using Pandas, NumPy, Matplotlib, and Scikit-learn; Generated dendrograms, cluster profiles, scatter plots, and heat maps to interpret and present findings; Developed Streamlit-based web interface for interactive cluster exploration

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Identified four distinct patient clusters representing unique clinical and symptom profiles, including elderly patients with advanced-stage cancer and high symptom burden, middle-aged patients with moderate treatment response, and younger patients with early-stage cancer and better prognosis | Established breast cancer as the most prevalent diagnosis (45+ cases), followed by prostate, cervical, colon, and oesophagus cancers | Revealed female patients comprised 57.9% of cases with age distribution peaking at 60–70 years | Clustering analysis exposed unexpected survival patterns including 100% mortality in some early-stage clusters suggesting potential gaps in treatment access or follow-up | Validated and cleaned over 500 patient records improving data reliability for cancer reporting | Developed visual dashboards and summary charts highlighting patient trends to support evidence-based discussions during departmental review meetings | Demonstrated feasibility of using AI-driven segmentation to enhance patient care planning, optimize resource allocation, and strengthen data-driven decision-making in palliative care services | Enhanced practical skills in oncology data management, unsupervised machine learning techniques, and result interpretation for public health applications

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

An Analysis of Obesity and Its Determinants Among Young Adults Aged 18–30 Years

BACKGROUND & METHODOLOGY

Obesity is a growing global health challenge associated with increased risks of chronic diseases such as diabetes, cardiovascular disorders, and hypertension. This study utilizes a synthetic obesity dataset of 2,111 records containing demographic, dietary, and lifestyle variables to analyze patterns and develop predictive models for obesity classification among young adults aged 18–30 years. Exploratory data analysis was conducted to examine obesity level distributions, while three machine learning models — logistic regression, decision tree, and random forest — were compared for predictive performance. Key findings highlight that family history of overweight, frequent consumption of high-calorie foods, low vegetable intake, physical inactivity, prolonged screen time, passive transport modes, and frequent alcohol consumption were the most significant risk factors for obesity. The Random Forest Classifier demonstrated the best performance across all obesity categories and was selected as the final model, confirming the potential of predictive analytics to inform targeted public health interventions for young adults.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a Random Forest Classifier to predict obesity categories among young adults aged 18–30 years using the UCI Machine Learning Repository obesity dataset of 2,111 records, incorporating demographic, dietary, and lifestyle variables including age, gender, height, weight, caloric food intake, vegetable consumption, physical activity, screen time, alcohol use, smoking, transport mode, and family history. The Random Forest outperformed logistic regression and decision tree models, achieving the highest accuracy and balanced performance across all obesity classes, and was deployed as a live Streamlit web application at obesity-predictio.streamlit.app. Key protective factors identified included regular physical activity, higher vegetable consumption, and active transport modes, with recommendations for health promotion programs targeting modifiable lifestyle behaviors and the adoption of predictive tools in healthcare settings for early identification of at-risk individuals.

INTERNSHIP Internship Experience

Furaha Centre — Developmental Profiles Analysis of Children Admitted for Occupational Therapy | Machine Learning for Early Childhood Intervention Planning | AI for Child Development Analytics

Dataset Used: FurahaAdmissionsAllClinics 2024–2025 dataset from Furaha Centre Meru (236 validated records with 96 variables; 144 records from 2024, 92 from 2025, compiled from clinic records and Cognito forms). Dataset containing child developmental profiles for occupational therapy admissions with variables including age, gender, residence, medical conditions, antenatal/perinatal history, developmental milestones (gross motor, fine motor, sensory), therapy needs, diagnosis impressions (DXOTImpression), and caregiver engagement data, used to identify trends in developmental delays, analyze geographic and demographic patterns, and develop predictive models to support early intervention and evidence-based occupational therapy planning for vulnerable pediatric populations

METHODOLOGY

Data entry and validation of over 230 client admission records from FurahaAdmissionsAllClinics dataset; cleaned and organized datasets by correcting inconsistencies, handling missing values, standardizing variable formats (age groups, diagnosis codes, milestone categories); performed descriptive analysis to summarize demographic and medical information; generated visualizations using bar charts, pie charts, and stacked comparisons to display trends by year, diagnosis, county, and activity type; applied Random Forest classifier to explore relationships between antenatal/perinatal complications and developmental conditions such as cerebral palsy and autism; used Python (Google Colab environment with Pandas, NumPy, Scikit-learn, Matplotlib) and Excel for data processing, modeling, and visualization; developed summary reports and dashboards to support supervisors in monitoring therapy progress and planning interventions

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Validated and cleaned over 230 client records, improving data accuracy and completeness for occupational therapy reporting; identified majority of children admitted were aged 1–4 years with high admissions from North Imenti, Laisamis, and Tigania East sub-counties; established mothers as primary caregivers bringing children for therapy, reflecting cultural caregiving norms; revealed gross motor delays were common, especially in walking (66.2% delayed) compared to sitting and crawling; identified cerebral palsy, Autism/ASD/ADHD, and developmental delays as most frequent diagnoses across both years with consistent patterns; Random Forest classifier achieved 81% accuracy, 50% sensitivity, and 94% specificity for predicting developmental delay risk based on antenatal/perinatal factors; developed interactive data visualization charts and summary reports that enhanced data interpretation and supported evidence-based decision-making; demonstrated feasibility of applying predictive analytics to pediatric occupational therapy data for early intervention planning; enhanced practical skills in health data management, machine learning modeling, and result interpretation for child development programs

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

The Impact of Screentime on Sleep Quality and Mental Health

BACKGROUND & METHODOLOGY

Excessive recreational screen time among adolescents and young adults, averaging over seven hours daily, has emerged as a pressing public health problem. This study examines the impact of screen exposure on physical health, mental wellbeing, social development, and cognitive performance using a large dataset of 100,000 records. Variables analyzed include daily screen time hours, social media platforms used, TikTok hours, sleep hours, stress levels, and mood scores. Regression analysis revealed that screen time and TikTok hours were significant positive predictors of stress ($R=0.748$) and negative predictors of mood ($R=-0.570$), while sleep duration emerged as the strongest protective factor. Classification models achieved 78–82% accuracy in predicting mental health risk groups. Key findings confirm that 68% of respondents spend 4–8 hours daily on screens, with increased use strongly correlated with higher stress and lower mood scores. The study concludes that excessive screen time poses serious risks to adolescent wellbeing across health, social, and cognitive domains, and recommends collective action from parents, educators, and policymakers to promote balanced digital engagement.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed regression and classification models to predict mental health outcomes — specifically stress levels and mood scores — from digital usage patterns using a dataset of 100,000 records. The stress prediction model achieved an R of 0.748, explaining approximately 75% of variance, with screen time hours, TikTok usage, and sleep hours identified as the most significant predictors. Classification models achieved 78–82% accuracy in categorizing individuals into low, moderate, and high mental health risk groups. The models were deployed via a Streamlit web application allowing users to input age, screen hours, and lifestyle variables to receive real-time predictions of BMI risk, sleep disturbance probability, and screen behaviour cluster assignments, with recommendations to launch awareness campaigns, integrate screen time guidelines into educational programs, and develop targeted interventions for high-risk urban youth aged 18–24.

INTERNSHIP Internship Experience

Meru University Hospital — Using Data Analytics to Understand Disease Patterns and Patient Demographics | AI for Outpatient Health Surveillance | Descriptive Analytics for Hospital Management

Dataset Used: Outpatient department records from Meru University Hospital (May 2025, 500+ validated patient records containing demographic, diagnostic, and clinical variables including age, gender, diagnosis, and visit status). Project applying data analytics techniques to explore outpatient records and uncover trends in disease distribution and patient demographics at a university-affiliated health facility. Dataset contains patient demographic, diagnostic, and clinical variables used to identify frequently diagnosed conditions, gender and age distributions, and patient re-visit rates through descriptive statistics and Python-based visualization to support evidence-based hospital management and public health decision-making.

METHODOLOGY

Internship Centre Work: Data entry and updating patient registers in the outpatient department | Cleaning and validating patient information to correct inconsistencies and missing values | Preparing and submitting monthly statistical summaries to the hospital records office | Supporting maternal and child health (MCH) record reviews and report generation | Conducting laboratory reports analysis to ensure proper data documentation | Capstone Project: Loaded and inspected outpatient dataset using Python (Pandas library) | Identified and handled missing values using `dropna()` | Standardized variable formats and validated categorical values (Gender, Status, Diagnosis) | Performed descriptive statistics and generated visualizations (gender distribution, age histograms, boxplots, diagnosis frequency charts, revisit patterns) | Analyzed disease patterns by age group and gender using cross-tabulations | Used Python (Pandas, Matplotlib), Google Colab, and Microsoft Excel for analysis and visualization

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully cleaned and validated over 500 outpatient records, reducing missing data entries through standardized input formats | Supported timely compilation and submission of monthly statistical reports to hospital records department | Enhanced turnaround time in data verification and health statistics preparation | Descriptive analysis revealed balanced gender distribution with majority of patients aged 20–40 years | Identified infectious and respiratory illnesses as most frequent diagnoses, followed by musculoskeletal and reproductive health cases | Established noticeable proportion of repeat visits indicating chronic conditions or follow-up consultations | Achieved zero missing values after data cleaning (`data.isna().sum() = 0`), confirming successful preprocessing | Visualizations clearly illustrated gender differences in diagnoses and age distribution patterns to guide resource allocation and preventive health strategies | Enhanced practical competencies in Python, data visualization, and health information management | Demonstrated practical value of AI and data science in improving healthcare reporting efficiency and evidence-based hospital management

CURRENT STATUS

Information scientist/ Archivist · Unemployed — Actively searching

Sector relevance: Partially related

SKILLS IN USE

Data collection · Data cleaning & wrangling · Data visualization · Model evaluation · Public health data analysis · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Prediction of Cervical Cancer Risk on Behavioral, Social, and Psychological Features

BACKGROUND & METHODOLOGY

Cervical cancer remains a leading cause of cancer-related deaths among women, particularly in low- and middle-income countries such as Kenya. This study applies a supervised machine learning approach to predict the likelihood of cervical cancer using behavioral, psychosocial, and empowerment-related variables from the UCI Machine Learning Repository Cervical Cancer dataset. Features analyzed include sexual risk behaviors, personal hygiene practices, perceptions of vulnerability and severity, motivation levels, social support, and empowerment indicators, with `ca_cervix` as the target variable. The dataset comprised 69 rows and 20 columns, split 80/20 for training and testing. A Random Forest Classifier achieved an accuracy of 92%, precision of 88%, recall of 86%, and an F1-score of 92%, demonstrating strong predictive potential. The top predictors identified were `behavior_sexualRisk`, `perception_vulnerability`, `empowerment_knowledge`, `motivation_strength`, and `socialSupport_emotionality`. Findings suggest that addressing behavioral risks, increasing empowerment through education, and enhancing social support could significantly improve early detection and reduce cervical cancer mortality, particularly in resource-constrained settings.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Random Forest Classifier to predict cervical cancer risk using behavioral, psychosocial, and empowerment variables from a dataset of 69 records sourced from the UCI Machine Learning Repository, achieving an accuracy of 92%, precision of 88%, recall of 86%, and an F1-score of 92%. The model identified sexual risk behavior, perceived vulnerability, and empowerment knowledge as the strongest predictors of cervical cancer risk, with minimal false negatives ensuring high sensitivity for identifying at-risk individuals. The model was deployed as a Streamlit web application providing High Risk or Low Risk classifications with recommendations for further medical screening, with recommendations to validate the model on larger and more diverse datasets, integrate it into community health screening programs, and enhance public health education targeting risky behaviors and low health literacy in resource-limited settings

INTERNSHIP Internship Experience

Furaha Centre — Developmental Profiles Analysis of Children Admitted for Occupational Therapy | Machine Learning for Early Childhood Intervention Planning | AI for Child Development Analytics

Dataset Used: FurahaAdmissionsAllClinics 2024–2025 dataset from Furaha Centre Meru (236 validated records with 96 variables; 144 records from 2024, 92 from 2025, compiled from clinic records and Cognito forms). Dataset containing child developmental profiles for occupational therapy admissions with variables including age, gender, residence, medical conditions, antenatal/perinatal history, developmental milestones (gross motor, fine motor, sensory), therapy needs, diagnosis impressions (DXOTImpression), and caregiver engagement data, used to identify trends in developmental delays, analyze geographic and demographic patterns, and develop predictive models to support early intervention and evidence-based occupational therapy planning for vulnerable pediatric populations

METHODOLOGY

Data entry and validation of over 230 client admission records from FurahaAdmissionsAllClinics dataset; cleaned and organized datasets by correcting inconsistencies, handling missing values, standardizing variable formats (age groups, diagnosis codes, milestone categories); performed descriptive analysis to summarize demographic and medical information; generated visualizations using bar charts, pie charts, and stacked comparisons to display trends by year, diagnosis, county, and activity type; applied Random Forest classifier to explore relationships between antenatal/perinatal complications and developmental conditions such as cerebral palsy and autism; used Python (Google Colab environment with Pandas, NumPy, Scikit-learn, Matplotlib) and Excel for data processing, modeling, and visualization; developed summary reports and dashboards to support supervisors in monitoring therapy progress and planning interventions

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Validated and cleaned over 230 client records, improving data accuracy and completeness for occupational therapy reporting; identified majority of children admitted were aged 1–4 years with high admissions from North Imenti, Laisamis, and Tigania East sub-counties; established mothers as primary caregivers bringing children for therapy, reflecting cultural caregiving norms; revealed gross motor delays were common, especially in walking (66.2% delayed) compared to sitting and crawling; identified cerebral palsy, Autism/ASD/ADHD, and developmental delays as most frequent diagnoses across both years with consistent patterns; Random Forest classifier achieved 81% accuracy, 50% sensitivity, and 94% specificity for predicting developmental delay risk based on antenatal/perinatal factors; developed interactive data visualization charts and summary reports that enhanced data interpretation and supported evidence-based decision-making; demonstrated feasibility of applying predictive analytics to pediatric occupational therapy data for early intervention planning; enhanced practical skills in health data management, machine learning modeling, and result interpretation for child development programs

CURRENT STATUS

Not currently working

Sector relevance: Not related

SKILLS IN USE

Python · Not employed

Confidence applying ENGAGE skills: 4 / 5



CAPSTONE Capstone Project

Analyzing and Predicting the Role of Antenatal Care in Maternal Mortality and Complications

BACKGROUND & METHODOLOGY

Maternal mortality remains high especially in low- and middle-income countries, with many deaths preventable through timely and effective antenatal care (ANC). This study analyzes a hospital-based dataset of 387 mothers to evaluate the impact of ANC visits on maternal mortality and complications, using exploratory data analysis and logistic regression modeling. Variables include ANC visits, complications, socio-demographic factors such as age, education, marital status, parity, and gravida, as well as clinical factors including booking status, mode of delivery, and hospital stay duration. Key findings reveal that women with fewer or no ANC visits had significantly higher rates of maternal death and complications, while increased ANC attendance improved survival outcomes. The logistic regression model achieved an accuracy of 71%, correctly identifying 32 true negatives and 23 true positives, with a Mean Absolute Error of 0.29. The study concludes that early, consistent, and high-quality ANC reduces maternal mortality, with quality and timing of care being just as critical as the number of visits.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a logistic regression model to predict maternal death risk using a hospital-based dataset of 387 mothers, incorporating ANC visits, complications, booking status, education, marital status, parity, gravida, mode of delivery, and hospital stay days as predictors. The model achieved an accuracy of 71%, an MSE of 0.29, and an MAE of 0.29, correctly classifying 32 survivors and 23 maternal deaths while identifying 12 dangerous missed cases. Analysis confirmed a strong protective association between higher ANC attendance and maternal survival, with women having 0–1 visits showing the highest mortality rates and those with 5+ visits showing significantly better outcomes. The model was deployed as a Streamlit web application providing real-time maternal death risk predictions — classified as Low, Medium, or High — with confidence scores, supporting evidence-based clinical decision-making and early intervention planning.

INTERNSHIP Internship Experience

Miathene Subcounty Hospital — Predicting Maternal Health Risk Using Antenatal Data Through Machine Learning Techniques | AI for Maternal Health Analytics

Dataset Used: Maternal health records from Miathene Subcounty Hospital (August–September 2025) containing variables: maternal age, blood loss, parity, birth weight, gestation period, mode of delivery, marital status, and ANC visits. Dataset containing maternal health variables used to predict maternal risk categories (low-risk vs high-risk) using logistic regression and linear regression models. The project aimed to identify key clinical and demographic factors influencing maternal health outcomes to support early risk identification and evidence-based obstetric care planning in resource-limited settings.

METHODOLOGY

Data entry and validation of maternal health records; Conducting data quality audits to ensure completeness and consistency of antenatal data; Assisting in compiling monthly reports; Cleaning and preprocessing maternal data using Python; Performing exploratory data analysis and visualization to identify trends; Training logistic regression and linear regression models to classify maternal risk; Generating visualizations (pie charts, histograms, correlation plots) using Python libraries; Evaluating model performance using accuracy, precision, recall, and F1-score metrics; Deploying predictive model as Streamlit web application for real-time clinical decision support

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Logistic regression model achieved 78% accuracy with precision of 0.80, recall of 0.74, and F1-score of 0.77; Identified key predictors of maternal risk: maternal age (>35 years), blood loss (>250ml), and regular ANC visits as protective factors; Established that mothers with limited ANC attendance (<4 visits) and C/S delivery mode showed higher risk profiles; Successfully deployed Streamlit web app (<https://bloddloss-4juh5cwwk9yxhnumtv2kat.streamlit.app/>) providing clinicians with real-time risk stratification; Enhanced practical skills in health data management, machine learning modeling, and result interpretation; Demonstrated feasibility of using routine antenatal data for predictive analytics in rural hospital settings; Provided evidence-based recommendations for integrating predictive tools into maternal health workflows to support early intervention and targeted clinical care

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Prediction of Covid-19 Vaccine Willingness Uptake Based on Risk Perception

BACKGROUND & METHODOLOGY

The COVID-19 pandemic posed unprecedented global health and socio-economic challenges, with vaccination emerging as a critical preventive measure. This study investigates the determinants of COVID-19 vaccine willingness by combining survey-based data with machine learning techniques. Primary survey data was complemented with secondary sources including WHO reports and peer-reviewed studies. Data preprocessing and analysis were conducted using Python, with predictive models including Logistic Regression, Random Forest, and clustering approaches applied to identify key drivers of vaccine willingness. Among models tested, Logistic Regression achieved the best performance with an accuracy of 80.8%, precision of 80.6%, recall of 80.8%, and F1-score of 80.2%. Analysis revealed that demographic factors such as age, education, and occupation, alongside individual risk perception of COVID-19, strongly influenced vaccine willingness, with younger and more educated respondents showing higher willingness. Findings provide evidence-based insights for policymakers to design targeted public health campaigns that address vaccine hesitancy in specific demographic groups.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Random Forest Classifier to predict COVID-19 vaccine willingness using demographic and risk perception variables including age, gender, education, occupation, income, health status, prior vaccination history, and awareness level, with a 70/30 train-test split. The model achieved an accuracy of 80.8%, precision of 80.6%, recall of 80.8%, and an F1-score of 80.2%, confirming that risk perception and demographic characteristics are reliable predictors of vaccine willingness. Heatmap correlation analysis revealed a significant positive relationship between risk perception and willingness to vaccinate. The model was deployed as a live Streamlit web application at covidvaccineuptake.streamlit.app, providing probability-based willingness predictions with visual dashboards for behavioral segmentation, with recommendations for policymakers to leverage behavioral segmentation for targeted awareness campaigns and to expand research to include post-vaccination behaviors and institutional trust variables.

INTERNSHIP Internship Experience

Miathene Subcounty Hospital — Predicting Maternal Health Risk Using Antenatal Data Through Machine Learning Techniques | AI for Maternal Health Analytics

Dataset Used: Maternal health records from Miathene Subcounty Hospital (August-September 2025) containing variables: maternal age, blood loss, parity, birth weight, gestation period, mode of delivery, marital status, and ANC visits. Dataset containing maternal health variables used to predict maternal risk categories (low-risk vs high-risk) using logistic regression and linear regression models. The project aimed to identify key clinical and demographic factors influencing maternal health outcomes to support early risk identification and evidence-based obstetric care planning in resource-limited settings.

METHODOLOGY

Data entry and validation of maternal health records; Conducting data quality audits to ensure completeness and consistency of antenatal data; Assisting in compiling monthly reports; Cleaning and preprocessing maternal data using Python; Performing exploratory data analysis and visualization to identify trends; Training logistic regression and linear regression models to classify maternal risk; Generating visualizations (pie charts, histograms, correlation plots) using Python libraries; Evaluating model performance using accuracy, precision, recall, and F1-score metrics; Deploying predictive model as Streamlit web application for real-time clinical decision support

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Logistic regression model achieved 78% accuracy with precision of 0.80, recall of 0.74, and F1-score of 0.77; Identified key predictors of maternal risk: maternal age (>35 years), blood loss (>250ml), and regular ANC visits as protective factors; Established that mothers with limited ANC attendance (<4 visits) and C/S delivery mode showed higher risk profiles; Successfully deployed Streamlit web app (<https://bloddloss-4juh5cwwk9yxnumtv2kat.streamlit.app/>) providing clinicians with real-time risk stratification; Enhanced practical skills in health data management, machine learning modeling, and result interpretation; Demonstrated feasibility of using routine antenatal data for predictive analytics in rural hospital settings; Provided evidence-based recommendations for integrating predictive tools into maternal health workflows to support early intervention and targeted clinical care

CURRENT STATUS

Not yet reported



CAPSTONE **Capstone Project** *Predicting Patient Adoption of Telemedicine*

BACKGROUND & METHODOLOGY

Despite the growing importance of telemedicine, many patients remain hesitant to adopt it due to barriers such as technology literacy, access, and trust in virtual healthcare. This study conducted a quantitative analysis using a patient dataset covering demographic factors, healthcare access, and telemedicine perceptions to predict patient adoption. Data preprocessing involved cleaning, handling missing values, and encoding categorical variables. Machine learning models including linear regression and Random Forest were applied, with model performance evaluated using MSE, RMSE, and R-. Key findings revealed that age, internet accessibility, perceived convenience, and trust in healthcare providers significantly influenced telemedicine adoption. The Random Forest model achieved an accuracy of 79%, outperforming linear regression by capturing non-linear relationships and interaction effects. Geographic variation was also evident, with California recording the highest adoption rate at over 25% and Michigan the lowest at approximately 15%, while temporal analysis showed a surge in adoption during early COVID-19 waves followed by stabilization.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Random Forest classification model to predict patient adoption of telemedicine across three categories — low, medium, and high — using demographic, geographic, and temporal variables from the telemedicine_use.csv dataset. The model achieved an accuracy of 79%, outperforming the linear regression baseline, with age group, state of residence, and pandemic phase identified as the most significant predictors of adoption. Geographic analysis revealed California as the highest-adopting state and Michigan as the lowest, highlighting infrastructure and policy disparities. Deployment was envisioned through a Streamlit web application and dashboard interface for real-time prediction, with recommendations for healthcare providers to target digital literacy campaigns at older populations and for policymakers to invest in digital infrastructure in underserved regions to reduce healthcare access disparities.

INTERNSHIP **Internship Experience** *Muthara Sub County Hospital — Predictive Analytics for Maternal Complications Using Antenatal Visits and Clinical Data | Machine Learning for Maternal Health Risk Prediction*

Dataset Used: Maternal delivery records (MOH 333) from Muthara Sub County Hospital registers (200 validated records containing age, marital status, Sub County, gestation weeks, antenatal visits, gravidae, duration of labor, vaginal examination, blood loss, child birth weight, child sex, delivery complications, and maternal outcome). Project applying predictive analytics on antenatal and clinical data to identify risk factors for maternal delivery complications at Muthara Sub County Hospital. Variables assessed included age, marital status, Sub County, gestation weeks, antenatal visits, gravidae, duration of labor, vaginal examination, blood loss (ml), child birth weight (g), child sex, status of baby, delivery complications and mother status after delivery. Logistic regression and decision tree models were applied to classify outcomes into "complication" and "none" to support early risk identification and timely clinical interventions in rural maternal health settings.

METHODOLOGY

Internship Centre Work: Assisted in data entry and verification of maternal records at the MCH unit; Participated in antenatal clinic (ANC) sessions observing clinical indicators such as parity, blood pressure, and gestational age; Conducted discussions with healthcare staff on challenges in managing maternal complications | Capstone Project: Extracted 200 delivery records from hospital registers; Cleaned and recoded variables (age, marital status, Sub County, gestation weeks, antenatal visits, gravidae, duration of labor, vaginal examination, blood loss, child birth weight, child sex, delivery complications, mother status); Conducted descriptive analysis using pie charts, bar charts, and histograms; Applied logistic regression and decision tree models in Python to predict complications; Evaluated model performance using accuracy metrics

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Found that 34% of mothers experienced delivery complications, with perineal tears (20%), episiotomy (10%), and postpartum hemorrhage (4%) being most common | Mothers with <4 antenatal visits had 40% complication rate vs 28% for ≥4 visits, confirming ANC as protective factor | Prolonged labor (>8 hours) increased complication likelihood by 2.5x; blood loss >300ml significantly increased risks | Logistic Regression achieved 70% accuracy, Decision Tree achieved 62% accuracy | Identified antenatal visits, blood loss, and labor duration as strongest predictors of maternal complications | Demonstrated feasibility of using routine clinical data for predictive risk stratification in rural hospital settings | Provided evidence-based recommendations for promoting consistent ANC attendance, adopting early risk detection tools, digitizing maternal records, building staff capacity in data-driven care, and strengthening community-referral networks

CURRENT STATUS

Unemployed · Unemployed — Actively searching

Sector relevance: Not related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · Data visualization · ML modeling · Model evaluation · MLOps / Model deployment · Deep Learning · Freelancing in data science / building a portfolio · Public health data analysis · Communication & teamwork · I'm just self learning since I don't have a job

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project *Predicting Patient Adoption of Telemedicine*

BACKGROUND & METHODOLOGY

Despite the growing importance of telemedicine, many patients remain hesitant to adopt it due to barriers such as technology literacy, access, and trust in virtual healthcare. This study conducted a quantitative analysis using a patient dataset covering demographic factors, healthcare access, and telemedicine perceptions to predict patient adoption. Data preprocessing involved cleaning, handling missing values, and encoding categorical variables. Machine learning models including linear regression and Random Forest were applied, with model performance evaluated using MSE, RMSE, and R-. Key findings revealed that age, internet accessibility, perceived convenience, and trust in healthcare providers significantly influenced telemedicine adoption. The Random Forest model achieved an accuracy of 79%, outperforming linear regression by capturing non-linear relationships and interaction effects. Geographic variation was also evident, with California recording the highest adoption rate at over 25% and Michigan the lowest at approximately 15%, while temporal analysis showed a surge in adoption during early COVID-19 waves followed by stabilization.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Random Forest classification model to predict patient adoption of telemedicine across three categories — low, medium, and high — using demographic, geographic, and temporal variables from the telemedicine_use.csv dataset. The model achieved an accuracy of 79%, outperforming the linear regression baseline, with age group, state of residence, and pandemic phase identified as the most significant predictors of adoption. Geographic analysis revealed California as the highest-adopting state and Michigan as the lowest, highlighting infrastructure and policy disparities. Deployment was envisioned through a Streamlit web application and dashboard interface for real-time prediction, with recommendations for healthcare providers to target digital literacy campaigns at older populations and for policymakers to invest in digital infrastructure in underserved regions to reduce healthcare access disparities.

INTERNSHIP Internship Experience

Nyambene Level 5 Hospital — Maternal and Neonatal Outcomes of Teenage Pregnancies in Igembe Constituencies | Data-Driven Insights for Adolescent Reproductive Health Programs

Dataset Used: Primary health facility records and maternity registers from Igembe, Meru County, Kenya (teenage pregnancies aged 13-19 years) containing variables: maternal age, parity, estimated blood loss, neonatal birth weight, ANC visits, mode of delivery, and sub-county of residence. Dataset presenting maternal and neonatal health data related to teenage pregnancies among individuals aged 13 to 19 years in Igembe, Meru County, Kenya. Key variables include demographic characteristics (maternal age, parity, sub-county), maternal outcomes (estimated blood loss, mode of delivery), and neonatal indicators (birth weight, ANC attendance). Used to analyze correlations between maternal age and birth outcomes, evaluate antenatal care utilization, and identify geographical disparities in maternal healthcare services to inform evidence-based strategies for improving maternal and child health.

METHODOLOGY

Extracted primary data from health facility records and maternity registers; Entered and managed data using Excel sheets; Cleaned, coded, and analyzed dataset using descriptive statistics (frequencies, percentages, means); Generated visualizations through bar charts to display trends; Conducted comparative analyses across sub-counties (Igembe Central, North, South) to assess regional disparities; Performed correlation analyses to explore relationships between ANC attendance, birth weight, and maternal outcomes such as blood loss; Supported data quality checks and validation processes

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Established that teenage pregnancies are predominantly first-time pregnancies (primigravidae); Most deliveries had normal blood loss (<500ml) though some postpartum hemorrhage cases were reported; Significant proportion of newborns had low birth weight (<2.5kg) indicating elevated neonatal risk; Many teenage mothers attended fewer than the recommended four ANC visits, highlighting gaps in antenatal care uptake; Most teenagers delivered vaginally with some requiring cesarean sections due to complications; Health outcomes varied across sub-counties indicating unequal access to healthcare services; Inadequate ANC visits were associated with increased maternal and neonatal complications; Enhanced practical data management and research skills through hands-on application; Identified limited computing resources as a key challenge during data processing; Provided actionable recommendations for strengthening adolescent reproductive health programs, improving ANC attendance, and addressing sub-county disparities in Meru County

CURRENT STATUS

No job so far · Unemployed — Actively searching

Sector relevance: Not related

SKILLS IN USE

Freelancing in data science / building a portfolio

Confidence applying ENGAGE skills: **4 / 5**



CAPSTONE Capstone Project

Prevalence of Hearing Loss, Hearing Difficulties and Access to Healthcare

BACKGROUND & METHODOLOGY

Hearing loss is a prevalent but often under-recognized public health issue, particularly among individuals aged 18–55 exposed to modern lifestyle risk factors such as prolonged earphone use, noisy environments, and untreated ear infections. This cross-sectional survey study assessed the prevalence of hearing difficulties, awareness of hearing health, exposure to risk factors, and access to hearing healthcare among 150 adults aged 18–55 using structured questionnaires. Descriptive statistics were applied to analyze prevalence and trends, while associations between risk factors and hearing challenges were identified. Key findings revealed that over 40% of participants showed signs of mild to moderate hearing impairment, with early detection rates remaining low as most cases were identified only after significant communication challenges emerged. Fewer than 30% of respondents had ever undergone a hearing test, and awareness of preventive measures was notably low. Common barriers to accessing care included lack of awareness, cost of healthcare, and limited screening opportunities. The study highlights significant gaps in hearing wellness awareness and preventive practices, underscoring the need for targeted public health interventions including community-based screenings, safe listening education, and improved access to affordable hearing care.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully conducted a cross-sectional survey study among 150 participants aged 18–55 to assess hearing loss prevalence, risk factors, and healthcare access, with data analyzed using Python-powered descriptive statistics and a logistic regression model incorporating demographics, clinical information, and self-reported hearing status as predictors. Key findings established that over 40% of participants showed mild to moderate hearing impairment, noise-induced hearing loss was most prevalent among younger adults in industrial or agricultural settings, and a strong positive association was identified between belief in early hearing care and willingness to invest in hearing solutions. Age, percentage of correct test responses, and average reaction time emerged as the most important predictive features. The model was deployed as a web-based tool accessible via QR code, with recommendations for public awareness campaigns on early signs of hearing loss, workplace and community screening programs, and education on safe listening practices to reduce the burden of untreated hearing loss.

INTERNSHIP Internship Experience

Beacon of Hope Hospital — Predicting H. pylori Infection Risk Using Routine Clinical Data | Machine Learning for Gastrointestinal Infection Risk Assessment | AI for Early Detection and Targeted Interventions

Dataset Used: Routinely collected outpatient register data from Beacon of Hope Hospital (800+ validated patient records containing age, sex, clinical diagnosis, location/village, and H. pylori stool test results). Capstone project focused on predicting Helicobacter pylori infection risk using routinely collected outpatient data to support early detection and targeted interventions for gastrointestinal infections. Dataset includes demographic variables (age, sex), clinical diagnosis, geographic location, and binary H. pylori test outcomes (positive/negative), used to train a logistic regression classifier for risk stratification and resource optimization in diagnostic testing.

METHODOLOGY

Internship Centre Work: Cleaned and standardized 800+ outpatient records from Excel | Encoded categorical variables and handled missing data | Created visualizations to explore infection patterns by age, sex, diagnosis, and location | Supported reporting and dashboard development for Community Health Workers | Capstone Project: Loaded and explored H. pylori dataset in Google Colab | Engineered features from clinical diagnosis and location fields | Trained logistic regression model with 80/20 train-test split | Evaluated model using accuracy, ROC AUC, confusion matrix, and classification report | Saved model using joblib and built Streamlit web application for real-time deployment | Generated visualizations (bar charts, histograms, probability distributions) using Python, Pandas, Matplotlib, and Scikit-learn

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Logistic regression model achieved 77.3% accuracy with ROC AUC of 0.81, demonstrating strong discriminative ability for H. pylori risk classification | Identified key predictors: clinical diagnosis (gastritis, H. pylori-induced gastritis, amoebiasis), age (peak risk 15-25 years), and geographic location (RONGAI village emerged as infection hotspot with ~50 positive cases) | Visual analysis revealed females comprised 56.4% of tested population with slightly higher positive rates | Model prediction probabilities showed bimodal distribution indicating confident classification of low vs. high-risk patients | Successfully deployed Streamlit web application enabling frontline health workers to input patient data and receive real-time risk predictions | Demonstrated feasibility of applying predictive analytics to routine outpatient data for optimizing diagnostic resource allocation | Provided actionable recommendations for data standardization, targeted community education in high-risk villages, and integration of predictive tools into CHW workflows | Enhanced practical skills in data cleaning, feature engineering, model evaluation, and web deployment for public health applications

CURRENT STATUS

ICT · Unemployed — Actively searching

Sector relevance: Partially related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · Data visualization · ML modeling · Model evaluation · Public health data analysis · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Prediction of Covid-19 Vaccine Willingness Uptake Based on Risk Perception

BACKGROUND & METHODOLOGY

The COVID-19 pandemic posed unprecedented global health and socio-economic challenges, with vaccination emerging as a critical preventive measure. This study investigates the determinants of COVID-19 vaccine willingness by combining survey-based data with machine learning techniques. Primary survey data was complemented with secondary sources including WHO reports and peer-reviewed studies. Data preprocessing and analysis were conducted using Python, with predictive models including Logistic Regression, Random Forest, and clustering approaches applied to identify key drivers of vaccine willingness. Among models tested, Logistic Regression achieved the best performance with an accuracy of 80.8%, precision of 80.6%, recall of 80.8%, and F1-score of 80.2%. Analysis revealed that demographic factors such as age, education, and occupation, alongside individual risk perception of COVID-19, strongly influenced vaccine willingness, with younger and more educated respondents showing higher willingness. Findings provide evidence-based insights for policymakers to design targeted public health campaigns that address vaccine hesitancy in specific demographic groups.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Random Forest Classifier to predict COVID-19 vaccine willingness using demographic and risk perception variables including age, gender, education, occupation, income, health status, prior vaccination history, and awareness level, with a 70/30 train-test split. The model achieved an accuracy of 80.8%, precision of 80.6%, recall of 80.8%, and an F1-score of 80.2%, confirming that risk perception and demographic characteristics are reliable predictors of vaccine willingness. Heatmap correlation analysis revealed a significant positive relationship between risk perception and willingness to vaccinate. The model was deployed as a live Streamlit web application at covidvaccineuptake.streamlit.app, providing probability-based willingness predictions with visual dashboards for behavioral segmentation, with recommendations for policymakers to leverage behavioral segmentation for targeted awareness campaigns and to expand research to include post-vaccination behaviors and institutional trust variables.

INTERNSHIP Internship Experience

Nyambene Level 5 Hospital — Maternal and Neonatal Outcomes of Teenage Pregnancies in Igembe Constituencies | Data-Driven Insights for Adolescent Reproductive Health Programs

Dataset Used: Primary health facility records and maternity registers from Igembe, Meru County, Kenya (teenage pregnancies aged 13-19 years) containing variables: maternal age, parity, estimated blood loss, neonatal birth weight, ANC visits, mode of delivery, and sub-county of residence. Dataset presenting maternal and neonatal health data related to teenage pregnancies among individuals aged 13 to 19 years in Igembe, Meru County, Kenya. Key variables include demographic characteristics (maternal age, parity, sub-county), maternal outcomes (estimated blood loss, mode of delivery), and neonatal indicators (birth weight, ANC attendance). Used to analyze correlations between maternal age and birth outcomes, evaluate antenatal care utilization, and identify geographical disparities in maternal healthcare services to inform evidence-based strategies for improving maternal and child health.

METHODOLOGY

Extracted primary data from health facility records and maternity registers; Entered and managed data using Excel sheets; Cleaned, coded, and analyzed dataset using descriptive statistics (frequencies, percentages, means); Generated visualizations through bar charts to display trends; Conducted comparative analyses across sub-counties (Igembe Central, North, South) to assess regional disparities; Performed correlation analyses to explore relationships between ANC attendance, birth weight, and maternal outcomes such as blood loss; Supported data quality checks and validation processes

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Established that teenage pregnancies are predominantly first-time pregnancies (primigravidae); Most deliveries had normal blood loss (<500ml) though some postpartum hemorrhage cases were reported; Significant proportion of newborns had low birth weight (<2.5kg) indicating elevated neonatal risk; Many teenage mothers attended fewer than the recommended four ANC visits, highlighting gaps in antenatal care uptake; Most teenagers delivered vaginally with some requiring cesarean sections due to complications; Health outcomes varied across sub-counties indicating unequal access to healthcare services; Inadequate ANC visits were associated with increased maternal and neonatal complications; Enhanced practical data management and research skills through hands-on application; Identified limited computing resources as a key challenge during data processing; Provided actionable recommendations for strengthening adolescent reproductive health programs, improving ANC attendance, and addressing sub-county disparities in Meru County

CURRENT STATUS

A technologist · Unemployed — Actively searching

Sector relevance: Not related

SKILLS IN USE

Python

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Predicting Breast Cancer Survival Outcomes Using Patients' Characteristics and Tumor Features

BACKGROUND & METHODOLOGY

Breast cancer is a leading cause of death among women globally, and current survival prediction methods lack personalization. This study developed a logistic regression model to predict breast cancer survival outcomes using a Kaggle dataset containing patient demographics and tumor features including age, race, marital status, tumor stage, hormone receptor status, tumor size, lymph node involvement, and survival months. Data preprocessing included exploratory analysis, label encoding, and a stratified 80/20 train-test split. The model achieved an accuracy of 89%, with tumor size, lymph node status, and hormone receptor status identified as the most influential predictors of survival. The confusion matrix showed 663 true positives for alive cases, 55 true positives for dead cases, 68 false positives, and 19 false negatives. The model was deployed as a Streamlit web application enabling clinicians to obtain real-time survival predictions with probability scores, demonstrating its clinical potential as a personalized, data-driven decision-support tool.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a logistic regression classifier to predict breast cancer survival outcomes (alive/dead) using clinical and pathological features from a Kaggle breast cancer dataset, with an 80/20 stratified train-test split. The model achieved an accuracy of 89%, correctly classifying 663 alive and 55 dead cases, with tumor size, lymph node involvement, and estrogen and progesterone receptor status identified as the most significant predictors through correlation analysis. The model was deployed as a Streamlit web application allowing clinicians to input patient characteristics and receive real-time survival predictions with probability scores, with recommendations to incorporate treatment and molecular data, validate externally, and explore advanced models such as Random Forest and neural networks to improve predictive power for real-world clinical application.

INTERNSHIP Internship Experience

Meru Hospice — Data-Driven Segmentation of Cancer Patients Receiving Palliative Care Using K-Means Clustering | Unsupervised Machine Learning for Patient Stratification in Oncology Care

★ LED TO A JOB OFFER

Dataset Used: Palliative care patient records from Meru Hospice cancer registry containing demographic, clinical, and psychosocial variables (age, gender, diagnosis type, cancer site, HIV status, pain score, mental status). Study applying unsupervised machine learning (K-Means clustering) to segment cancer patients based on demographic, clinical, and psychosocial variables to identify distinct patient subgroups with unique clinical and symptom profiles ranging from elderly patients with advanced disease and high pain levels to younger patients with stable mental health and lower symptom burden, supporting personalized palliative care delivery and evidence-based resource allocation

METHODOLOGY

Internship Centre Work: Data entry and updating of cancer patient registers; Validating and cleaning patient health data by checking for missing values, duplicates, and inconsistencies within electronic health records; Compiling and preparing monthly cancer surveillance and treatment reports for hospital management and county health department; Supporting data review meetings to discuss trends in cancer cases, treatment outcomes, and palliative care needs | Capstone Project: Data collection, cleaning, encoding, and normalization of palliative care patient data; Applied K-Means clustering algorithm using Python to group patients based on shared characteristics; Performed descriptive statistics and data visualization using Pandas, NumPy, Matplotlib, and Scikit-learn; Generated dendrograms, cluster profiles, scatter plots, and heat maps to interpret and present findings; Developed Streamlit-based web interface for interactive cluster exploration

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Identified four distinct patient clusters representing unique clinical and symptom profiles, including elderly patients with advanced-stage cancer and high symptom burden, middle-aged patients with moderate treatment response, and younger patients with early-stage cancer and better prognosis | Established breast cancer as the most prevalent diagnosis (45+ cases), followed by prostate, cervical, colon, and oesophagus cancers | Revealed female patients comprised 57.9% of cases with age distribution peaking at 60-70 years | Clustering analysis exposed unexpected survival patterns including 100% mortality in some early-stage clusters suggesting potential gaps in treatment access or follow-up | Validated and cleaned over 500 patient records improving data reliability for cancer reporting | Developed visual dashboards and summary charts highlighting patient trends to support evidence-based discussions during departmental review meetings | Demonstrated feasibility of using AI-driven segmentation to enhance patient care planning, optimize resource allocation, and strengthen data-driven decision-making in palliative care services | Enhanced practical skills in oncology data management, unsupervised machine learning techniques, and result interpretation for public health applications

CURRENT STATUS

Site technician · Internship

Sector relevance: Partially related

SKILLS IN USE

Data visualization · Ethics & bias in AI · Freelancing in data science / building a portfolio · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Prevalence, Complications and Preventive Strategies of Asthma Among Children 0–14 Years

BACKGROUND & METHODOLOGY

Asthma is one of the most common chronic respiratory diseases affecting children aged 0–14 years, with a global prevalence of 5–15%. Despite being treatable, asthma remains a major public health challenge due to complications including recurrent hospitalizations, impaired school performance, and psychosocial distress, with the burden compounded in low- and middle-income countries by underdiagnosis and limited healthcare resources. This study utilized secondary data from WHO, GINA reports, hospital records, and the Kaggle Asthma dataset to establish prevalence patterns and associated complications. Descriptive and comparative statistical techniques were employed alongside thematic analysis to identify educational and psychosocial impacts. A logistic regression predictive model was developed and evaluated using accuracy, recall, and F1-score, achieving an overall accuracy of 74.8%, with strong performance in identifying non-asthmatic children (precision: 0.96, recall: 0.77) but limited sensitivity for asthma-positive cases (recall: 0.25), reflecting class imbalance in the dataset. The model was deployed via a Streamlit interface to estimate asthma risk and complication probabilities based on inputs including child age, gender, environmental exposures, family history, and symptoms.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a logistic regression classifier to predict asthma diagnosis risk among children aged 0–14 years using secondary data from the Kaggle Asthma dataset, with a 718-record test set. The model achieved an overall accuracy of 74.8%, with high precision of 0.96 and recall of 0.77 for non-asthmatic cases, though limited recall of 0.25 for asthma-positive cases due to significant class imbalance (32 positive vs. 686 negative cases). The model was deployed as a Streamlit web application accepting inputs of age, gender, environmental exposures, family history, and symptoms to generate real-time asthma risk predictions and complication probability scores, with recommendations to improve performance through data balancing techniques, incorporation of additional clinical and environmental features, and integration with school and community-based screening programs to complement data-driven early detection efforts.

INTERNSHIP Internship Experience

Furaha Centre — Developmental Profiles Analysis of Children Admitted for Occupational Therapy | Machine Learning for Early Childhood Intervention Planning | AI for Child Development Analytics

Dataset Used: FurahaAdmissionsAllClinics 2024–2025 dataset from Furaha Centre Meru (236 validated records with 96 variables; 144 records from 2024, 92 from 2025, compiled from clinic records and Cognito forms). Dataset containing child developmental profiles for occupational therapy admissions with variables including age, gender, residence, medical conditions, antenatal/perinatal history, developmental milestones (gross motor, fine motor, sensory), therapy needs, diagnosis impressions (DXOTImpression), and caregiver engagement data, used to identify trends in developmental delays, analyze geographic and demographic patterns, and develop predictive models to support early intervention and evidence-based occupational therapy planning for vulnerable pediatric populations

METHODOLOGY

Data entry and validation of over 230 client admission records from FurahaAdmissionsAllClinics dataset; cleaned and organized datasets by correcting inconsistencies, handling missing values, standardizing variable formats (age groups, diagnosis codes, milestone categories); performed descriptive analysis to summarize demographic and medical information; generated visualizations using bar charts, pie charts, and stacked comparisons to display trends by year, diagnosis, county, and activity type; applied Random Forest classifier to explore relationships between antenatal/perinatal complications and developmental conditions such as cerebral palsy and autism; used Python (Google Colab environment with Pandas, NumPy, Scikit-learn, Matplotlib) and Excel for data processing, modeling, and visualization; developed summary reports and dashboards to support supervisors in monitoring therapy progress and planning interventions

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Validated and cleaned over 230 client records, improving data accuracy and completeness for occupational therapy reporting; identified majority of children admitted were aged 1–4 years with high admissions from North Imenti, Laisamis, and Tigania East sub-counties; established mothers as primary caregivers bringing children for therapy, reflecting cultural caregiving norms; revealed gross motor delays were common, especially in walking (66.2% delayed) compared to sitting and crawling; identified cerebral palsy, Autism/ASD/ADHD, and developmental delays as most frequent diagnoses across both years with consistent patterns; Random Forest classifier achieved 81% accuracy, 50% sensitivity, and 94% specificity for predicting developmental delay risk based on antenatal/perinatal factors; developed interactive data visualization charts and summary reports that enhanced data interpretation and supported evidence-based decision-making; demonstrated feasibility of applying predictive analytics to pediatric occupational therapy data for early intervention planning; enhanced practical skills in health data management, machine learning modeling, and result interpretation for child development programs

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Prediction of Cervical Cancer Risk on Behavioral, Social, and Psychological Features

BACKGROUND & METHODOLOGY

Cervical cancer remains a leading cause of cancer-related deaths among women, particularly in low- and middle-income countries such as Kenya. This study applies a supervised machine learning approach to predict the likelihood of cervical cancer using behavioral, psychosocial, and empowerment-related variables from the UCI Machine Learning Repository Cervical Cancer dataset. Features analyzed include sexual risk behaviors, personal hygiene practices, perceptions of vulnerability and severity, motivation levels, social support, and empowerment indicators, with `ca_cervix` as the target variable. The dataset comprised 69 rows and 20 columns, split 80/20 for training and testing. A Random Forest Classifier achieved an accuracy of 92%, precision of 88%, recall of 86%, and an F1-score of 92%, demonstrating strong predictive potential. The top predictors identified were `behavior_sexualRisk`, `perception_vulnerability`, `empowerment_knowledge`, `motivation_strength`, and `socialSupport_emotionality`. Findings suggest that addressing behavioral risks, increasing empowerment through education, and enhancing social support could significantly improve early detection and reduce cervical cancer mortality, particularly in resource-constrained settings.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Random Forest Classifier to predict cervical cancer risk using behavioral, psychosocial, and empowerment variables from a dataset of 69 records sourced from the UCI Machine Learning Repository, achieving an accuracy of 92%, precision of 88%, recall of 86%, and an F1-score of 92%. The model identified sexual risk behavior, perceived vulnerability, and empowerment knowledge as the strongest predictors of cervical cancer risk, with minimal false negatives ensuring high sensitivity for identifying at-risk individuals. The model was deployed as a Streamlit web application providing High Risk or Low Risk classifications with recommendations for further medical screening, with recommendations to validate the model on larger and more diverse datasets, integrate it into community health screening programs, and enhance public health education targeting risky behaviors and low health literacy in resource-limited settings

INTERNSHIP Internship Experience

Muthara Sub County Hospital — Predictive Analytics for Maternal Complications Using Antenatal Visits and Clinical Data | Machine Learning for Maternal Health Risk Prediction

Dataset Used: Maternal delivery records (MOH 333) from Muthara Sub County Hospital registers (200 validated records containing age, marital status, Sub County, gestation weeks, antenatal visits, gravidae, duration of labor, vaginal examination, blood loss, child birth weight, child sex, delivery complications, and maternal outcome). Project applying predictive analytics on antenatal and clinical data to identify risk factors for maternal delivery complications at Muthara Sub County Hospital. Variables assessed included age, marital status, Sub County, gestation weeks, antenatal visits, gravidae, duration of labor, vaginal examination, blood loss (ml), child birth weight (g), child sex, status of baby, delivery complications and mother status after delivery. Logistic regression and decision tree models were applied to classify outcomes into "complication" and "none" to support early risk identification and timely clinical interventions in rural maternal health settings.

METHODOLOGY

Internship Centre Work: Assisted in data entry and verification of maternal records at the MCH unit; Participated in antenatal clinic (ANC) sessions observing clinical indicators such as parity, blood pressure, and gestational age; Conducted discussions with healthcare staff on challenges in managing maternal complications | Capstone Project: Extracted 200 delivery records from hospital registers; Cleaned and recoded variables (age, marital status, Sub County, gestation weeks, antenatal visits, gravidae, duration of labor, vaginal examination, blood loss, child birth weight, child sex, delivery complications, mother status); Conducted descriptive analysis using pie charts, bar charts, and histograms; Applied logistic regression and decision tree models in Python to predict complications; Evaluated model performance using accuracy metrics

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Found that 34% of mothers experienced delivery complications, with perineal tears (20%), episiotomy (10%), and postpartum hemorrhage (4%) being most common | Mothers with <4 antenatal visits had 40% complication rate vs 28% for ≥4 visits, confirming ANC as protective factor | Prolonged labor (>8 hours) increased complication likelihood by 2.5x; blood loss >300ml significantly increased risks | Logistic Regression achieved 70% accuracy, Decision Tree achieved 62% accuracy | Identified antenatal visits, blood loss, and labor duration as strongest predictors of maternal complications | Demonstrated feasibility of using routine clinical data for predictive risk stratification in rural hospital settings | Provided evidence-based recommendations for promoting consistent ANC attendance, adopting early risk detection tools, digitizing maternal records, building staff capacity in data-driven care, and strengthening community-referral networks

CURRENT STATUS

Not yet reported



CAPSTONE **Capstone Project**

Predicting Breast Cancer Survival Outcomes Using Patients' Characteristics and Tumor Features

BACKGROUND & METHODOLOGY

Breast cancer is a leading cause of death among women globally, and current survival prediction methods lack personalization. This study developed a logistic regression model to predict breast cancer survival outcomes using a Kaggle dataset containing patient demographics and tumor features including age, race, marital status, tumor stage, hormone receptor status, tumor size, lymph node involvement, and survival months. Data preprocessing included exploratory analysis, label encoding, and a stratified 80/20 train-test split. The model achieved an accuracy of 89%, with tumor size, lymph node status, and hormone receptor status identified as the most influential predictors of survival. The confusion matrix showed 663 true positives for alive cases, 55 true positives for dead cases, 68 false positives, and 19 false negatives. The model was deployed as a Streamlit web application enabling clinicians to obtain real-time survival predictions with probability scores, demonstrating its clinical potential as a personalized, data-driven decision-support tool.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a logistic regression classifier to predict breast cancer survival outcomes (alive/dead) using clinical and pathological features from a Kaggle breast cancer dataset, with an 80/20 stratified train-test split. The model achieved an accuracy of 89%, correctly classifying 663 alive and 55 dead cases, with tumor size, lymph node involvement, and estrogen and progesterone receptor status identified as the most significant predictors through correlation analysis. The model was deployed as a Streamlit web application allowing clinicians to input patient characteristics and receive real-time survival predictions with probability scores, with recommendations to incorporate treatment and molecular data, validate externally, and explore advanced models such as Random Forest and neural networks to improve predictive power for real-world clinical application.

INTERNSHIP **Internship Experience**

Nyambene Level 5 Hospital — Maternal and Neonatal Outcomes of Teenage Pregnancies in Igembe Constituencies | Data-Driven Insights for Adolescent Reproductive Health Programs

Dataset Used: Primary health facility records and maternity registers from Igembe, Meru County, Kenya (teenage pregnancies aged 13-19 years) containing variables: maternal age, parity, estimated blood loss, neonatal birth weight, ANC visits, mode of delivery, and sub-county of residence. Dataset presenting maternal and neonatal health data related to teenage pregnancies among individuals aged 13 to 19 years in Igembe, Meru County, Kenya. Key variables include demographic characteristics (maternal age, parity, sub-county), maternal outcomes (estimated blood loss, mode of delivery), and neonatal indicators (birth weight, ANC attendance). Used to analyze correlations between maternal age and birth outcomes, evaluate antenatal care utilization, and identify geographical disparities in maternal healthcare services to inform evidence-based strategies for improving maternal and child health.

METHODOLOGY

Extracted primary data from health facility records and maternity registers; Entered and managed data using Excel sheets; Cleaned, coded, and analyzed dataset using descriptive statistics (frequencies, percentages, means); Generated visualizations through bar charts to display trends; Conducted comparative analyses across sub-counties (Igembe Central, North, South) to assess regional disparities; Performed correlation analyses to explore relationships between ANC attendance, birth weight, and maternal outcomes such as blood loss; Supported data quality checks and validation processes

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Established that teenage pregnancies are predominantly first-time pregnancies (primigravidae); Most deliveries had normal blood loss (<500ml) though some postpartum hemorrhage cases were reported; Significant proportion of newborns had low birth weight (<2.5kg) indicating elevated neonatal risk; Many teenage mothers attended fewer than the recommended four ANC visits, highlighting gaps in antenatal care uptake; Most teenagers delivered vaginally with some requiring cesarean sections due to complications; Health outcomes varied across sub-counties indicating unequal access to healthcare services; Inadequate ANC visits were associated with increased maternal and neonatal complications; Enhanced practical data management and research skills through hands-on application; Identified limited computing resources as a key challenge during data processing; Provided actionable recommendations for strengthening adolescent reproductive health programs, improving ANC attendance, and addressing sub-county disparities in Meru County

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Analyzing and Predicting the Role of Antenatal Care in Maternal Mortality and Complications

BACKGROUND & METHODOLOGY

Maternal mortality remains high especially in low- and middle-income countries, with many deaths preventable through timely and effective antenatal care (ANC). This study analyzes a hospital-based dataset of 387 mothers to evaluate the impact of ANC visits on maternal mortality and complications, using exploratory data analysis and logistic regression modeling. Variables include ANC visits, complications, socio-demographic factors such as age, education, marital status, parity, and gravida, as well as clinical factors including booking status, mode of delivery, and hospital stay duration. Key findings reveal that women with fewer or no ANC visits had significantly higher rates of maternal death and complications, while increased ANC attendance improved survival outcomes. The logistic regression model achieved an accuracy of 71%, correctly identifying 32 true negatives and 23 true positives, with a Mean Absolute Error of 0.29. The study concludes that early, consistent, and high-quality ANC reduces maternal mortality, with quality and timing of care being just as critical as the number of visits.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a logistic regression model to predict maternal death risk using a hospital-based dataset of 387 mothers, incorporating ANC visits, complications, booking status, education, marital status, parity, gravida, mode of delivery, and hospital stay days as predictors. The model achieved an accuracy of 71%, an MSE of 0.29, and an MAE of 0.29, correctly classifying 32 survivors and 23 maternal deaths while identifying 12 dangerous missed cases. Analysis confirmed a strong protective association between higher ANC attendance and maternal survival, with women having 0–1 visits showing the highest mortality rates and those with 5+ visits showing significantly better outcomes. The model was deployed as a Streamlit web application providing real-time maternal death risk predictions — classified as Low, Medium, or High — with confidence scores, supporting evidence-based clinical decision-making and early intervention planning.

INTERNSHIP Internship Experience

Meru University Hospital — Analysis of Laboratory Test Patterns and Diagnostic Outcomes | Descriptive Analytics for Routine Screening | AI for Laboratory Data Quality Improvement

Dataset Used: Routine laboratory test records from Meru University Hospital (September–October 2025, 3,620 test records from 571 unique patients). Dataset containing routine laboratory test data including VDRL/RPR, urinalysis, malaria RDT, HIV, pregnancy tests, and stool microscopy with variables such as test name, specimen type, result remarks, and testing dates, used to describe testing patterns, evaluate diagnostic outcomes, calculate positivity rates, and identify data quality gaps to support evidence-based laboratory service delivery and public health planning

METHODOLOGY

Internship Centre Work: Entered laboratory and clinic data into digital registers | Updated and validated patient records in laboratory logbooks | Cleaned and verified health records to identify inconsistencies | Prepared monthly reports covering laboratory outputs and MCH indicators | Supported data review meetings to discuss trends and discrepancies | Capstone Project Work: Extracted 3,620 laboratory records from facility registers | Removed duplicate entries and handled missing dates | Standardized variable names (test codes, remarks, specimen types) | Applied descriptive statistics to summarize test frequencies and result distributions | Calculated positivity rates for common tests | Generated bar charts and pie charts using Python (pandas, matplotlib) and Excel for visualization | Provided recommendations for improving data quality and reporting accuracy

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully cleaned and validated 3,620 laboratory test records from 571 unique patients, improving data reliability for reporting | Identified urinalysis and serological tests as accounting for the largest proportion of laboratory workload | Established top three most frequently conducted tests: VDRL/RPR, urinalysis (glucose/proteins), and malaria RDT representing over 60% of laboratory activities | Calculated positivity rates: Malaria RDT 7.7%, Stool Microscopy 4.6%, Pregnancy Tests 4.5%, VDRL/RPR/HIV/Widal 2–3% | Identified and addressed data quality issues including duplicate entries, non-standard result codes, and missing specimen details through cleaning and standardization | Developed simple dashboards and charts visualizing testing trends for departmental review meetings | Enhanced technical skills in data cleaning, descriptive analysis, and visualization using Python and Excel | Provided actionable recommendations for standardizing data entry codes, automating laboratory dashboards, and linking laboratory data with clinical information to improve decision-making and public health planning

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Prevalence, Complications and Preventive Strategies of Asthma Among Children 0–14 Years

BACKGROUND & METHODOLOGY

Asthma is one of the most common chronic respiratory diseases affecting children aged 0–14 years, with a global prevalence of 5–15%. Despite being treatable, asthma remains a major public health challenge due to complications including recurrent hospitalizations, impaired school performance, and psychosocial distress, with the burden compounded in low- and middle-income countries by underdiagnosis and limited healthcare resources. This study utilized secondary data from WHO, GINA reports, hospital records, and the Kaggle Asthma dataset to establish prevalence patterns and associated complications. Descriptive and comparative statistical techniques were employed alongside thematic analysis to identify educational and psychosocial impacts. A logistic regression predictive model was developed and evaluated using accuracy, recall, and F1-score, achieving an overall accuracy of 74.8%, with strong performance in identifying non-asthmatic children (precision: 0.96, recall: 0.77) but limited sensitivity for asthma-positive cases (recall: 0.25), reflecting class imbalance in the dataset. The model was deployed via a Streamlit interface to estimate asthma risk and complication probabilities based on inputs including child age, gender, environmental exposures, family history, and symptoms.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a logistic regression classifier to predict asthma diagnosis risk among children aged 0–14 years using secondary data from the Kaggle Asthma dataset, with a 718-record test set. The model achieved an overall accuracy of 74.8%, with high precision of 0.96 and recall of 0.77 for non-asthmatic cases, though limited recall of 0.25 for asthma-positive cases due to significant class imbalance (32 positive vs. 686 negative cases). The model was deployed as a Streamlit web application accepting inputs of age, gender, environmental exposures, family history, and symptoms to generate real-time asthma risk predictions and complication probability scores, with recommendations to improve performance through data balancing techniques, incorporation of additional clinical and environmental features, and integration with school and community-based screening programs to complement data-driven early detection efforts.

INTERNSHIP Internship Experience

Consolata Hospital Nkubu — Development of a Risk Prediction Tool for Low Birth Weight Using Comprehensive Maternal Data | Machine Learning for Maternal Health Analytics | AI for Neonatal Risk Assessment

Dataset Used: Maternal and Child Health (MCH) books from Consolata Hospital Nkubu containing maternal health records with variables such as mother's age, blood pressure, BMI, gestational age, number of antenatal visits, hemoglobin levels, and baby's birth weight. Dataset containing maternal health records used to develop a machine learning-based prediction model for newborn birth weight. The model helps identify pregnancies at risk of low birth weight, which is a key factor in neonatal health outcomes. Variables include mother's age, body mass index (BMI), blood pressure, gestational age, number of antenatal visits, hemoglobin levels, and other maternal health indicators used to train regression models for real-time birth weight estimation and early risk detection in maternal healthcare settings.

METHODOLOGY

Internship Centre Work: Receiving and organizing patient files from various departments | Recording patient diagnosis and clinical assessments | File auditing for completeness and compliance with documentation standards | Abstracting key data points (age, delivery mode, birth weight) for analysis | Scanning and renaming birth notification forms using standardized naming conventions | Visualizing departmental data through charts and dashboards | Capstone Project: Extracted maternal health data from MCH books | Data Cleaning: Handled missing values via mean/median imputation, removed duplicates, corrected data types, detected outliers using IQR method, normalized continuous variables | Descriptive Analysis: Computed mean, median, standard deviation, performed correlation analysis to identify features influencing birth weight | Visualization: Generated bar charts, scatter plots, correlation heatmaps using Python | Model Development: Trained and compared Linear Regression, Decision Tree Regressor, and Random Forest Regressor models | Model Evaluation: Assessed performance using R² score, Mean Squared Error (MSE), and Mean Absolute Error (MAE) | Deployment: Developed and deployed interactive Streamlit web application for real-time birth weight prediction

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Logistic regression model achieved 75% accuracy, precision of 0.75, and F1-score of 0.6 for birth weight prediction | Identified key predictors of birth weight: gestational age, BMI, blood pressure, and maternal age as most influential factors | Successfully deployed Streamlit web application (<https://birthweightpred-xq5arrlg7nyoqneqmrhwu.streamlit.app/>) providing healthcare workers with real-time birth weight estimation tool for early risk identification | Demonstrated feasibility of applying machine learning to routine MCH data for predictive analytics in maternal healthcare | Enhanced practical skills in data cleaning, regression modeling, visualization, and web application deployment | Provided actionable recommendations for integrating predictive tools into antenatal care workflows, expanding dataset diversity, and conducting continuous model retraining | Highlighted importance of data quality improvement and digital health system adoption for reliable predictive modeling in resource-limited settings

CURRENT STATUS

Not yet reported



CAPSTONE **Capstone Project**

Analyzing and Predicting the Role of Antenatal Care in Maternal Mortality and Complications

BACKGROUND & METHODOLOGY

Maternal mortality remains high especially in low- and middle-income countries, with many deaths preventable through timely and effective antenatal care (ANC). This study analyzes a hospital-based dataset of 387 mothers to evaluate the impact of ANC visits on maternal mortality and complications, using exploratory data analysis and logistic regression modeling. Variables include ANC visits, complications, socio-demographic factors such as age, education, marital status, parity, and gravida, as well as clinical factors including booking status, mode of delivery, and hospital stay duration. Key findings reveal that women with fewer or no ANC visits had significantly higher rates of maternal death and complications, while increased ANC attendance improved survival outcomes. The logistic regression model achieved an accuracy of 71%, correctly identifying 32 true negatives and 23 true positives, with a Mean Absolute Error of 0.29. The study concludes that early, consistent, and high-quality ANC reduces maternal mortality, with quality and timing of care being just as critical as the number of visits.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a logistic regression model to predict maternal death risk using a hospital-based dataset of 387 mothers, incorporating ANC visits, complications, booking status, education, marital status, parity, gravida, mode of delivery, and hospital stay days as predictors. The model achieved an accuracy of 71%, an MSE of 0.29, and an MAE of 0.29, correctly classifying 32 survivors and 23 maternal deaths while identifying 12 dangerous missed cases. Analysis confirmed a strong protective association between higher ANC attendance and maternal survival, with women having 0–1 visits showing the highest mortality rates and those with 5+ visits showing significantly better outcomes. The model was deployed as a Streamlit web application providing real-time maternal death risk predictions — classified as Low, Medium, or High — with confidence scores, supporting evidence-based clinical decision-making and early intervention planning.

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

The Impact of Screentime on Sleep Quality and Mental Health

BACKGROUND & METHODOLOGY

Excessive recreational screen time among adolescents and young adults, averaging over seven hours daily, has emerged as a pressing public health problem. This study examines the impact of screen exposure on physical health, mental wellbeing, social development, and cognitive performance using a large dataset of 100,000 records. Variables analyzed include daily screen time hours, social media platforms used, TikTok hours, sleep hours, stress levels, and mood scores. Regression analysis revealed that screen time and TikTok hours were significant positive predictors of stress ($R=0.748$) and negative predictors of mood ($R=-0.570$), while sleep duration emerged as the strongest protective factor. Classification models achieved 78–82% accuracy in predicting mental health risk groups. Key findings confirm that 68% of respondents spend 4–8 hours daily on screens, with increased use strongly correlated with higher stress and lower mood scores. The study concludes that excessive screen time poses serious risks to adolescent wellbeing across health, social, and cognitive domains, and recommends collective action from parents, educators, and policymakers to promote balanced digital engagement.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed regression and classification models to predict mental health outcomes — specifically stress levels and mood scores — from digital usage patterns using a dataset of 100,000 records. The stress prediction model achieved an R - of 0.748, explaining approximately 75% of variance, with screen time hours, TikTok usage, and sleep hours identified as the most significant predictors. Classification models achieved 78–82% accuracy in categorizing individuals into low, moderate, and high mental health risk groups. The models were deployed via a Streamlit web application allowing users to input age, screen hours, and lifestyle variables to receive real-time predictions of BMI risk, sleep disturbance probability, and screen behaviour cluster assignments, with recommendations to launch awareness campaigns, integrate screen time guidelines into educational programs, and develop targeted interventions for high-risk urban youth aged 18–24.

INTERNSHIP Internship Experience

Consolata Hospital Nkubu — Development of a Risk Prediction Tool for Low Birth Weight Using Comprehensive Maternal Data | Machine Learning for Maternal Health Analytics | AI for Neonatal Risk Assessment

★ LED TO A JOB OFFER

Dataset Used: Maternal and Child Health (MCH) books from Consolata Hospital Nkubu containing maternal health records with variables such as mother's age, blood pressure, BMI, gestational age, number of antenatal visits, hemoglobin levels, and baby's birth weight. Dataset containing maternal health records used to develop a machine learning-based prediction model for newborn birth weight. The model helps identify pregnancies at risk of low birth weight, which is a key factor in neonatal health outcomes. Variables include mother's age, body mass index (BMI), blood pressure, gestational age, number of antenatal visits, hemoglobin levels, and other maternal health indicators used to train regression models for real-time birth weight estimation and early risk detection in maternal healthcare settings.

METHODOLOGY

Internship Centre Work: Receiving and organizing patient files from various departments | Recording patient diagnosis and clinical assessments | File auditing for completeness and compliance with documentation standards | Abstracting key data points (age, delivery mode, birth weight) for analysis | Scanning and renaming birth notification forms using standardized naming conventions | Visualizing departmental data through charts and dashboards | Capstone Project: Extracted maternal health data from MCH books | Data Cleaning: Handled missing values via mean/median imputation, removed duplicates, corrected data types, detected outliers using IQR method, normalized continuous variables | Descriptive Analysis: Computed mean, median, standard deviation, performed correlation analysis to identify features influencing birth weight | Visualization: Generated bar charts, scatter plots, correlation heatmaps using Python | Model Development: Trained and compared Linear Regression, Decision Tree Regressor, and Random Forest Regressor models | Model Evaluation: Assessed performance using R - score, Mean Squared Error (MSE), and Mean Absolute Error (MAE) | Deployment: Developed and deployed interactive Streamlit web application for real-time birth weight prediction

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Logistic regression model achieved 75% accuracy, precision of 0.75, and F1-score of 0.6 for birth weight prediction | Identified key predictors of birth weight: gestational age, BMI, blood pressure, and maternal age as most influential factors | Successfully deployed Streamlit web application (<https://birthweightpred-xq5arrlg7nyoqneqmvrrhwu.streamlit.app/>) providing healthcare workers with real-time birth weight estimation tool for early risk identification | Demonstrated feasibility of applying machine learning to routine MCH data for predictive analytics in maternal healthcare | Enhanced practical skills in data cleaning, regression modeling, visualization, and web application deployment | Provided actionable recommendations for integrating predictive tools into antenatal care workflows, expanding dataset diversity, and conducting continuous model retraining | Highlighted importance of data quality improvement and digital health system adoption for reliable predictive modeling in resource-limited settings

CURRENT STATUS

Customer service · Unemployed — Actively searching

Sector relevance: Yes — Fully related

SKILLS IN USE

Ethics & bias in AI · Public health data analysis · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

An Analysis of Obesity and Its Determinants Among Young Adults Aged 18–30 Years

BACKGROUND & METHODOLOGY

Obesity is a growing global health challenge associated with increased risks of chronic diseases such as diabetes, cardiovascular disorders, and hypertension. This study utilizes a synthetic obesity dataset of 2,111 records containing demographic, dietary, and lifestyle variables to analyze patterns and develop predictive models for obesity classification among young adults aged 18–30 years. Exploratory data analysis was conducted to examine obesity level distributions, while three machine learning models — logistic regression, decision tree, and random forest — were compared for predictive performance. Key findings highlight that family history of overweight, frequent consumption of high-calorie foods, low vegetable intake, physical inactivity, prolonged screen time, passive transport modes, and frequent alcohol consumption were the most significant risk factors for obesity. The Random Forest Classifier demonstrated the best performance across all obesity categories and was selected as the final model, confirming the potential of predictive analytics to inform targeted public health interventions for young adults.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a Random Forest Classifier to predict obesity categories among young adults aged 18–30 years using the UCI Machine Learning Repository obesity dataset of 2,111 records, incorporating demographic, dietary, and lifestyle variables including age, gender, height, weight, caloric food intake, vegetable consumption, physical activity, screen time, alcohol use, smoking, transport mode, and family history. The Random Forest outperformed logistic regression and decision tree models, achieving the highest accuracy and balanced performance across all obesity classes, and was deployed as a live Streamlit web application at obesity-predictio.streamlit.app. Key protective factors identified included regular physical activity, higher vegetable consumption, and active transport modes, with recommendations for health promotion programs targeting modifiable lifestyle behaviors and the adoption of predictive tools in healthcare settings for early identification of at-risk individuals.

INTERNSHIP Internship Experience

Consolata Hospital Nkubu — Development of a Risk Prediction Tool for Low Birth Weight Using Comprehensive Maternal Data | Machine Learning for Maternal Health Analytics | AI for Neonatal Risk Assessment

Dataset Used: Maternal and Child Health (MCH) books from Consolata Hospital Nkubu containing maternal health records with variables such as mother's age, blood pressure, BMI, gestational age, number of antenatal visits, hemoglobin levels, and baby's birth weight. Dataset containing maternal health records used to develop a machine learning-based prediction model for newborn birth weight. The model helps identify pregnancies at risk of low birth weight, which is a key factor in neonatal health outcomes. Variables include mother's age, body mass index (BMI), blood pressure, gestational age, number of antenatal visits, hemoglobin levels, and other maternal health indicators used to train regression models for real-time birth weight estimation and early risk detection in maternal healthcare settings.

METHODOLOGY

Internship Centre Work: Receiving and organizing patient files from various departments | Recording patient diagnosis and clinical assessments | File auditing for completeness and compliance with documentation standards | Abstracting key data points (age, delivery mode, birth weight) for analysis | Scanning and renaming birth notification forms using standardized naming conventions | Visualizing departmental data through charts and dashboards | Capstone Project: Extracted maternal health data from MCH books | Data Cleaning: Handled missing values via mean/median imputation, removed duplicates, corrected data types, detected outliers using IQR method, normalized continuous variables | Descriptive Analysis: Computed mean, median, standard deviation, performed correlation analysis to identify features influencing birth weight | Visualization: Generated bar charts, scatter plots, correlation heatmaps using Python | Model Development: Trained and compared Linear Regression, Decision Tree Regressor, and Random Forest Regressor models | Model Evaluation: Assessed performance using R– score, Mean Squared Error (MSE), and Mean Absolute Error (MAE) | Deployment: Developed and deployed interactive Streamlit web application for real-time birth weight prediction

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Logistic regression model achieved 75% accuracy, precision of 0.75, and F1-score of 0.6 for birth weight prediction | Identified key predictors of birth weight: gestational age, BMI, blood pressure, and maternal age as most influential factors | Successfully deployed Streamlit web application (<https://birthweightpred-xq5arrlg7nyoqneqmrhwu.streamlit.app/>) providing healthcare workers with real-time birth weight estimation tool for early risk identification | Demonstrated feasibility of applying machine learning to routine MCH data for predictive analytics in maternal healthcare | Enhanced practical skills in data cleaning, regression modeling, visualization, and web application deployment | Provided actionable recommendations for integrating predictive tools into antenatal care workflows, expanding dataset diversity, and conducting continuous model retraining | Highlighted importance of data quality improvement and digital health system adoption for reliable predictive modeling in resource-limited settings

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Predictive Modeling of Babies' Birth Weight Outcomes Utilizing the Patients' Data Using Linear Regression Model

BACKGROUND & METHODOLOGY

This report outlines an analytical study aimed at identifying and understanding the key factors influencing birth weight, with a particular focus on predicting low birth weight (LBW), defined as less than 2,500 grams. The study employed descriptive statistics, exploratory data analysis, and predictive modeling using linear regression on the birthweight.csv dataset, which contains 642 records with 11 variables including maternal age, hypertension status, gestational weeks, infant sex, ethnicity, and birth weight. The model achieved an R² of 66% and an Adjusted R² of 65%, indicating moderate explanatory power. Gestational age emerged as the strongest predictor of birth weight, with no clear relationship found between maternal age and birth weight. The study provides a framework for identifying at-risk pregnancies and recommends targeted prenatal interventions, strengthened educational campaigns for expectant mothers, and further research into socio-economic and environmental factors influencing birth weight.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a linear regression model to predict infant birth weight outcomes using maternal and gestational data from a dataset of 642 records, incorporating variables such as maternal age, hypertension status, gestational weeks, infant sex, and ethnicity. The model achieved an R² of 66% and an Adjusted R² of 65%, demonstrating moderate predictive power with gestational age identified as the strongest predictor of birth weight. A web-based interface was proposed to allow healthcare providers to input patient characteristics and receive predicted birth weight probabilities, with recommendations to strengthen prenatal care programs, develop maternal health education campaigns, and explore advanced models such as Random Forest and Gradient Boosting for improved accuracy.

INTERNSHIP Internship Experience

Khairat Medical Center — Child Growth Monitoring and Height Prediction

Dataset Used: Child growth monitoring records extracted from Khairat Medical Center's routine child health registry (August 2024). Dataset of child health records containing serial number, date, age in months, gender, weight (kg), and height (cm), collected from routine growth monitoring clinic visits at Khairat Medical Center, used to analyze growth patterns and predict child height.

METHODOLOGY

Cleaned and preprocessed child health records using Python, handling missing values and formatting numeric columns; conducted descriptive statistical analysis and exploratory data analysis using Pandas, Matplotlib, and Seaborn; performed crosstabulation to examine relationships between gender, age group, and weight category; built and evaluated a linear regression model using an 80/20 train-test split; and deployed a Streamlit web application serialized with joblib to allow health workers to input age, weight, and gender for instant height predictions.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a linear regression model to predict child height using age, weight, and gender, achieving an R² of 0.85 and RMSE of 3.74 cm, explaining 85% of height variation in the dataset; identified age as the strongest predictor (correlation of 0.84 with height), followed by weight (0.80), while gender had minimal influence; regression equation established as $\text{Height} = 53.52 + (0.67 \times \text{Age}) + (1.21 \times \text{Weight}) + (0.31 \times \text{Gender})$; deployed a functional Streamlit web application for real-time height prediction at clinic level, with recommendations to integrate the tool into child health monitoring systems and expand the dataset with nutritional and parental height variables.

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Development of a Linear Regression Model for Predicting Child Weight Using Age and Sex

BACKGROUND & METHODOLOGY

Child health and nutrition are critical indicators of socio-economic development, especially in low- and middle-income countries. A persistent challenge in this field is the accurate determination of children's age and its relationship to growth parameters such as weight, which limits effective monitoring and timely intervention in cases of malnutrition. This study developed a linear regression model capable of predicting child weight based on age and sex, using secondary data from child health records guided by findings from the Demographic and Health Surveys (DHS) and Multiple Indicator Cluster Surveys (MICS). The dataset was preprocessed and analyzed using descriptive statistics and linear regression modeling, with child weight as the dependent variable and age and sex as independent variables. The model achieved an R^2 of 0.702 and an Adjusted R^2 of 0.696, with a Mean Absolute Error of 1.42 kg and a Root Mean Squared Error of 1.90 kg, confirming that age is the strongest predictor of child weight, with sex contributing modestly. The study concludes that predictive modeling can complement traditional survey approaches in health monitoring and recommends integration into public health systems and future expansion to include socio-economic variables.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a linear regression model to predict child weight using age (in months) and sex as independent variables, trained and tested on a child health dataset sourced from DHS and MICS records. The model achieved an R^2 of 0.702 and Adjusted R^2 of 0.696, explaining approximately 70% of variation in child weight, with a MAE of 1.42 kg and RMSE of 1.90 kg, indicating predictions accurate to within -1.5 – 2 kg. Key findings showed age increasing weight by 0.17 kg per month and males being approximately 0.45 kg heavier than females. The model was deployed as a Child Weight Prediction App allowing health practitioners to input a child's age and sex to obtain instant weight estimates, with recommendations to expand the model with socio-economic and health-related variables for improved accuracy.

INTERNSHIP Internship Experience

Mbungoni Catholic Health Dispensary — HIV Viral Load Outcome Prediction and Treatment Adherence Monitoring

Dataset Used: Facility-level electronic medical records from Kenya EMR system, Mbungoni Catholic Health Dispensary. HIV patient dataset containing clinical and treatment variables including sex, age, initial viral load results, number of Enhanced Adherence Counselling (EAC) sessions, turnaround time (TAT) for tests, and latest viral load results, used to predict viral load suppression status (suppressed vs unsuppressed) and support HIV program performance monitoring.

METHODOLOGY

Registered patients in the Kenya EMR system; entered and updated HIV testing and treatment data; validated and cleaned patient files; supported HIV Testing Services (HTS) by reviewing and organizing records; assisted with preparation of monthly program performance reports; cleaned and recoded categorical variables; applied descriptive statistics and Pearson correlation analysis; and developed regression-based predictive models using Python, Pandas, NumPy, Scikit-learn, and Matplotlib to predict latest viral load outcomes.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a logistic regression model to predict HIV viral load outcomes (suppressed vs unsuppressed) using clinical and adherence variables, achieving an overall accuracy of 71%, sensitivity of 86%, and specificity of 50%, demonstrating reliable detection of patients with high viral load who require targeted intervention; key findings revealed that 91.7% of patients were initiated on EAC in line with national guidelines, with initial viral load results and number of EAC sessions identified as the strongest predictors of latest viral load outcome; the model's high sensitivity makes it a practical tool for early identification of high-risk patients, with recommendations to strengthen facility-level data entry and validation, train clinical staff in basic data analysis, and scale the predictive modeling approach to other facilities for improved HIV program monitoring.

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Development of a Linear Regression Model for Predicting Child Weight Using Age and Sex

BACKGROUND & METHODOLOGY

Child health and nutrition are critical indicators of socio-economic development, especially in low- and middle-income countries. A persistent challenge in this field is the accurate determination of children's age and its relationship to growth parameters such as weight, which limits effective monitoring and timely intervention in cases of malnutrition. This study developed a linear regression model capable of predicting child weight based on age and sex, using secondary data from child health records guided by findings from the Demographic and Health Surveys (DHS) and Multiple Indicator Cluster Surveys (MICS). The dataset was preprocessed and analyzed using descriptive statistics and linear regression modeling, with child weight as the dependent variable and age and sex as independent variables. The model achieved an R^2 of 0.702 and an Adjusted R^2 of 0.696, with a Mean Absolute Error of 1.42 kg and a Root Mean Squared Error of 1.90 kg, confirming that age is the strongest predictor of child weight, with sex contributing modestly. The study concludes that predictive modeling can complement traditional survey approaches in health monitoring and recommends integration into public health systems and future expansion to include socio-economic variables.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a linear regression model to predict child weight using age (in months) and sex as independent variables, trained and tested on a child health dataset sourced from DHS and MICS records. The model achieved an R^2 of 0.702 and Adjusted R^2 of 0.696, explaining approximately 70% of variation in child weight, with a MAE of 1.42 kg and RMSE of 1.90 kg, indicating predictions accurate to within ± 1.5 –2 kg. Key findings showed age increasing weight by 0.17 kg per month and males being approximately 0.45 kg heavier than females. The model was deployed as a Child Weight Prediction App allowing health practitioners to input a child's age and sex to obtain instant weight estimates, with recommendations to expand the model with socio-economic and health-related variables for improved accuracy.

INTERNSHIP Internship Experience

Moving the Goalpost Kilifi — Development of a Data-Driven Participant Prediction App for Forecasting Attendance in Community Programs | Machine Learning for Community Engagement Optimization

Dataset Used: Attendance and demographic data from community programs at Moving the Goalpost Kilifi (January—August 2025, 780+ validated participant records). Dataset containing participant engagement records with variables including county location, activity type, activity date, and total participants reached, used to develop a predictive application that forecasts attendance levels for community-based programs using linear regression modeling and Streamlit-based interactive dashboards.

METHODOLOGY

Internship Centre Work: Supported data entry and management for community and health program records; conducted data cleaning and consistency checks using Excel and Python; assisted in preparing monthly reports and visual dashboards; participated in community engagement meetings to interpret data trends | Capstone Project: Cleaned and validated participant data by handling missing values, standardizing date formats, and recoding categorical variables (Activity area, Activity place, Participants reached); applied exploratory data analysis to identify attendance patterns by county, activity type, and time period; trained Linear Regression predictive model using Python (pandas, numpy, scikit-learn, matplotlib); developed Streamlit-based web application for real-time attendance forecasting and visualization; evaluated model performance using R^2 , MAE, and MSE metrics

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Validated over 780 participant records and improved monthly reporting data accuracy by 70% | Developed interactive dashboards showing participation trends by program type and region, revealing Kilifi County as dominant contributor (68.9% of participants) followed by Kwale (24.3%) | Identified Football League as highest-engagement activity (300+ participants) while Capacity Building sessions had lowest reach | Linear Regression model achieved $R^2 = 0.469$ (explaining ~47% of attendance variation), MAE = 7.42 participants, MSE = 148.68 | Key predictors identified: location/county, program type, and activity date | Successfully deployed Streamlit web app providing program coordinators with real-time attendance forecasts for improved planning and resource allocation | Enhanced technical skills in Python, data cleaning, predictive modeling, and app deployment while contributing actionable insights for community program optimization

CURRENT STATUS

Information technology assistant · Unemployed — Actively searching

Sector relevance: Yes – Fully related

SKILLS IN USE

Python · Data collection · Data cleaning & wrangling · Data visualization · ML modeling · Model evaluation · MLOps / Model deployment

Confidence applying ENGAGE skills: **5 / 5**



CAPSTONE Capstone Project

Cholera Test Result Analysis and Prediction

BACKGROUND & METHODOLOGY

This project aimed to build and deploy a machine learning model to predict cholera outbreaks using a dataset of 312 records and 10 features combining health, demographic, and environmental indicators. Variables included sex, age, sub-county of residence, water source, sanitation facility, income level, informal settlement status, and cholera test result. After cleaning and analyzing the data, a binary logistic regression model was trained and tested, with cholera test result as the dependent variable. The model produced reliable results with an accuracy of 79%, precision of 0.83, recall of 0.94, and an F1-score of 0.88. Key findings revealed that individuals in informal settlements had a positivity rate of 40% compared to 16% in formal settlements, and those using public wells and boreholes faced significantly higher infection risk. To make the system practical, the model was deployed as an interactive web application using Streamlit, enabling real-time cholera risk predictions to support early public health intervention and resource allocation.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a binary logistic regression model to predict cholera test results using health, environmental, and socio-economic variables from a dataset of 312 records across multiple sub-counties. The model achieved an accuracy of 79%, precision of 0.83, recall of 0.94, and an F1-score of 0.88, identifying water source, sanitation type, and age as the most significant predictors of cholera infection. The model was deployed as a user-friendly Streamlit web application enabling real-time risk predictions, with the tool providing actionable insights for public health teams to prioritize high-risk sub-counties, improve water safety and sanitation infrastructure, and direct preventive interventions toward the most vulnerable populations.

INTERNSHIP Internship Experience

Moving the Goalpost, Kilifi — Youth Program Participation, Attendance Prediction and Inclusion Monitoring

Dataset Used: Organization's routine program monitoring hard copy records digitized via Kobo Toolbox, covering activity sessions across Kilifi, Kwale, Mombasa, and Tana River counties. Program monitoring dataset containing participant attendance records disaggregated by activity type (football coaching, football leagues, peer education, mental health sessions, capacity building, climate action, safeguarding), age bands (7–10, 11–13, 14–25, 26+), sex, county of participation, and disability status, used to analyze program reach, coverage gaps, and attendance trends over time.

METHODOLOGY

Performed routine data entry from hard copy records into Kobo Toolbox; cleaned and validated monitoring data including date parsing, numeric coercion, harmonization of activity labels, standardization of county names, and filling missing counts; conducted descriptive and time-series analyses disaggregated by activity type, age band, county, and disability status using Python, pandas, and Matplotlib; produced five publication-ready visualizations and a dashboard template; prepared monthly CSV summaries and pivot tables for program teams; and built and evaluated a Random Forest Classifier to predict session attendance.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a Random Forest Classifier achieving approximately 75% accuracy in predicting session attendance, demonstrating reliable potential for data-driven program planning; key findings identified Kilifi as the dominant participation hub with over 12,000 participants, football-based activities as the strongest youth mobilizers, and youth aged 11–25 comprising nearly 70% of all participants; flagged critically low disability inclusion rates and underrepresentation in Mombasa and Tana River as priority gaps; produced an operational dashboard template and monthly summary reports to support M&E reporting, with recommendations to expand outreach to underserved counties, adopt predictive dashboards for real-time planning, and train staff in data use for evidence-based programming.

CURRENT STATUS

Not yet reported



CAPSTONE Capstone Project

Predictive Modeling of Bloodfilm Outcomes Utilizing Patient Data

BACKGROUND & METHODOLOGY

This project aimed to analyze a patient dataset to predict blood film results (positive or negative) as a proxy for malaria diagnosis. Using data from the maltreat.csv file containing demographic and health metric variables including sex, ethnicity, age, weight, and temperature, the project applied data cleaning, statistical exploration, and machine learning techniques. A linear regression model was built within a machine learning pipeline incorporating imputation, feature scaling, and target variable encoding. The model was designed to support early case detection and prioritization of confirmatory malaria testing, in line with WHO and UNICEF goals around reducing malaria mortality, particularly among children under five in sub-Saharan Africa. Key findings from exploratory analysis included a positive relationship between age and weight, a high concentration of younger patients in the dataset, and the identification of data leakage between correlated variables. The model was deployed as a Streamlit web application enabling real-time bloodfilm outcome predictions from patient inputs.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a machine learning pipeline to predict patient bloodfilm outcomes using demographic and clinical variables including age, weight, temperature, sex, and ethnicity, sourced from the maltreat.csv dataset of 652 records. The model incorporated missing value imputation, feature scaling via StandardScaler, and linear regression for binary classification, and was deployed as a user-friendly Streamlit web application for real-time predictions. Key analytical findings identified age and weight as positively correlated predictors and flagged data leakage between falcipasex and falcsex variables, improving model reliability. Future recommendations include transitioning to a classification model such as logistic regression for more appropriate evaluation metrics, and expanding the dataset to improve predictive accuracy and generalizability for malaria screening support.

INTERNSHIP Internship Experience

Beacon of Hope Hospital, Kajiado County — Outpatient Random Blood Sugar Risk Prediction and NCD Screening

Dataset Used: Excel-based outpatient records from Beacon of Hope Hospital, Kajiado County (over 700 patient records). Outpatient dataset of over 700 patient records containing age, sex, village, clinical diagnosis, and random blood sugar (RBS) test results, with a binary target variable (rbs_status) engineered to classify patients as high-risk or non-risk for elevated blood sugar, used to analyze NCD screening patterns and predict blood sugar risk.

METHODOLOGY

Supported data entry and validation of outpatient records; assisted in preparing monthly NCD screening reports; conducted data quality checks and reconciled missing values; cleaned and preprocessed Excel-based RBS dataset; encoded categorical variables including sex, village, and diagnosis; engineered binary target variable (rbs_status); trained a Random Forest Classifier with balanced class weights; generated visualizations of age distribution, sex distribution, top diagnoses, diabetes-hypertension co-occurrence, and monthly RBS trends; and deployed a Streamlit web application with interactive real-time prediction and downloadable patient reports.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a Random Forest Classifier to predict high blood sugar risk in outpatient settings, achieving approximately 76% accuracy with binary prediction and probability score outputs; key findings identified age (median 60 years for high RBS cases), Type 2 diabetes mellitus, and essential hypertension as the strongest predictors of elevated blood sugar; co-occurrence analysis revealed that 91 patients (43% of all diabetic cases) had both diabetes and hypertension, indicating significant NCD comorbidity burden; seasonal analysis showed peak high RBS cases in April and June; the deployed Streamlit app enables health workers to input patient details and receive real-time risk predictions with downloadable reports, with recommendations for cross-departmental data linkage, integrated NCD screening protocols, and age-specific screening exemptions for children under 5.

CURRENT STATUS

I did a course in Community health, but am unemployed · Unemployed — Actively searching
Sector relevance: Yes — Fully related

SKILLS IN USE

Python · Public health data analysis

Confidence applying ENGAGE skills: 5 / 5



CAPSTONE Capstone Project

Cholera Test Result Analysis and Prediction

BACKGROUND & METHODOLOGY

This project aimed to build and deploy a machine learning model to predict cholera outbreaks using a dataset of 312 records and 10 features combining health, demographic, and environmental indicators. Variables included sex, age, sub-county of residence, water source, sanitation facility, income level, informal settlement status, and cholera test result. After cleaning and analyzing the data, a binary logistic regression model was trained and tested, with cholera test result as the dependent variable. The model produced reliable results with an accuracy of 79%, precision of 0.83, recall of 0.94, and an F1-score of 0.88. Key findings revealed that individuals in informal settlements had a positivity rate of 40% compared to 16% in formal settlements, and those using public wells and boreholes faced significantly higher infection risk. To make the system practical, the model was deployed as an interactive web application using Streamlit, enabling real-time cholera risk predictions to support early public health intervention and resource allocation.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed and deployed a binary logistic regression model to predict cholera test results using health, environmental, and socio-economic variables from a dataset of 312 records across multiple sub-counties. The model achieved an accuracy of 79%, precision of 0.83, recall of 0.94, and an F1-score of 0.88, identifying water source, sanitation type, and age as the most significant predictors of cholera infection. The model was deployed as a user-friendly Streamlit web application enabling real-time risk predictions, with the tool providing actionable insights for public health teams to prioritize high-risk sub-counties, improve water safety and sanitation infrastructure, and direct preventive interventions toward the most vulnerable populations.

INTERNSHIP Internship Experience

Moving the Goalpost (MTG), Kilifi County — Community Engagement and Sports-Based Program Monitoring and Participation Analysis

Dataset Used: KoboToolbox submissions and online Excel monitoring sheet, MTG Community Engagement Department, Coastal Kenya (772 records across Kilifi, Kwale, Mombasa, and Tana River counties). Program monitoring dataset of 772 records and 35 variables capturing participant demographics, activity types, attendance figures, county, department, project name, age group, and gender, covering activities including football leagues, peer education sessions, mental health workshops, Tumanyane sessions, and capacity-building events across four Coastal counties.

METHODOLOGY

Performed data entry and validation of community activity records into KoboToolbox and an online Excel monitoring sheet; cross-checked participant totals, activity types, and session dates for accuracy; conducted data cleaning including removing duplicates, handling missing values, standardizing categorical variables, and validating numeric participant counts; generated descriptive statistics and visualizations of participant distribution by gender, county, department, and age group using Python (pandas, matplotlib, seaborn) and Microsoft Excel; and developed and evaluated a linear regression model to predict total participants reached based on activity type, department, and county.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully analyzed 772 community engagement records, revealing Kilifi County as the dominant hub with 68.9% of all participants, followed by Kwale at 24.2%, Mombasa at 5.0%, and Tana River at 1.8%; gender analysis showed female participants comprised 90.2% of total reach, with football activities attracting the highest female engagement; youth aged 14–25 accounted for the majority of participants, confirming alignment with the program's youth empowerment mandate; the linear regression model provided directional insights into key participation drivers despite limited predictive accuracy, with recommendations to explore advanced models such as Random Forest and Ridge Regression, automate KoboToolbox-Excel data synchronization, implement monthly data quality checks, and develop visual dashboards to support evidence-based program planning and resource allocation.

CURRENT STATUS

Not yet reported

**CAPSTONE** **Capstone Project***Report on Child Health Data Bungoma County***BACKGROUND & METHODOLOGY**

This study examined the social and economic factors affecting children's health in Bungoma County, Kenya, using household survey data from 2013–2014. Descriptive analysis and regression models were applied to understand patterns across 874 children and approximately 300 variables. Findings revealed that mothers' education, family wealth, and access to healthcare played the most significant roles in shaping children's health outcomes. Children from poorer families and those whose mothers had little or no education were more likely to face illnesses and poor nutrition. The regression model demonstrated strong predictive power with an R - of 0.7095 and an Adjusted R - of 0.7033, confirming the reliability of the results. The study concludes that children's health is influenced not only by medical care but also by the social and economic conditions they grow up in, and recommends increased investment in maternal education, family livelihoods, and equitable access to healthcare services.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a multiple linear regression model to analyze and predict child health outcomes in Bungoma County using household survey data from 2013–2014, incorporating predictors such as household wealth index, maternal education, child age and sex, area of residence, and children's weight. The model achieved a strong R - of 0.7095 and an Adjusted R - of 0.7033, identifying maternal education, household wealth, and access to healthcare as the most significant determinants of child wellbeing. A web-based dashboard prototype was proposed for local health stakeholders to enable localized, equity-focused analysis, with recommendations to invest in maternal education programs, poverty-reduction initiatives, and expanded access to child health services across underserved areas of Bungoma County.

INTERNSHIP **Internship Experience***Wananchi Nursing Home — Kilifi — Application of Random Forest Algorithm in Predicting Outpatient Diagnosis Outcomes*

Dataset Used: Outpatient register data from Wananchi Nursing Home, Kilifi (January–August 2025, 15,722 patient records). Dataset containing outpatient visit records with variables including age, gender, number of visits, and diagnosis outcomes (Confirmed or Suspected), used to predict diagnosis outcomes using machine learning techniques and improve outpatient reporting efficiency.

METHODOLOGY

Participated in data entry, validation, and preparation of outpatient records; updated registers, organized diagnosis records, and assisted in report compilation; supported monthly analysis of patient attendance and diagnosis trends; cleaned data by removing missing entries, encoding gender (Male=0, Female=1), and defining diagnosis stages; trained Random Forest Classifier using Python with 100 decision trees (70% training, 30% testing split); evaluated model using accuracy, precision, recall, and confusion matrix; created visualizations using Python, Pandas, Scikit-learn, Matplotlib, and Streamlit.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Achieved 99.9% model accuracy with perfect precision and recall (1.00) for Confirmed cases; identified Age and Number of Visits as key predictors of diagnosis outcomes; revealed data imbalance challenge (4,500 Confirmed vs. 4 Suspected cases) causing model bias toward majority class; demonstrated females represented 52.7% of patients with most being children and adults aged 25-40; identified seasonal variation with peak demand May–July; improved data science skills from data preprocessing and visualization to model evaluation and result interpretation; demonstrated how AI can improve efficiency in healthcare reporting and support evidence-based decision-making

CURRENT STATUS**No job · Unemployed — Actively searching**

Sector relevance: Yes – Fully related

Confidence applying ENGAGE skills: **4 / 5**



CAPSTONE Capstone Project

Predictive Modeling of Babies' Birth Weight Outcomes Utilizing the Patients' Data Using Linear Regression Model

BACKGROUND & METHODOLOGY

This report outlines an analytical study aimed at identifying and understanding the key factors influencing birth weight, with a particular focus on predicting low birth weight (LBW), defined as less than 2,500 grams. The study employed descriptive statistics, exploratory data analysis, and predictive modeling using linear regression on the birthweight.csv dataset, which contains 642 records with 11 variables including maternal age, hypertension status, gestational weeks, infant sex, ethnicity, and birth weight. The model achieved an R² of 66% and an Adjusted R² of 65%, indicating moderate explanatory power. Gestational age emerged as the strongest predictor of birth weight, with no clear relationship found between maternal age and birth weight. The study provides a framework for identifying at-risk pregnancies and recommends targeted prenatal interventions, strengthened educational campaigns for expectant mothers, and further research into socio-economic and environmental factors influencing birth weight.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a linear regression model to predict infant birth weight outcomes using maternal and gestational data from a dataset of 642 records, incorporating variables such as maternal age, hypertension status, gestational weeks, infant sex, and ethnicity. The model achieved an R² of 66% and an Adjusted R² of 65%, demonstrating moderate predictive power with gestational age identified as the strongest predictor of birth weight. A web-based interface was proposed to allow healthcare providers to input patient characteristics and receive predicted birth weight probabilities, with recommendations to strengthen prenatal care programs, develop maternal health education campaigns, and explore advanced models such as Random Forest and Gradient Boosting for improved accuracy.

INTERNSHIP Internship Experience

Beacon of Hope Hospital — Routine Immunization Coverage and Dropout Analysis

Dataset Used: Routine immunization register data from Beacon of Hope Hospital (September-October 2025). Dataset containing routine immunization records with variables including village location, gender, vaccine types (BCG, DPT1-3, PCV1-3, OPV1-3, ROTA1-3, MR1), and immunization completion status (Fully Immunized vs Not Fully Immunized), used to analyze coverage patterns, dropout rates, and geographic and demographic disparities in vaccine delivery.

METHODOLOGY

Cleaned and standardized vaccine columns using pandas; encoded dose completion as binary indicators (0 = not received, 1 = received); aggregated data by village, gender, and vaccine type; used matplotlib and seaborn for static visualizations including bar charts, pie charts, and coverage comparisons; deployed interactive Streamlit dashboards for real-time monitoring; created visualizations to highlight geographic disparities (Immunization Status by Village) and gender gaps (Immunization Status by Gender); addressed class imbalance in predictive modeling using class_weight='balanced'; debugged Streamlit prediction issues by verifying input encoding and feature alignment.

CURRENT STATUS

Still applying for a job · Unemployed — Actively searching

Sector relevance: Yes – Fully related

SKILLS IN USE

Python · Data cleaning & wrangling · Data visualization · ML modeling · MLOps / Model deployment · Deep Learning

Confidence applying ENGAGE skills: **4 / 5**



CAPSTONE Capstone Project

Predictive Modeling of Bloodfilm Outcomes Utilizing Patient Data

BACKGROUND & METHODOLOGY

This project aimed to analyze a patient dataset to predict blood film results (positive or negative) as a proxy for malaria diagnosis. Using data from the maltreat.csv file containing demographic and health metric variables including sex, ethnicity, age, weight, and temperature, the project applied data cleaning, statistical exploration, and machine learning techniques. A linear regression model was built within a machine learning pipeline incorporating imputation, feature scaling, and target variable encoding. The model was designed to support early case detection and prioritization of confirmatory malaria testing, in line with WHO and UNICEF goals around reducing malaria mortality, particularly among children under five in sub-Saharan Africa. Key findings from exploratory analysis included a positive relationship between age and weight, a high concentration of younger patients in the dataset, and the identification of data leakage between correlated variables. The model was deployed as a Streamlit web application enabling real-time bloodfilm outcome predictions from patient inputs.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a machine learning pipeline to predict patient bloodfilm outcomes using demographic and clinical variables including age, weight, temperature, sex, and ethnicity, sourced from the maltreat.csv dataset of 652 records. The model incorporated missing value imputation, feature scaling via StandardScaler, and linear regression for binary classification, and was deployed as a user-friendly Streamlit web application for real-time predictions. Key analytical findings identified age and weight as positively correlated predictors and flagged data leakage between falcipasex and falcsex variables, improving model reliability. Future recommendations include transitioning to a classification model such as logistic regression for more appropriate evaluation metrics, and expanding the dataset to improve predictive accuracy and generalizability for malaria screening support.

CURRENT STATUS

Not yet reported

**CAPSTONE** **Capstone Project***Report on Child Health Data Bungoma County***BACKGROUND & METHODOLOGY**

This study examined the social and economic factors affecting children's health in Bungoma County, Kenya, using household survey data from 2013–2014. Descriptive analysis and regression models were applied to understand patterns across 874 children and approximately 300 variables. Findings revealed that mothers' education, family wealth, and access to healthcare played the most significant roles in shaping children's health outcomes. Children from poorer families and those whose mothers had little or no education were more likely to face illnesses and poor nutrition. The regression model demonstrated strong predictive power with an R - of 0.7095 and an Adjusted R - of 0.7033, confirming the reliability of the results. The study concludes that children's health is influenced not only by medical care but also by the social and economic conditions they grow up in, and recommends increased investment in maternal education, family livelihoods, and equitable access to healthcare services.

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Successfully developed a multiple linear regression model to analyze and predict child health outcomes in Bungoma County using household survey data from 2013–2014, incorporating predictors such as household wealth index, maternal education, child age and sex, area of residence, and children's weight. The model achieved a strong R - of 0.7095 and an Adjusted R - of 0.7033, identifying maternal education, household wealth, and access to healthcare as the most significant determinants of child wellbeing. A web-based dashboard prototype was proposed for local health stakeholders to enable localized, equity-focused analysis, with recommendations to invest in maternal education programs, poverty-reduction initiatives, and expanded access to child health services across underserved areas of Bungoma County.

INTERNSHIP **Internship Experience***Mbungoni Catholic Health Dispensary — Predicting Latest Viral Load Results of HIV Patients Using Clinical and Treatment Data / Machine Learning for HIV Program Monitoring*

Dataset Used: Facility-level electronic medical records from Kenya EMR system at Mbungoni Catholic Health Dispensary (HIV patient registers with demographic, treatment, and viral load data). Project applying data science methods to predict latest viral load (VL) outcomes among HIV patients to support program monitoring by identifying patients at risk of high VL and strengthening adherence interventions. Dataset includes variables such as age, sex, initial VL results, number of Enhanced Adherence Counselling (EAC) sessions, and turnaround time for tests, used to stratify patients into suppressed versus unsuppressed categories for targeted clinical follow-up.

METHODOLOGY

Internship Centre Work: Registered patients in KenyaEMR system; entered and updated HIV testing and treatment data; validated and cleaned patient files; supported HTS services by reviewing and organizing records; assisted with preparation of monthly program performance reports | Capstone Project: Data cleaning (handled missing values, corrected inconsistencies, recoded categorical variables: sex, initial VL, EAC sessions); performed descriptive statistics and Pearson correlation analysis; developed logistic regression predictive model for VL outcomes (suppressed vs unsuppressed); generated visualizations using Python, Pandas, NumPy, Scikit-learn, and Matplotlib; evaluated model using accuracy, sensitivity, specificity, and confusion matrix

FINDINGS, CONCLUSIONS & RECOMMENDATIONS

Logistic regression model achieved 71% accuracy with high sensitivity (86%) for detecting patients with high viral load and moderate specificity (50%) for identifying suppressed cases | Identified initial high VL results and number of EAC sessions as significant predictors of latest VL outcome | Model demonstrated strong utility for flagging high-risk patients requiring enhanced adherence counselling and closer clinical follow-up | Enabled patient stratification to support early interventions and targeted resource allocation | Improved practical data science skills in public health settings including data cleaning, analysis, modeling, and interpretation | Contributed to strengthening HIV program reporting and patient monitoring at facility level | Highlighted importance of data quality, staff collaboration, and iterative model testing | Challenges encountered: missing/incomplete records, data inconsistencies, limited computing resources

CURRENT STATUS**I dont have a job · Unemployed — Actively searching**

Sector relevance: Yes — Fully related

SKILLS IN USE

Python · Data collection · Data visualization · Model evaluation · Communication & teamwork

Confidence applying ENGAGE skills: **5 / 5**